

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\124-C-NVS.pvdb

Date of Analysis : Apr 10,2026 4:31pm

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User

FileName : 124-C-NVS

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Vessel Design Summary:

Step: 34 4:31pm Apr 10,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1960.000 mm ID
Vessel Design Length, Tangent to Tangent 20.68 m
Specified Datum Line Distance 0.00 m
Shell Side Design Temperature 150 °C
Channel Side Design Temperature 82 °C
Shell Side Design Pressure 18.001 N/mm²
Channel Side Design Pressure 1.000 N/mm²
Shop Shell Side Test Pressure 23.401 N/mm²
Shop Channel Side Test Pressure 1.300 N/mm²
Wind Design Code ASCE/SEI 7-16
Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Flange	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-266 2	...	...	K03506	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-179	...	...	K01200	
Tubesheet	SA-266 2	...	...	K03506	
Jacket	SA-516 70	...	...	K02700	
Jkt Closure	SA-516 70	...	...	K02700	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (N/mm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Channel COV LEFT	1.000	0.10	2.010	3.0000	N/A	No
Channel FLG LEFT	1.000	0.10	1.740	3.0000	N/A	No
Channel - rear	1.019	0.10	2.620	3.0000	N/A	No
shell	18.012	0.00	18.300	3.0000	N/A	No
shell cone -rea	18.001	0.00	18.300	3.0000	N/A	No
Shell-2	18.012	0.00	18.300	3.0000	N/A	No
shell cone- fro	18.001	0.00	18.300	3.0000	N/A	No
shell- front	18.012	0.00	18.300	3.0000	N/A	No
Channel front	1.019	0.10	2.620	3.0000	N/A	No
Channel FLG RIGHT	1.000	0.10	1.720	3.0000	N/A	No
Channel COV RIGHT	1.000	0.00	1.560	3.0000	N/A	No



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Vessel Design Summary:

Step: 34

4:31pm Apr 10, 2026

Liquid Level: 1.96 m Dens.: 999.542 kg/m³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev m	Element Length m	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	0.2	0.2	60.0	150.0	104.3	...	1.00	1.00
Body Flg	0.3	0.2	60.0	150.0	133.6	...	1.00	1.00
Cylinder	2.3	2.0	22.0	22.0	10.3	...	1.00	1.00
Cylinder	4.7	2.0	145.0	145.0	142.2	...	1.00	1.00
Cylinder	5.4	0.7	145.0	145.0	142.1	...	1.00	1.00
Cylinder	15.3	9.9	145.0	145.0	142.2	...	1.00	1.00
Cylinder	16.0	0.7	145.0	145.0	142.1	...	1.00	1.00
Cylinder	18.0	2.0	145.0	145.0	142.2	...	1.00	1.00
Cylinder	20.3	2.0	22.0	22.0	10.3	...	1.00	1.00
Body Flg	20.5	0.2	22.0	150.0	134.1	...	1.00	1.00
Body Flg	20.7	0.2	22.0	150.0	133.8	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure N/mm²	External M.A.W.P. N/mm²
10	20	150	50.7492	0.10131	No Calc
20	30	150	126.619	0.10131	No Calc
30	Tsht	22	8.96634	0.10131	1.55
Tsht	40	22	No Calc	0.10131	No Calc
40	Jckt	145	No Calc	...	17.6
Jckt	Jckt	145	No Calc	...	17.6
Jckt	50	145	No Calc	...	17.6
50	60	145	No Calc	...	17.6
60	70	145	No Calc	...	17.6
70	80	145	No Calc	...	17.6
80	Jckt	145	No Calc	...	17.6
Jckt	Jckt	145	No Calc	...	17.6
Jckt	Tsht	145	No Calc	...	17.6
Tsht	90	145	No Calc	...	No Calc
90	100	22	8.98282	0.10131	1.54
100	110	150	127.152	0.10131	No Calc
110	120	150	99.7074	...	No Calc

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners m	Allowable Length Bet. Stiffeners m	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	1.99997	1676.51	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	Jckt	0.20003	No Calc	No Calc	No Calc
Jckt	Jckt	1.6	No Calc	No Calc	No Calc
Jckt	50	11.69	No Calc	No Calc	No Calc
50	60	11.69	No Calc	No Calc	No Calc
60	70	11.69	No Calc	No Calc	No Calc
70	80	11.69	No Calc	No Calc	No Calc
80	Jckt	11.69	No Calc	No Calc	No Calc
Jckt	Jckt	1.6	No Calc	No Calc	No Calc
Jckt	Tsht	0.19997	No Calc	No Calc	No Calc
Tsht	90	No Calc	No Calc	No Calc	No Calc



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90	100	2.00003	1544.68	No Calc	No Calc
100	110	No Calc	No Calc	No Calc	No Calc
110	120	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

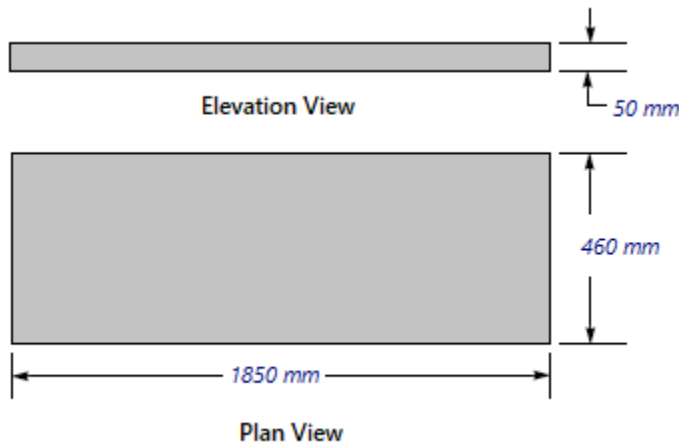
Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N2- Tube outle	WRC-297	0.988	Shell	Passed
N3- Shell inle	WRC-297	0.988	Shell	Passed
N4- Shell outl	WRC-297	0.988	Shell	Passed
<i>N1- Tube inLet</i>	<i>WRC-297</i>	<i>1.065</i>	<i>Shell</i>	<i>Failed</i>

Foundation/Support Summary

Saddle Parameters:

Saddle Width	225.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1550.000 mm
Wear Pad Width	254.000 mm
Wear Pad Thickness	16.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	3150.000 mm
Baseplate Length	1850.000 mm
Baseplate Thickness	50.000 mm
Baseplate Width	460.000 mm
Number of Ribs (including outside ribs)	4
Rib Thickness	15.000 mm
Web Thickness	15.000 mm
Height of Center Web	627.000 mm
Number of Bolts in Baseplate	6

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	2058870.38 N
Maximum Transverse Saddle Shear Load	89829.02 N
Maximum Longitudinal Saddle Shear Load	179658.05 N

Summary of Maximum Saddle Loads, Pressure Test Case:



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Maximum Vertical Saddle Load	1732076.50	N
Maximum Transverse Saddle Shear Load	5904.13	N
Maximum Longitudinal Saddle Shear Load	2412.32	N

Weights:

Fabricated - Bare w/o removable internals	285587.8	kgm
Shop Test - Fabricated + water (full)	342025.8	kgm
Shipping - Fab. + rem. intls.+ shipping app.	286001.3	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	289941.3	kgm
Empty - Fab. + intls. + details + wghts.	289941.3	kgm
Operating - Empty + operating liquid (no ca)	316157.7	kgm

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Nozzle Summary:

Step: 32 4:31pm Apr 10,2026

Nozzle Calculation Summary (N/mm² & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N2- Tube outlet	2.52	OK	...	OK	11.33	OK	Passed
N3- Shell inlet	15.7	...	...	OK	32.31	OK	Passed
N4- Shell outle	15.2	...	...	OK	33.90	OK	Passed
N1- Tube inlet	2.23	OK	...	OK	11.33	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 15.2 Nozzle : N4- Shell outle [Shellside]
 Minimum MAWP Nozzles : 2.23 Nozzle : N1- Tube inlet [Tubeside]

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N2- Tube outlet	1338.000	180.000	1500.000
40	N3- Shell inlet	3338.000	180.000	993.151
80	N4- Shell outle	16983.000	0.000	1047.101
90	N1- Tube inlet	19338.000	0.000	1536.000

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 31 4:31pm Apr 10,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
N3- Shell inlet	24.000 in	160	None	24.000	59.512	...	...	523.60	...
N4- Shell outle	24.000 in	120	None	24.000	46.025	...	...	529.02	...
N2- Tube outlet	30.000 in Actual		WNF	31.575	20.000	1302.00	24.00	357.80	150
N1- Tube inlet	30.000 in Actual		WNF	31.575	20.000	1162.00	24.00	357.80	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N3- Shell i	SA-266 2	50.000	50.000	...	...	...
N4- Shell o	SA-266 2	50.000	50.000	...	...	...
N2- Tube ou	SA-106 B	14.000	20.000	20.000	24.000	...
N1- Tube in	SA-106 B	14.000	20.000	20.000	24.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum m	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N3- Shell inlet	3.693	180.0	250.00	0.00	shell
N4- Shell outle	16.983	0.0	250.00	0.00	shell- front
N2- Tube outlet	1.338	180.0	250.00	0.00	Channel - rear
N1- Tube inlet	19.338	0.0	250.00	0.00	Channel front

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Bill of Material:

Step: 30 4:31pm Apr 10,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 150.0mm THK X 2300.0mm OD	SA-266 2
1	WELD NECK FLANGE: 150.0mm THK X 2300.0mm OD	SA-266 2
2	CYLINDER: 145.0mm THK X 1960.0mm ID X 2.0m	SA-516 70
2	CYLINDER: 145.0mm THK X 1960.0mm ID X 0.7m	SA-516 70
1	CYLINDER: 22.0mm THK X 1960.0mm ID X 2.0m	SA-516 70
1	WELD NECK FLANGE: 150.0mm THK X 2300.0mm OD	SA-516 70
1	BLIND FLANGE: 150.0mm THK X 2300.0mm OD	SA-516 70
2	INSULATION: 2000mm X 50mm THK	PP
22	BAFFLES: 1824.0mm DIA X 10.0mm THK	SA-516 70
12	TIE RODS: 16mm DIA X 16000mm	SA-193 B7
2	TUBESHEET: 2250.0mm X 355.0mm THK	SA-266 2
4267	TUBES: 16000.0mm X 19.0mm DIA X 2.1mm THK	SA-179
2	JACKET CYL: 1600.0mm LONG X 2532.0mm DIA X 185.0mm THK	SA-516 70
2	JACKET SEALING RING: 210.0mm THK	SA-516 70
2	SADDLE: 225.0mm X 120 DEG	SA-516 70
2	GASKET: 2002.0mm OD X 1976.0mm ID	...
88	BODY FLANGE BOLTS: 38.1mm DIA	SA-193 B7
176	NUTS FOR BODY FLANGE BOLTS: 38.1mm DIA	...
2	CYLINDRICAL SEGMENT 22.0mm THK X 2445.9mm X 2.0m	SA-516 70
1	CYLINDRICAL SEGMENT 22.0mm THK X 1265.7mm X 2.0m	SA-516 70
1	CYLINDRICAL SEGMENT 145.0mm THK X 5286.7mm X 9.9m	SA-516 70
1	CYLINDRICAL SEGMENT 145.0mm THK X 870.7mm X 9.9m	SA-516 70
1	NAMEPLATE	...



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MDMT Summary:

Step: 33 4:31pm Apr 10,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Left Saddle [24]	B	-31						No
Sdl 2 Fr60 [24]	B	-31						No
Channel COV L[11]	B	-7	-19		0.773	37.500	1.00	No
Channel FLG L[11]	B	-21	-45	-29	0.590	22.000	1.00	No
Channel FLG R[11]	B	-5	-28	-29	0.590	22.000	1.00	No
Channel COV R[11]	B	10	4		0.894	37.500	1.00	No
Channel - rear[8]	B	-5	-28	-29	0.587	22.000	1.00	No
Channel front [8]	B	-5	-28	-29	0.587	22.000	1.00	No
N2- Tube outle[1]	B	-5	-29	-29	0.594	22.000	1.00	No
Nozzle Flg [5]	A	-18	-29					
N1- Tube inlet[1]	B	-5	-29	-29	0.594	22.000	1.00	No
Nozzle Flg [5]	A	-18	-29					
Tubesheet: CS[14]	B	29	-104		0.184	88.750	1.00	No
Bolting [21]		-48						
Bolting [21]		-48						
Warmest MDMT:		29	4					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Channel/Tube	4.0	-28.9	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and
 UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
 The Basic MDMT includes the (30F) PWHT credit if applicable.



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Warnings and Errors:

Step: 0 4:31pm Apr 10,2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

Note:

For nozzles that are welded to both the Core and Jacket of jacketed vessels, the 2021 ASME Code requires that the opening in the jacket also be checked. This is accomplished by modelling an additional nozzle in the same location as the nozzle that penetrates through to the vessel core. The flange specification if any can be removed if needed. Other items such as weld sizes, pad specification etc. should also be addressed.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

Step: 1 4:31pm Apr 10,2026

Units used in this Analysis (SI):

Name	System Unit	Constant	User Unit
Length	Feet	0.3048	m
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	0.0006	m ²
Moment	ft. lbs.	1.3563	N-m
Stress	lbs./sq.in.	6.8946	kPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	N/mm ²
Elast. Modulus	lbs./sq.in.	0.0069	N/mm ²
Pipe Density	lbs./cu.in.	27680.0000	kg/m ³
Ins. Density	lbs./cu.ft.	16.0183	kg/m ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	0.4470	m/sec
Tray Weight	lbs./sq.ft.	47.8790	N/m ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	0.3048	m
Volume	in.3	0.1600E-04	m ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	18.001	N/mm ²
Channel Side Design Pressure	1	N/mm ²
Shell Side Design Temperature	150.0	°C
Channel Side Design Temperature	82.0	°C
Radiography, Shell Side	None	
Radiography, Channel Side	None	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-28.9	°C
MDMT (CET), Tube Side	-28.9	°C
User defined MAWP, Shell Side	0	N/mm ²
User defined MAWP, Channel Side	0	N/mm ²
User defined MAPnc, Shell Side	0	N/mm ²
User defined MAPnc, Channel Side	0	N/mm ²
User defined Test Pres., Shell Side	0	N/mm ²
User defined Test Pres., Channel Side	0	N/mm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No



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Input Echo:

Step: 1 4:31pm Apr 10, 2026

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	27 m/sec
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0.3 m
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 m
Distance Upwind of Crest Lh	0 m
Distance from Crest to the Vessel x	0 m
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 m

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P. + Static Head to Element Bottom	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year



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Input Echo:

Step: 1 4:31pm Apr 10,2026

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	Yes
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/11
Element From Node	10
Element To Node	20
Element Type	Flange
Description	Channel COV LEFT
Distance "FROM" to "TO"	0.15 m
Flange Outside Diameter	2300 mm
Element Thickness	150 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	60 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1 N/mm ²
Design Temperature Internal Pressure	82 °C
Design External Pressure	0.1013 N/mm ²
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	137990 kPa
Allowable Stress, Operating	137990 kPa
Allowable Stress, Hydrotest	224983 kPa
Material Density	7750.4 kg/m ³
P Number Thickness	32.004 mm
Yield Stress, Operating	230070 kPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	2/11
Element From Node	20
Element To Node	30
Element Type	Flange
Description	Channel FLG LEFT
Distance "FROM" to "TO"	0.185 m
Flange Inside Diameter	1960 mm
Element Thickness	150 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	60 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1 N/mm ²
Design Temperature Internal Pressure	82 °C
Design External Pressure	0.1013 N/mm ²



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Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	3/11
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Channel - rear
Distance "FROM" to "TO"	2 m
Inside Diameter	1960 mm
Element Thickness	22 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	22 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1 N/mm ²
Design Temperature Internal Pressure	82 °C
Design External Pressure	0.1013 N/mm ²
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	137990 kPa
Allowable Stress, Operating	137990 kPa
Allowable Stress, Hydrotest	233982 kPa
Material Density	7750.4 kg/m ³
P Number Thickness	32.004 mm
Yield Stress, Operating	242583 kPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Liquid	1.96 m
Liquid Density	999.54 kg/m ³

Element From Node	30
Detail Type	Insulation
Detail ID	Insulation- Tub
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Insulation	2 m
Thickness of Insulation	50 mm
Density	320.37 kg/m ³

Element From Node	30
Detail Type	Nozzle
Detail ID	N2- Tube outlet
Dist. from "FROM" Node / Offset dist	1 m
Nozzle Diameter	30 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	3208.7 N
Grade of Attached Flange	GR 1.1



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Input Echo:

Step: 1 4:31pm Apr 10,2026

Nozzle Matl

SA-106 B

Element#	4/11
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	shell
Distance "FROM" to "TO"	2 m
Inside Diameter	1960 mm
Element Thickness	145 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	145 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	18.001 N/mm ²
Design Temperature Internal Pressure	150 °C
Design External Pressure	0 N/mm ²
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Liquid	1.96 m
Liquid Density	565.74 kg/m ³

Element From Node	40
Detail Type	Nozzle
Detail ID	N3- Shell inlet
Dist. from "FROM" Node / Offset dist	1 m
Nozzle Diameter	24 in.
Nozzle Schedule	160
Nozzle Class	None
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	22408 N
Grade of Attached Flange	None
Nozzle Matl	SA-266 2

Element From Node	40
Detail Type	Weight
Detail ID	Internal pipe w
Dist. from "FROM" Node / Offset dist	1 m
Miscellaneous Weight	14905 N
Offset from Element Centerline	1000 mm

Element#	5/11
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	shell cone -rea
Distance "FROM" to "TO"	0.7 m
Inside Diameter	1960 mm
Element Thickness	145 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	145 mm
External Corrosion Allowance	0 mm



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Input Echo:

Step: 1 4:31pm Apr 10,2026

Design Internal Pressure	18.001	N/mm ²
Design Temperature Internal Pressure	150	°C
Design External Pressure	0	N/mm ²
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element#	6/11	
Element From Node	60	
Element To Node	70	
Element Type	Cylinder	
Description	Shell-2	
Distance "FROM" to "TO"	9.89	m
Inside Diameter	1960	mm
Element Thickness	145	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	145	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	18.001	N/mm ²
Design Temperature Internal Pressure	150	°C
Design External Pressure	0	N/mm ²
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	0.45	m
Width of Saddle	225	mm
Height of Saddle at Bottom	1550	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	254	mm
Thickness of Wear Plate	16	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	m

Element From Node	60	
Detail Type	Saddle	
Detail ID	Sdl 2 Fr60	
Dist. from "FROM" Node / Offset dist	9.44	m
Width of Saddle	225	mm
Height of Saddle at Bottom	1550	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	254	mm
Thickness of Wear Plate	16	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.4000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm



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Input Echo:

Step: 1 4:31pm Apr 10,2026

Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 m
Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Liquid	1.96 m
Liquid Density	565.74 kg/m ³
Element From Node	60
Detail Type	Weight
Detail ID	Fireproofing
Dist. from "FROM" Node / Offset dist	0.45 m
Miscellaneous Weight	4412.7 N
Offset from Element Centerline	1000 mm
Element From Node	60
Detail Type	Weight
Detail ID	Fireproofing
Dist. from "FROM" Node / Offset dist	9.44 m
Miscellaneous Weight	4412.7 N
Offset from Element Centerline	1000 mm

Element#	7/11
Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	shell cone- fro
Distance "FROM" to "TO"	0.7 m
Inside Diameter	1960 mm
Element Thickness	145 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	145 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	18.001 N/mm ²
Design Temperature Internal Pressure	150 °C
Design External Pressure	0 N/mm ²
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element#	8/11
Element From Node	80
Element To Node	90
Element Type	Cylinder
Description	shell- front
Distance "FROM" to "TO"	2 m
Inside Diameter	1960 mm
Element Thickness	145 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	145 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	18.001 N/mm ²
Design Temperature Internal Pressure	150 °C
Design External Pressure	0 N/mm ²
Design Temperature External Pressure	150 °C



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Input Echo:

Step: 1 4:31pm Apr 10,2026

Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	80
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Liquid	1.96 m
Liquid Density	565.74 kg/m ³
Element From Node	80
Detail Type	Nozzle
Detail ID	N4- Shell outle
Dist. from "FROM" Node / Offset dist	1 m
Nozzle Diameter	24 in.
Nozzle Schedule	120
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	21752 N
Grade of Attached Flange	None
Nozzle Matl	SA-266 2
Element From Node	80
Detail Type	Weight
Detail ID	Internal pipe w
Dist. from "FROM" Node / Offset dist	1 m
Miscellaneous Weight	14905 N
Offset from Element Centerline	1000 mm

Element#	9/11
Element From Node	90
Element To Node	100
Element Type	Cylinder
Description	Channel front
Distance "FROM" to "TO"	2 m
Inside Diameter	1960 mm
Element Thickness	22 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	22 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1 N/mm ²
Design Temperature Internal Pressure	82 °C
Design External Pressure	0.1013 N/mm ²
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	90
Detail Type	Liquid
Detail ID	Liquid: 80
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Liquid	1.96 m
Liquid Density	999.54 kg/m ³
Element From Node	90
Detail Type	Insulation



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Input Echo: Step: 1 4:31pm Apr 10,2026

Detail ID	Tube side- Ins.0
Dist. from "FROM" Node / Offset dist	0 m
Height/Length of Insulation	2 m
Thickness of Insulation	50 mm
Density	320.37 kg/m ³
Element From Node	90
Detail Type	Nozzle
Detail ID	N1- Tube inlet
Dist. from "FROM" Node / Offset dist	1 m
Nozzle Diameter	30 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	2714.6 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element#	10/11
Element From Node	100
Element To Node	110
Element Type	Flange
Description	Channel FLG RIGHT
Distance "FROM" to "TO"	0.185 m
Flange Inside Diameter	1960 mm
Element Thickness	150 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	22 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1 N/mm ²
Design Temperature Internal Pressure	82 °C
Design External Pressure	0.1013 N/mm ²
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	11/11
Element From Node	110
Element To Node	120
Element Type	Flange
Description	Channel COV RIGHT
Distance "FROM" to "TO"	0.15 m
Flange Outside Diameter	2300 mm
Element Thickness	150 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	22 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1 N/mm ²
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 N/mm ²
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No



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XY Coordinate Calculations: Step: 2 4:31pm Apr 10,2026

XY Coordinate Calculations:

From	To	X (Horiz.) m	Y (Vert.) m	DX (Horiz.) m	DY (Vert.) m
Channel COV LEF		0.153	...	0.15	...
Channel FLG LEF		0.338	...	0.185	...
Channel - rear		2.338	...	2	...
shell		4.693	...	2	...
shell cone -rea		5.393	...	0.7	...
Shell-2		15.283	...	9.89	...
shell cone- fro		15.983	...	0.7	...
shell- front		17.983	...	2	...
Channel front		20.338	...	2	...
Channel FLG RIG		20.526	...	0.185	...
Channel COV RIG		20.676	...	0.15	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flg Calc [Int P]: CH COV FG

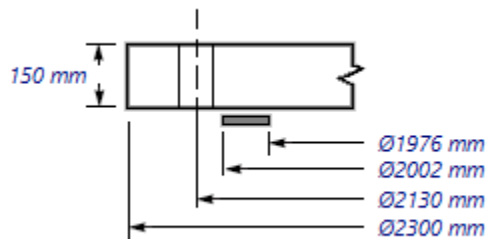
Flng: 9 4:31pm Apr 10,2026

Flange Input Data Values

Description: CH COV FG :

Channel COV LEFT

Flange Type		Blind
Design Pressure	P	1.00 N/mm ²
Design Temperature		82 °C
Internal Corrosion Allowance	ci	3.0000 mm
External Corrosion Allowance	ce	0.0000 mm
Use Corrosion Allowance in Thickness Calcs.		Yes
Flange Outside Diameter	A	2300.000 mm
Flange Thickness	t	150.0000 mm
Flange Material		SA-266 2
Flange Material UNS number		K03506
Flange Allowable Stress At Temperature	Sfo	137989.50 kPa
Flange Allowable Stress At Ambient	Sfa	137989.50 kPa
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	171986.91 kPa
Bolt Allowable Stress At Ambient	Sa	171986.91 kPa
Perimeter along the Center of the Bolts	L	6691.592 mm
Diameter of Bolt Circle	C	2130.000 mm
Nominal Bolt Diameter	a	38.1000 mm
Type of Threads		TEMA Thread Series
Number of Bolts		44
Flange Face Outside Diameter	Fod	2027.000 mm
Flange Face Inside Diameter	Fid	1960.000 mm
Flange Facing Sketch		Sketch 1a
Gasket Outside Diameter	Go	2002.000 mm
Gasket Inside Diameter	Gi	1976.000 mm
Gasket Factor	m	3.7500
Gasket Design Seating Stress	y	62051.40 kPa
Gasket Thickness	tg	3.0000 mm
Raised face depth	Fd	0.0000 mm



ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Go - Gi)/2$	13.000 mm
Basic Gasket Width,	$bo = N/2$	6.500 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1989.151 mm

Basic Flange and Bolt Loads:



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Flg Calc [Int P]: CH COV FG

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Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1989^2 \cdot 1$$

$$= 3107518.500 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 1989 \cdot 3.75 \cdot 1$$

$$= 301099.812 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(3107518 + 301100 + 0, 0)$$

$$= 3408618.250 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 62051 \cdot 6.42 \cdot \pi \cdot 1989 + 0 \cdot 0 \cdot 0$$

$$= 2491155.250 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((3408618 + 0 + 4 \cdot 0 / 1989) / 171987, 2491155 / 171987)$$

$$= 0.020 \text{ m}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	0.020	0.040	
Radial Distance between Bolts and Edge:	38.100	85.000	
Circ. Spacing between the Bolts:	82.550	151.952	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 0.04 \cdot 171987 / (62051 \cdot \pi (2002 + 1976))$$

$$= 8.846 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 171987 (0.02 + 0.04) / 2$$

$$= 5133961.50 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 3408618.25 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2130 - 1989) / 2$$

$$= 70.4245 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 3408618 \cdot 70.4 \cdot 1 / (147^2 \cdot 1989) + 0.3 \cdot 1 \cdot 1 \cdot 1989^2 / 147^2$$

$$= 65542.80 \text{ kPa}$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 5133961 \cdot 70.4 \cdot 1 / (147^2 \cdot 1989)$$



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Flg Calc [Int P]: CH COV FG

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= 15982.29 kPa

Bolt Stress, Operating [BS_o]:

= W_{m1} / A_b

= 3408618/0.04

= 85466.06 kPa

Bolt Stress, Seating [BS_a]:

= (W_{m2} / A_b)

= (2491155/0.04)

= 62462.02 kPa

Flange Stress Analysis Results: kPa

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	65542.80	137989.50	15982.29	137989.50
Bolting	85466.06	171986.91	62462.02	171986.91

Reqd. Blind Flange Thickness at Center 104.311 mm

Reqd. Blind Flange Thickness at Gasket 53.028 mm

Estimated M.A.W.P. (Operating) 2.01 N/mm²

Estimated Finished Weight of Flange at given Thk. 4830.2 kgm

Estimated Unfinished Weight of Forging at given Thk 4830.2 kgm

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.77, Temperature Reduction per Fig. UCS 66.1 = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 10 °C

Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -7 °C

Min Metal Temp. at Required thickness (UCS 66.1) -19 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kPa

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	62947.31	137989.50	15349.40	137989.50
Bolting	85466.06	171986.91	62462.02	171986.91

Minimum Req'd Thickness, ambient, no Corrosion: 101.321 mm

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Flg Calc [Int P]: CH FL LFT

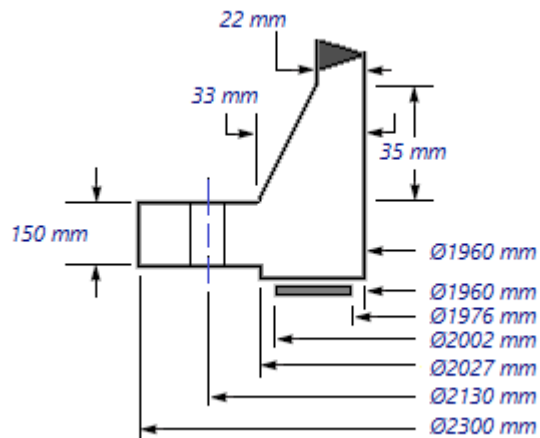
Flng: 10 4:31pm Apr 10,2026

Flange Input Data Values

Description: CH FL LFT :

Channel FLG LEFT

Flange Type	Integral Weld Neck		
Design Pressure	P	1.00	N/mm ²
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1960.000	mm
Flange Outside Diameter	A	2300.000	mm
Flange Thickness	t	150.0000	mm
Thickness of Hub at Small End	go	22.0000	mm
Thickness of Hub at Large End	gl	33.0000	mm
Length of Hub	h	35.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	137989.50	kPa
Flange Allowable Stress At Ambient	Sfa	137989.50	kPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	171986.91	kPa
Bolt Allowable Stress At Ambient	Sa	171986.91	kPa
Diameter of Bolt Circle	C	2130.000	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		44	
Flange Face Outside Diameter	Fod	2027.000	mm
Flange Face Inside Diameter	Fid	1960.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2002.000	mm
Gasket Inside Diameter	Gi	1976.000	mm
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	62051.40	kPa
Gasket Thickness	tg	3.0000	mm
Raised face depth	Fd	0.0000	mm





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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1 \cdot (1960/2 + 3)) / (137990 \cdot 1 - 0.6 \cdot 1) + Ca$$

$$= 10.1548 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137990 \cdot 1 \cdot 19) / (983 + 0.6 \cdot 19)$$

$$= 2.637 \text{ N/mm}^2$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137990 \cdot 1 \cdot 22) / (980 + 0.6 \cdot 22)$$

$$= 3.057 \text{ N/mm}^2$$

Corroded Flange Thickness, $t_c = T - c_i$	147.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1966.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	30.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	19.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	52.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width, $b_o = N/2$	6.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1989.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1989^2 \cdot 1$$

$$= 3107518.500 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 1989 \cdot 3.75 \cdot 1$$

$$= 301099.812 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1966^2 \cdot 1$$

$$= 3035604.500 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 3107518 - 3035604$$

$$= 71913.875 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o, 1 + W_o, 2 + H'p, 0)$$

$$= \max(3107518 + 301100 + 0, 0)$$

$$= 3408618.250 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 62051 \cdot 6.42 \cdot \pi \cdot 1989 + 0 \cdot 0 \cdot 0$$

$$= 2491155.250 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((W_o + Fa + 4Me/G)/S_b, W_g/S_a)$$

$$= \max((3408618 + 0 + 4 \cdot 0/1989)/171987, 2491155/171987)$$

$$= 0.020 \text{ m}^2$$



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Flg Calc [Int P]: CH FL LFT

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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 147/(3.75 + 0.5)$$

$$= 283.729 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2130 \cdot \sin(\pi / 44)$$

$$= 151.952 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{152/(2 \cdot 38.1 + 147)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	0.020	0.040	
Radial Distance between Hub and Bolts:	50.800	52.000	
Radial Distance between Bolts and Edge:	38.100	85.000	
Circ. Spacing between the Bolts:	82.550	151.952	283.729

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 0.04 \cdot 171987 / (62051 \cdot \pi (2002 + 1976))$$

$$= 8.846 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa (Am + Ab) / 2$$

$$= 171987 (0.02 + 0.04) / 2$$

$$= 5133961.50 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 3408618 - 3107518$$

$$= 301099.84 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2130 - 1989) / 2$$

$$= 70.4245 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((2130 - 1966) / 2 + 70.4)$$

$$= 76.2123 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2130 - 1966 - 30) / 2$$

$$= 67.0000 \text{ mm}$$

Summary of Moments for Internal Pressure: (N-m)

Loading	Force	Distance	Bolt Corr	Moment
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Flg Calc [Int P]: CH FL LFT Flng: 10 4:31pm Apr 10,2026

End Pressure, Md	3035604.	67.0000	1.0000	203468.
Face Pressure, Mt	71914.	76.2123	1.0000	5483.
Gasket Load, Mg	301100.	70.4245	1.0000	21213.
Gasket Seating, Matm	5133962.	70.4245	1.0000	361703.

Total Moment for Operation, Mo 230164. N-m
Total Moment for Gasket seating, Mg 361703. N-m

Effective Hub Length, ho = sqrt(Bcor*goCor) 193.272 mm
Hub Ratio, h/h0 = HL / H0 0.181
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.579

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.902
Factor V per 4.16.5 0.413
Factor f per 4.16.5 1.733

Stress factor from Table 4.16.4 [K]:

= A / B
= 2300/1966
= 1.170

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169 (K² • log10(K))) / (K² - 1))/(K - 1)
= (0.66845 + (5.7169 (1.17² • log10(1.17))) / (1.17² - 1))/(1.17 - 1)
= 12.448

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1)/((1.0472 + 1.9448 • K²) • (K - 1))
= (1.17²(1 + 8.55246 • log10(1.17)) - 1)/((1.0472 + 1.9448 • 1.17²) • (1.17 - 1))
= 1.851

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1)/(1.36136(K² - 1) • (K - 1))
= (1.17²(1 + 8.55246 • log10(1.17)) - 1)/(1.36136(1.17² - 1) • (1.17 - 1))
= 13.679

Stress factor from Table 4.16.4 [Z]:

= (K² + 1)/(K² - 1)
= (1.17² + 1)/(1.17² - 1)
= 6.425

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 13.7 • 19² • 193/0.41
= 2313022.250 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.9/193
= 0.005 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (147 • 0.0047 + 1)/1.85 + 147³/2313022
= 2.284

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.73•230164/1966)/(2.28•30²)
= 98634.27 kPa

Longitudinal Hub Stress, Seating [SHa]:



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$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g1^2)$$

$$= (1.73 \cdot 361703 / 1966) / (2.28 \cdot 30^2)$$

$$= 155003.67 \text{ kPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (1.91 \cdot 230164 / 1966) / (2.28 \cdot 147^2)$$

$$= 4538.99 \text{ kPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (1.91 \cdot 361703 / 1966) / (2.28 \cdot 147^2)$$

$$= 7133.02 \text{ kPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (12.4 \cdot 230164 / (147^2 \cdot 1966)) - 6.43 \cdot 4539$$

$$= 38251.67 \text{ kPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (12.4 \cdot 361703 / (147^2 \cdot 1966)) - 6.43 \cdot 7133$$

$$= 60112.47 \text{ kPa}$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (98634 + \max(4539, 38252)) / 2$$

$$= 68442.98 \text{ kPa}$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (155004 + \max(7133, 60112)) / 2$$

$$= 107558.07 \text{ kPa}$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 3408618 / 0.04$$

$$= 85466.06 \text{ kPa}$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (2491155 / 0.04)$$

$$= 62462.02 \text{ kPa}$$

Flange Stress Analysis Results: kPa

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	98634.27	206984.23	155003.67	206984.23
Radial Flange	4538.99	137989.50	7133.02	137989.50
Tangential Flange	38251.67	137989.50	60112.47	137989.50
Maximum Average	68442.98	137989.50	107558.07	137989.50
Bolting	85466.06	171986.91	62462.02	171986.91

Minimum Required Flange Thickness [Rigidity] 133.629 mm

Estimated M.A.W.P. (Operating) 1.74 N/mm²

Estimated Finished Weight of Flange at given Thk. 1369.1 kgm

Estimated Unfinished Weight of Forging at given Thk 1631.1 kgm

Flange Stress results per WRC-538 at computed S_{fmax} (kPa):



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Flg Calc [Int P]: CH FL LFT

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		Actual	Operating Allowed	stress ratio	Gasket Seating Actual Allowed		stress ratio
Longitudinal Hub	Sh	499961.97	499961.97	1.000	499961.97	499961.97	1.000
Tangential Flange	St	195022.69	374971.50	0.520	195022.69	374971.50	0.520
Tangential Flange	Sto	85639.61	249980.98	0.343	85639.61	249980.98	0.343
Combined	Sh/f + St	485273.94	749943.00	0.647	485273.94	749943.00	0.647
Combined	Sr + St	218164.33	749943.00	0.291	218164.33	749943.00	0.291
Combined	Sh + Sto	585601.56	749943.00	0.781	585601.56	749943.00	0.781
Radial Flange	Sr	23141.64	499961.97	0.046	23141.64	499961.97	0.046
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f_2

	f	f_2	f_3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g_1}{g_0} \right) + e \left(\frac{h}{h_0} \right) + g \left(\frac{g_1}{g_0} \right)^2 + i \left(\frac{h}{h_0} \right)^2 + k \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}{\left[1 + b \left(\frac{g_1}{g_0} \right) + d \left(\frac{h}{h_0} \right) + f \left(\frac{g_1}{g_0} \right)^2 + h \left(\frac{h}{h_0} \right)^2 + j \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.8/1.04)

= 1.7225

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h_0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.27/0.9) if $h/h_0 < 0.35$, else max (0.25; 0.3/0.8)

= 0.2951

In this case, the hub ratio h/h_0 , 0.1811 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.



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Flg Calc [Int P]: CH FL LFT

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Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507470.6 kPa
Minimum permissible bolt stress	Sbmin	289983.2 kPa
Max. permissible bolt stress prior to flange damage	Sfmax	417626.9 kPa
Target assembly gasket stress	SgT	206988.0 kPa
Maximum permissible gasket stress	Sgmax	275984.0 kPa
Minimum gasket operating stress	Sgmin-O	96994.4 kPa
Minimum gasket seating stress	Sgmin-S	139991.9 kPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	0.0009 m ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	249981.0 kPa
Flange Material Operating Yield Stress	Syo	230069.6 kPa
Single flange rotation at Sfmax	Thetafmax	0.542 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (2002^2 - 1976^2)$$

$$= 0.1 \text{ m}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Abs)$$

$$= 206988 \cdot 0.081 / (44 \cdot 0.00091)$$

$$= 421576.2 \text{ kPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(421576; 507471)$$

$$= 421576.2 \text{ kPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(421576; 289983)$$

$$= 421576.2 \text{ kPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(421576; 417627)$$

$$= 417626.9 \text{ kPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 417627 \geq 139992 (0.081 / (44 \cdot 0.00091))$$

$$:: 417626.9 \geq 285124.0 \text{ kPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 417627 \geq (96994 \cdot 0.081 + \pi/4 \cdot 1 \cdot 1976^2) / (0.7 \cdot 0.00091 \cdot 44)$$

$$:: 417626.9 \geq 392056.8 \text{ kPa} \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 417627 \leq 275984 (0.081 / (44 \cdot 0.00091))$$

$$:: 417626.9 \leq 562101.6 \text{ kPa} \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$:: 417627 \leq 417627 (1/0.54)$$



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:: 417626.9 ≤ 770007.9 kPa Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

= Sbsel • K • Abs • φb

= 417627 • 0.2 • 0.00091 • 38.1

= 2885.7 N-m

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118113 kPa

Bolt Allowable Stress for: SA-193 B7 Sb 171986 kPa

Note: Since Sb ≥ SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)

= 52.14•361703/1•999622•0.41/(1.83•202701232•19²•193•0.3)

= 1.000 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)

= 52.14•230164/1•999622•0.41/(1.83•199214080•19²•193•0.3)

= 0.647 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)

= 52.14•361703/1•999622•0.41/(2.28•202701232•19²•193•0.3)

= 0.802 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)

= 52.14•230164/1•999622•0.41/(2.28•199214080•19²•193•0.3)

= 0.519 (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.59, Temperature Reduction per Fig. UCS 66.1 = 23 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C

Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -21 °C

Min Metal Temp. at Required thickness (UCS 66.1) -45 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	90153.80	206984.23	138762.45	206984.23
Radial Flange	5063.04	137989.50	7792.90	137989.50
Tangential Flange	33148.16	137989.50	51020.82	137989.50
Maximum Average	61650.98	137989.50	94891.64	137989.50
Bolting	85466.06	171986.91	62462.02	171986.91

Minimum Req'd Thickness, ambient, no Corrosion: 116.942 mm

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Jacket/VB Calcs: Jacket

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Jacket Design per ASME VIII, Division 2, 4.11.2: Jacket

Design per: Figure 4.11.1 Type 1

Attachment per: Table 4.11.1, Detail 6, Figure: (a)

Input Values:

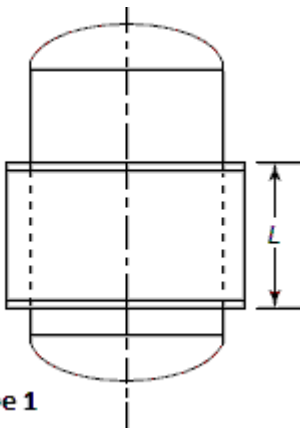
Jacket Design Internal Pressure	Pj	18.00	N/mm ²
Jacket Design Temperature (Internal)		150	C
Jacket Design External Pressure	Pje	0.00	N/mm ²
Jacket Design Temperature (External)		150	C
Static fluid Pressure in this Jacket	pStatic	0.01	N/mm ²

Closure Bar Material		SA-516 70	
Closure Bar Design Stress	[S or f]	137989.50	kPa
Closure Bar Thickness New	tc	210.0000	mm
Closure Bar Corrosion Allowance	cc	3.0000	mm

Inner Shell Outside Diameter	Dso	2250.000	mm
Inner Shell Thickness New	ts	145.0000	mm
Corroded Local Shell Thickness at Closure	ts,c	142.0000	mm
Inner Shell Corrosion Allow. Inside	csi	3.0000	mm
Inner Shell Corrosion Allow. Outside	cso	0.0000	mm
Inner Shell Vacuum Pressure		0.0000	N/mm ²

Jacket Material		SA-516 70	
Jacket Design Stress	Sj	137989.50	kPa
Jacket Inside Diameter New	Dji	2532.000	mm
Jacket Thickness New	tj	185.0000	mm
Jacket Corrosion Allowance Inside	cji	3.0000	mm
Jacket Length	L	1600.00	mm
Jacket Design Length for External Pressure	SL	1600.00	mm
Specific Gravity of contents in Jacket	sg	0.000	
Jacket Long. Weld Joint Efficiency	E	1.00	

This has been specified as a vapor/distribution belt.



Type 1



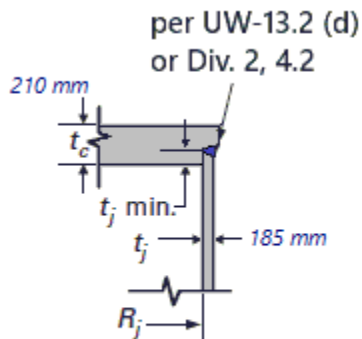
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FileName : 124-C-NVS

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Jacket/VB Calcs: Jacket

Step: 5 4:31pm Apr 10, 2026



Div. 2 Detail 6 Fig. (a)

Results:

Jacket Inside Radius [Rj]:

$$= (D_{ji} + 2 \cdot c_i) / 2$$

$$= (2532 + 2 \cdot 3) / 2$$

$$= 1269.000 \text{ mm}$$

Jacket Corroded Thickness [tjc]:

$$= t_j - c_{ji}$$

$$= 185 - 3$$

$$= 182.000 \text{ mm}$$

Shell Outside Radius [Rs]:

$$= (D_{so} - 2 \cdot c_{so}) / 2$$

$$= (2250 - 2 \cdot 0) / 2$$

$$= 1125.000 \text{ mm}$$

Shell Corroded Thickness [tsc]:

$$= t_s - c_{si} - c_{so}$$

$$= 145 - 3 - 0$$

$$= 142.000 \text{ mm}$$

Jacket Space Width [j]:

$$= (D_{ji} + 2 \cdot c_{ji} - D_{so} + 2 \cdot c_{so}) / 2$$

$$= (2532 + 2 \cdot 3 - 2250 + 2 \cdot 0) / 2$$

$$= 144.000 \text{ mm}$$

Jacket Thickness Calculations for Cylinder:

Longitudinal Joint: User Defined

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (18 \cdot 1269) / (137990 \cdot 1 - 0.6 \cdot 18)$$

$$= 179.7082 + 3.0000 = 182.7082 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.011 N/mm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137990 \cdot 1 \cdot 182) / (1269 + 0.6 \cdot 182)$$

$$= 18.2 - 0.011 = 18.2 \text{ N/mm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137990 \cdot 1 \cdot 185) / (1266 + 0.6 \cdot 185)$$

$$= 18.5 \text{ N/mm}^2$$



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Jacket/VB Calcs: Jacket

Step: 5 4:31pm Apr 10, 2026

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (18 \cdot (1269 + 0.6 \cdot 182)) / (1 \cdot 182)$$

$$= 136387.984 \text{ kPa}$$

Jacket Minimum Design Metal Temperature Calculations:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Required Closure Thickness [trc]:

$$= 1.414 \cdot \sqrt{(P_j \cdot R_s \cdot j / S) + c}$$

$$= 1.414 \cdot \sqrt{(18 \cdot 1125 \cdot 144 / 137990) + 3}$$

$$= 208.614 \text{ mm}$$

Maximum Allowable Working Pressure, Closure Bar [CBMawp]:

$$= S / (j \cdot R) \cdot ((tr - c) / 1.414)^2 - p_{\text{Static}}$$

$$= 137990 / (144 \cdot 1125) \cdot ((210 - 3) / 1.414)^2 - 0.011$$

$$= 18.245 \text{ N/mm}^2$$

Forming Strain/Elongation Results:

Jacket Forming Strain, Cylinders formed from Plate ($50 \cdot t_{\text{nom}} / R_f(1 - R_f/R_o)$): 6.809 %

Note: Please check Code requirements as the computed elongation is > 5%.

External Pressure results for the Outer Jacket Cylinder:

Maximum External Pressure per Code Case 2286:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 1600 / \sqrt{(1451 \cdot 182)}$$

$$= 3.114$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 1.5 \text{ and } M_x < 13, Ch = 0.92 / (M_x - 0.579)$$

$$= 0.92 / (3.11 - 0.579)$$

$$= 0.3630$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.36 \cdot 195074 (182 / 2902)$$

$$= 7104960.000 \text{ kPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 7104960.0 / 231984.0$$

$$= 30.627$$

Predicted Buckling Stress [Fic]:

$$\text{Since } Fr \geq 2.439, Fic = F_y$$

$$= 231983.984 \text{ kPa}$$

Factor of Safety [FS]:

$$\text{Since } Fic [231984] \geq F_y [231984], FS = 1.667$$

Allowable Circumferential Compressive Stress [Fha]:

$$\text{Since } Fr > 2.439, Fha = F_y / FS$$

$$= 231983.984 / 1.667$$

$$= 139162.547 \text{ kPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha (t / D_o)$$



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Jacket/VB Calcs: Jacket

Step: 5 4:31pm Apr 10, 2026

$$= 2 \cdot 139 (182/2902)$$

$$= 17.456 \text{ N/mm}^2$$

Summary of Results (mm):

	Required	Actual	Result
Jacket Thk Internal	182.708	185.000	Passed
Closure Bar Thickness	208.614	210.000	Passed

MAWP Results for the Jacket/Vapor/Distribution Belt and Closure Bar:

Jacket Maximum Allowable Working Pressure	18.22	N/mm ²
Closure Bar Maximum Allowable Working Pressure	18.24	N/mm ²
Overall MAWP of Jacket Components	18.22	N/mm ²

Jacket Cylinder External Maximum Allowable Working Pressure:

Jacket Ext. Maximum Allowable Working Pressure	17.46	N/mm ²
--	-------	-------------------

Jacket Hydrotest Pressure Calculations:

Jacket Hydrotest Pressure per UG-99(b) Note 35:

$$= 1.3 \cdot P \cdot \text{Stress Ratio}$$

$$= 1.3 \cdot 18 \cdot 1$$

$$= 23.401 \text{ N/mm}^2$$

Jacket Hydrotest Pressure per UG-99(b):

$$= 1.3 \cdot \min(\text{Core EMAWP, Jacket MAWP}) \cdot \text{Stress Ratio}$$

$$= 1.3 \cdot \min(689, 18.2) \cdot 1$$

$$= 23.690 \text{ N/mm}^2$$

Note:

For vessels/jackets constructed of different materials, be sure to check the stresses due to differential thermal expansion, especially for vessels at moderate to high temperatures.

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Jacket/VB Calcs: Jacket

Step: 6 4:31pm Apr 10, 2026

Jacket Design per ASME VIII, Division 2, 4.11.2: Jacket

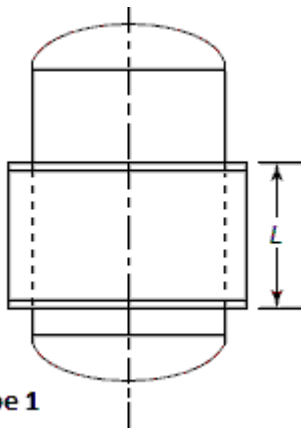
Design per: Figure 4.11.1 Type 1

Attachment per: Table 4.11.1, Detail 6, Figure: (a)

Input Values:

Jacket Design Internal Pressure	Pj	18.00	N/mm ²
Jacket Design Temperature (Internal)		150	C
Jacket Design External Pressure	Pje	0.00	N/mm ²
Jacket Design Temperature (External)		150	C
Static fluid Pressure in this Jacket	pStatic	0.01	N/mm ²
Closure Bar Material		SA-516 70	
Closure Bar Design Stress	[S or f]	137989.50	kPa
Closure Bar Thickness New	tc	210.0000	mm
Closure Bar Corrosion Allowance	cc	3.0000	mm
Inner Shell Outside Diameter	Dso	2250.000	mm
Inner Shell Thickness New	ts	145.0000	mm
Corroded Local Shell Thickness at Closure	ts,c	142.0000	mm
Inner Shell Corrosion Allow. Inside	csi	3.0000	mm
Inner Shell Corrosion Allow. Outside	cso	0.0000	mm
Inner Shell Vacuum Pressure		0.0000	N/mm ²
Jacket Material		SA-516 70	
Jacket Design Stress	Sj	137989.50	kPa
Jacket Inside Diameter New	Dji	2532.000	mm
Jacket Thickness New	tj	185.0000	mm
Jacket Corrosion Allowance Inside	cji	3.0000	mm
Jacket Length	L	1600.00	mm
Jacket Design Length for External Pressure	SL	1600.00	mm
Specific Gravity of contents in Jacket	sg	0.000	
Jacket Long. Weld Joint Efficiency	E	1.00	

This has been specified as a vapor/distribution belt.



Type 1



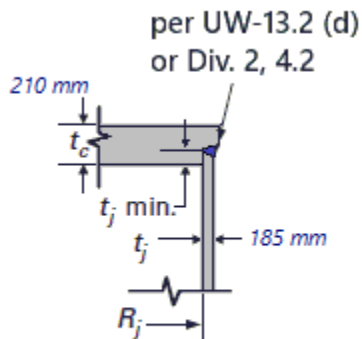
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Jacket/VB Calcs: Jacket

Step: 6 4:31pm Apr 10, 2026



Div. 2 Detail 6 Fig. (a)

Results:

Jacket Inside Radius [Rj]:

$$= (D_{ji} + 2 \cdot c_i) / 2$$

$$= (2532 + 2 \cdot 3) / 2$$

$$= 1269.000 \text{ mm}$$

Jacket Corroded Thickness [tjc]:

$$= t_j - c_{ji}$$

$$= 185 - 3$$

$$= 182.000 \text{ mm}$$

Shell Outside Radius [Rs]:

$$= (D_{so} - 2 \cdot c_{so}) / 2$$

$$= (2250 - 2 \cdot 0) / 2$$

$$= 1125.000 \text{ mm}$$

Shell Corroded Thickness [tsc]:

$$= t_s - c_{si} - c_{so}$$

$$= 145 - 3 - 0$$

$$= 142.000 \text{ mm}$$

Jacket Space Width [j]:

$$= (D_{ji} + 2 \cdot c_{ji} - D_{so} + 2 \cdot c_{so}) / 2$$

$$= (2532 + 2 \cdot 3 - 2250 + 2 \cdot 0) / 2$$

$$= 144.000 \text{ mm}$$

Jacket Thickness Calculations for Cylinder:

Longitudinal Joint: User Defined

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (18 \cdot 1269) / (137990 \cdot 1 - 0.6 \cdot 18)$$

$$= 179.7082 + 3.0000 = 182.7082 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.011 N/mm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137990 \cdot 1 \cdot 182) / (1269 + 0.6 \cdot 182)$$

$$= 18.2 - 0.011 = 18.2 \text{ N/mm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137990 \cdot 1 \cdot 185) / (1266 + 0.6 \cdot 185)$$

$$= 18.5 \text{ N/mm}^2$$



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Jacket/VB Calcs: Jacket

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Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (18 \cdot (1269 + 0.6 \cdot 182)) / (1 \cdot 182) \\
 &= 136387.984 \text{ kPa}
 \end{aligned}$$

Jacket Minimum Design Metal Temperature Calculations:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Required Closure Thickness [trc]:

$$\begin{aligned}
 &= 1.414 \cdot \sqrt{(P_j \cdot R_s \cdot j / S) + c} \\
 &= 1.414 \cdot \sqrt{(18 \cdot 1125 \cdot 144 / 137990) + 3} \\
 &= 208.614 \text{ mm}
 \end{aligned}$$

Maximum Allowable Working Pressure, Closure Bar [CBMawp]:

$$\begin{aligned}
 &= S / (j \cdot R) \cdot ((t_r - c) / 1.414)^2 - p_{\text{Static}} \\
 &= 137990 / (144 \cdot 1125) \cdot ((210 - 3) / 1.414)^2 - 0.011 \\
 &= 18.245 \text{ N/mm}^2
 \end{aligned}$$

Forming Strain/Elongation Results:

Jacket Forming Strain, Cylinders formed from Plate ($50 \cdot t_{\text{nom}} / R_f (1 - R_f / R_o)$): 6.809 %

Note: Please check Code requirements as the computed elongation is > 5%.

External Pressure results for the Outer Jacket Cylinder:

Maximum External Pressure per Code Case 2286:

Determine Parameter [Mx]:

$$\begin{aligned}
 &= L / \sqrt{(R_o \cdot t)} \\
 &= 1600 / \sqrt{(1451 \cdot 182)} \\
 &= 3.114
 \end{aligned}$$

Compute the Factor [Ch]:

$$\begin{aligned}
 &\text{Since } M_x > 1.5 \text{ and } M_x < 13, Ch = 0.92 / (M_x - 0.579) \\
 &= 0.92 / (3.11 - 0.579) \\
 &= 0.3630
 \end{aligned}$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$\begin{aligned}
 &= 1.6 \cdot Ch \cdot E_y (t / D_o) \\
 &= 1.6 \cdot 0.36 \cdot 195074 (182 / 2902) \\
 &= 7104960.000 \text{ kPa}
 \end{aligned}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$\begin{aligned}
 &= F_{he} / F_y \\
 &= 7104960.0 / 231984.0 \\
 &= 30.627
 \end{aligned}$$

Predicted Buckling Stress [Fic]:

$$\begin{aligned}
 &\text{Since } F_r \geq 2.439, F_{ic} = F_y \\
 &= 231983.984 \text{ kPa}
 \end{aligned}$$

Factor of Safety [FS]:

$$\text{Since } F_{ic} [231984] \geq F_y [231984], FS = 1.667$$

Allowable Circumferential Compressive Stress [Fha]:

$$\begin{aligned}
 &\text{Since } F_r > 2.439, F_{ha} = F_y / FS \\
 &= 231983.984 / 1.667 \\
 &= 139162.547 \text{ kPa}
 \end{aligned}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$



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Jacket/VB Calcs: Jacket

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$$= 2 \cdot 139 (182/2902)$$

$$= 17.456 \text{ N/mm}^2$$

Summary of Results (mm):

	Required	Actual	Result
Jacket Thk Internal	182.708	185.000	Passed
Closure Bar Thickness	208.614	210.000	Passed

MAWP Results for the Jacket/Vapor/Distribution Belt and Closure Bar:

Jacket Maximum Allowable Working Pressure	18.22	N/mm ²
Closure Bar Maximum Allowable Working Pressure	18.24	N/mm ²
Overall MAWP of Jacket Components	18.22	N/mm ²

Jacket Cylinder External Maximum Allowable Working Pressure:

Jacket Ext. Maximum Allowable Working Pressure	17.46	N/mm ²
--	-------	-------------------

Jacket Hydrotest Pressure Calculations:

Jacket Hydrotest Pressure per UG-99(b) Note 35:

$$= 1.3 \cdot P \cdot \text{Stress Ratio}$$

$$= 1.3 \cdot 18 \cdot 1$$

$$= 23.401 \text{ N/mm}^2$$

Jacket Hydrotest Pressure per UG-99(b):

$$= 1.3 \cdot \min(\text{Core EMAWP, Jacket MAWP}) \cdot \text{Stress Ratio}$$

$$= 1.3 \cdot \min(689, 18.2) \cdot 1$$

$$= 23.690 \text{ N/mm}^2$$

Note:

For vessels/jackets constructed of different materials, be sure to check the stresses due to differential thermal expansion, especially for vessels at moderate to high temperatures.

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Flg Calc [Int P]: CH COV FG

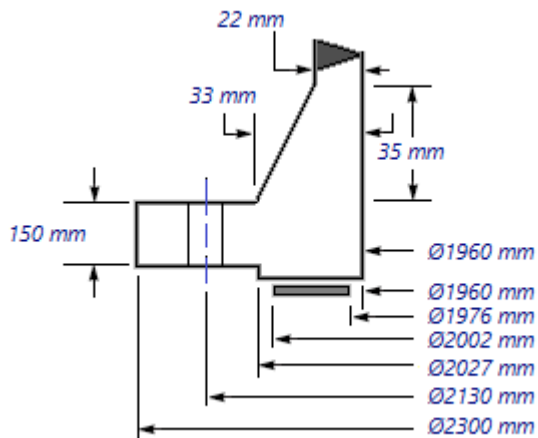
Flng: 11 4:31pm Apr 10,2026

Flange Input Data Values

Description: CH COV FG :

Channel FLG RIGHT

Flange Type	Integral Weld Neck		
Design Pressure	P	1.00	N/mm ²
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1960.000	mm
Flange Outside Diameter	A	2300.000	mm
Flange Thickness	t	150.0000	mm
Thickness of Hub at Small End	go	22.0000	mm
Thickness of Hub at Large End	gl	33.0000	mm
Length of Hub	h	35.0000	mm
Flange Material		SA-516 70	
Flange Material UNS number		K02700	
Flange Allowable Stress At Temperature	Sfo	137989.50	kPa
Flange Allowable Stress At Ambient	Sfa	137989.50	kPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	171986.91	kPa
Bolt Allowable Stress At Ambient	Sa	171986.91	kPa
Diameter of Bolt Circle	C	2130.000	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		44	
Flange Face Outside Diameter	Fod	2027.000	mm
Flange Face Inside Diameter	Fid	1960.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2002.000	mm
Gasket Inside Diameter	Gi	1976.000	mm
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	62051.40	kPa
Gasket Thickness	tg	3.0000	mm
Raised face depth	Fd	0.0000	mm





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Flg Calc [Int P]: CH COV FG

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ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1 \cdot (1960/2 + 3)) / (137990 \cdot 1 - 0.6 \cdot 1) + Ca$$

$$= 10.1548 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137990 \cdot 1 \cdot 19) / (983 + 0.6 \cdot 19)$$

$$= 2.637 \text{ N/mm}^2$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137990 \cdot 1 \cdot 22) / (980 + 0.6 \cdot 22)$$

$$= 3.057 \text{ N/mm}^2$$

Corroded Flange Thickness, $t_c = T - c_i$	147.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1966.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	30.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	19.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	52.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width, $bo = N/2$	6.500 mm
Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1989.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1989^2 \cdot 1$$

$$= 3107518.500 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 1989 \cdot 3.75 \cdot 1$$

$$= 301099.812 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1966^2 \cdot 1$$

$$= 3035604.500 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 3107518 - 3035604$$

$$= 71913.875 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(3107518 + 301100 + 0, 0)$$

$$= 3408618.250 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 62051 \cdot 6.42 \cdot \pi \cdot 1989 + 0 \cdot 0 \cdot 0$$

$$= 2491155.250 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((3408618 + 0 + 4 \cdot 0/1989)/171987, 2491155/171987)$$

$$= 0.020 \text{ m}^2$$



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Flg Calc [Int P]: CH COV FG

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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 147/(3.75 + 0.5)$$

$$= 283.729 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2130 \cdot \sin(\pi / 44)$$

$$= 151.952 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{152/(2 \cdot 38.1 + 147)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	0.020	0.040	
Radial Distance between Hub and Bolts:	50.800	52.000	
Radial Distance between Bolts and Edge:	38.100	85.000	
Circ. Spacing between the Bolts:	82.550	151.952	283.729

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 0.04 \cdot 171987 / (62051 \cdot \pi (2002 + 1976))$$

$$= 8.846 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm

Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa (Am + Ab) / 2$$

$$= 171987 (0.02 + 0.04) / 2$$

$$= 5133961.50 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 3408618 - 3107518$$

$$= 301099.84 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2130 - 1989) / 2$$

$$= 70.4245 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((2130 - 1966) / 2 + 70.4)$$

$$= 76.2123 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2130 - 1966 - 30) / 2$$

$$= 67.0000 \text{ mm}$$

Summary of Moments for Internal Pressure: (N-m)

Loading	Force	Distance	Bolt Corr	Moment
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Flg Calc [Int P]: CH COV FG Flng: 11 4:31pm Apr 10, 2026

End Pressure, Md	3035604.	67.0000	1.0000	203468.
Face Pressure, Mt	71914.	76.2123	1.0000	5483.
Gasket Load, Mg	301100.	70.4245	1.0000	21213.
Gasket Seating, Matm	5133962.	70.4245	1.0000	361703.

Total Moment for Operation, Mo 230164. N-m
Total Moment for Gasket seating, Mg 361703. N-m

Effective Hub Length, ho = sqrt(Bcor*goCor) 193.272 mm
Hub Ratio, h/h0 = HL / H0 0.181
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.579

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.902
Factor V per 4.16.5 0.413
Factor f per 4.16.5 1.733

Stress factor from Table 4.16.4 [K]:

= A / B
= 2300/1966
= 1.170

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169 (K² • log10(K))) / (K² - 1)) / (K - 1)
= (0.66845 + (5.7169 (1.17² • log10(1.17))) / (1.17² - 1)) / (1.17 - 1)
= 12.448

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
= (1.17²(1 + 8.55246 • log10(1.17)) - 1) / ((1.0472 + 1.9448 • 1.17²) • (1.17 - 1))
= 1.851

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
= (1.17²(1 + 8.55246 • log10(1.17)) - 1) / (1.36136(1.17² - 1) • (1.17 - 1))
= 13.679

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.17² + 1) / (1.17² - 1)
= 6.425

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 13.7 • 19² • 193 / 0.41
= 2313022.250 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.9 / 193
= 0.005 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³ / d
= (147 • 0.0047 + 1) / 1.85 + 147³ / 2313022
= 2.284

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.73 • 230164 / 1966) / (2.28 • 30²)
= 98634.27 kPa

Longitudinal Hub Stress, Seating [SHa]:



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$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g1^2)$$

$$= (1.73 \cdot 361703 / 1966) / (2.28 \cdot 30^2)$$

$$= 155003.67 \text{ kPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (1.91 \cdot 230164 / 1966) / (2.28 \cdot 147^2)$$

$$= 4538.99 \text{ kPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (1.91 \cdot 361703 / 1966) / (2.28 \cdot 147^2)$$

$$= 7133.02 \text{ kPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (12.4 \cdot 230164 / (147^2 \cdot 1966)) - 6.43 \cdot 4539$$

$$= 38251.67 \text{ kPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (12.4 \cdot 361703 / (147^2 \cdot 1966)) - 6.43 \cdot 7133$$

$$= 60112.47 \text{ kPa}$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (98634 + \max(4539, 38252)) / 2$$

$$= 68442.98 \text{ kPa}$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (155004 + \max(7133, 60112)) / 2$$

$$= 107558.07 \text{ kPa}$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 3408618 / 0.04$$

$$= 85466.06 \text{ kPa}$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (2491155 / 0.04)$$

$$= 62462.02 \text{ kPa}$$

Flange Stress Analysis Results: kPa

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	98634.27	206984.23	155003.67	206984.23
Radial Flange	4538.99	137989.50	7133.02	137989.50
Tangential Flange	38251.67	137989.50	60112.47	137989.50
Maximum Average	68442.98	137989.50	107558.07	137989.50
Bolting	85466.06	171986.91	62462.02	171986.91

Minimum Required Flange Thickness [Rigidity] 134.087 mm

Estimated M.A.W.P. (Operating) 1.72 N/mm²

Estimated Finished Weight of Flange at given Thk. 1369.1 kgm

Estimated Unfinished Weight of Forging at given Thk 1631.1 kgm

Flange Stress results per WRC-538 at computed S_{fmax} (kPa):



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		Actual	Operating Allowed	stress ratio	Gasket Seating		stress ratio
					Actual	Allowed	
Longitudinal Hub	Sh	519960.34	519960.41	1.000	519960.34	519960.41	1.000
Tangential Flange	St	202823.55	389970.31	0.520	202823.55	389970.31	0.520
Tangential Flange	Sto	89065.17	259980.20	0.343	89065.17	259980.20	0.343
Combined	Sh/f + St	504684.81	779940.62	0.647	504684.81	779940.62	0.647
Combined	Sr + St	226890.86	779940.62	0.291	226890.86	779940.62	0.291
Combined	Sh + Sto	609025.56	779940.62	0.781	609025.56	779940.62	0.781
Radial Flange	Sr	24067.30	519960.41	0.046	24067.30	519960.41	0.046
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f_2

	f	f_2	f_3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g_1}{g_0} \right) + e \left(\frac{h}{h_0} \right) + g \left(\frac{g_1}{g_0} \right)^2 + i \left(\frac{h}{h_0} \right)^2 + k \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}{\left[1 + b \left(\frac{g_1}{g_0} \right) + d \left(\frac{h}{h_0} \right) + f \left(\frac{g_1}{g_0} \right)^2 + h \left(\frac{h}{h_0} \right)^2 + j \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.8/1.04)

= 1.7225

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h_0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.27/0.9) if $h/h_0 < 0.35$, else max (0.25; 0.3/0.8)

= 0.2951

In this case, the hub ratio h/h_0 , 0.1811 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.



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Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507470.6 kPa
Minimum permissible bolt stress	Sbmin	289983.2 kPa
Max. permissible bolt stress prior to flange damage	Sfmax	434331.9 kPa
Target assembly gasket stress	SgT	206988.0 kPa
Maximum permissible gasket stress	Sgmax	275984.0 kPa
Minimum gasket operating stress	Sgmin-O	96994.4 kPa
Minimum gasket seating stress	Sgmin-S	139991.9 kPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	0.0009 m ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	259980.2 kPa
Flange Material Operating Yield Stress	Syo	242583.1 kPa
Single flange rotation at Sfmax	Thetafmax	0.568 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (2002^2 - 1976^2)$$

$$= 0.1 \text{ m}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Abs)$$

$$= 206988 \cdot 0.081 / (44 \cdot 0.00091)$$

$$= 421576.2 \text{ kPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(421576; 507471)$$

$$= 421576.2 \text{ kPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(421576; 289983)$$

$$= 421576.2 \text{ kPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(421576; 434332)$$

$$= 421576.2 \text{ kPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 421576 \geq 139992 (0.081 / (44 \cdot 0.00091))$$

$$:: 421576.2 \geq 285124.0 \text{ kPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 421576 \geq (96994 \cdot 0.081 + \pi/4 \cdot 1 \cdot 1976^2) / (0.7 \cdot 0.00091 \cdot 44)$$

$$:: 421576.2 \geq 392056.8 \text{ kPa} \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 421576 \leq 275984 (0.081 / (44 \cdot 0.00091))$$

$$:: 421576.2 \leq 562101.6 \text{ kPa} \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$:: 421576 \leq 434332 (1/0.57)$$



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:: 421576.2 ≤ 764769.8 kPa Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

= Sbsel • K • Abs • φb

= 421576 • 0.2 • 0.00091 • 38.1

= 2913.0 N-m

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118113 kPa

Bolt Allowable Stress for: SA-193 B7 Sb 171986 kPa

Note: Since Sb ≥ SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)

= 52.14 • 361703 / 1 • 999622 • 0.41 / (1.84 • 201322320 • 19² • 193 • 0.3)

= 1.001 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)

= 52.14 • 230164 / 1 • 999622 • 0.41 / (1.84 • 197835184 • 19² • 193 • 0.3)

= 0.648 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)

= 52.14 • 361703 / 1 • 999622 • 0.41 / (2.28 • 201322320 • 19² • 193 • 0.3)

= 0.808 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)

= 52.14 • 230164 / 1 • 999622 • 0.41 / (2.28 • 197835184 • 19² • 193 • 0.3)

= 0.523 (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.59, Temperature Reduction per Fig. UCS 66.1 = 23 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C

Min Metal Temp. at Required thickness (UCS 66.1) -28 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	90153.80	206984.23	138762.45	206984.23
Radial Flange	5063.04	137989.50	7792.90	137989.50
Tangential Flange	33148.16	137989.50	51020.82	137989.50
Maximum Average	61650.98	137989.50	94891.64	137989.50
Bolting	85466.06	171986.91	62462.02	171986.91

Minimum Req'd Thickness, ambient, no Corrosion: 117.500 mm

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Flg Calc [Int P]: New Flange

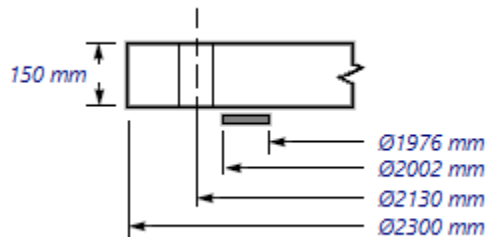
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Flange Input Data Values

Description: New Flange :

Channel COV RIGHT

Flange Type	TEMA Channel Cover		
Design Pressure	P	1.00	N/mm ²
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	2300.000	mm
Flange Thickness	t	150.0000	mm
Flange Material		SA-516 70	
Flange Material UNS number		K02700	
Flange Allowable Stress At Temperature	Sfo	137989.50	kPa
Flange Allowable Stress At Ambient	Sfa	137989.50	kPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	171986.91	kPa
Bolt Allowable Stress At Ambient	Sa	171986.91	kPa
Perimeter along the Center of the Bolts	L	6691.592	mm
Youngs Modulus for Blind Flange Matl.	Sy	197835184.00	kPa
Allowed Channel Cover Deflection	dfAllow	2.4500	mm
Diameter of Bolt Circle	C	2130.000	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		44	
Flange Face Outside Diameter	Fod	2027.000	mm
Flange Face Inside Diameter	Fid	1960.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2002.000	mm
Gasket Inside Diameter	Gi	1976.000	mm
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	62051.40	kPa
Gasket Thickness	tg	3.0000	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (Go - Gi)/2$	13.000 mm
Basic Gasket Width,	$bo = N/2$	6.500 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1989.151 mm



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Flg Calc [Int P]: New Flange

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Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1989^2 \cdot 1$$

$$= 3107518.500 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 1989 \cdot 3.75 \cdot 1$$

$$= 301099.812 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(3107518 + 301100 + 0, 0)$$

$$= 3408618.250 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 62051 \cdot 6.42 \cdot \pi \cdot 1989 + 0 \cdot 0 \cdot 0$$

$$= 2491155.250 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((3408618 + 0 + 4 \cdot 0 / 1989) / 171987, 2491155 / 171987)$$

$$= 0.020 \text{ m}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	0.020	0.040	
Radial Distance between Bolts and Edge:	38.100	85.000	
Circ. Spacing between the Bolts:	82.550	151.952	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 0.04 \cdot 171987 / (62051 \cdot \pi (2002 + 1976))$$

$$= 8.846 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a (A_m + A_b) / 2$$

$$= 171987 (0.02 + 0.04) / 2$$

$$= 5133961.50 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o$$

$$= 3408618.25 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2130 - 1989) / 2$$

$$= 70.4245 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot W_{m1} \cdot h_G \cdot B_{sc} / (t^2 \cdot G) + C \cdot Z \cdot P_{eq} \cdot G^2 / t^2$$

$$= 1.9 \cdot 3408618 \cdot 70.4 \cdot 1 / (147^2 \cdot 1989) + 0.3 \cdot 1 \cdot 1 \cdot 1989^2 / 147^2$$

$$= 65542.80 \text{ kPa}$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:



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Flg Calc [Int P]: New Flange

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$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 5133961 \cdot 70.4 \cdot 1 / (147^2 \cdot 1989)$$

$$= 15982.29 \text{ kPa}$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 3408618 / 0.04$$

$$= 85466.06 \text{ kPa}$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (2491155 / 0.04)$$

$$= 62462.02 \text{ kPa}$$

Flange Stress Analysis Results: kPa

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	65542.80	137989.50	15982.29	137989.50
Bolting	85466.06	171986.91	62462.02	171986.91

TEMA Channel Cover Deflection Calculations:

Actual deflection at center per TEMA formula	1.8482 mm
Allowed deflection at center of Cover	2.4500 mm

Reqd. Blind Flange Thickness at Center	104.311 mm
Reqd. Blind Flange Thickness at Gasket	53.028 mm
Min. Required Flange Thickness per TEMA	133.817 mm

Estimated M.A.W.P. (Operating)	1.56 N/mm ²
Estimated Finished Weight of Flange at given Thk.	4830.2 kgm
Estimated Unfinished Weight of Forging at given Thk	4830.2 kgm

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.89, Temperature Reduction per Fig. UCS 66.1 = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	10 °C
Min Metal Temp. at Required thickness (UCS 66.1)	4 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kPa

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	62947.31	137989.50	15349.40	137989.50
Bolting	85466.06	171986.91	62462.02	171986.91

Minimum Req'd Thickness, ambient, no Corrosion: 133.045 mm

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Internal Pressure Calculations: Step: 9 4:31pm Apr 10,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd N/mm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kPa	Element#
Channel COV LEFT		1	60	3	2300	137990	1
Channel FLG LEFT		1	60	3	1960	137990	2
Channel - rear		1.0193	22	3	1960	137990	3
shell		18.012	145	3	1960	137990	4
shell cone -rea		18.001	145	3	1960	137990	5
Shell-2		18.012	145	3	1960	137990	6
shell cone- fro		18.001	145	3	1960	137990	7
shell- front		18.012	145	3	1960	137990	8
Channel front		1.0193	22	3	1960	137990	9
Channel FLG RIGH		1	22	3	1960	137990	10
Channel COV RIGH		1	22	3	2300	137990	11

Element Required Thickness and MAWP :

From	To	Design Pressure N/mm ²	M.A.W.P. Corroded N/mm ²	M.A.P. New & Cold N/mm ²	Minimum Thickness mm	Required Thickness mm	Element#
Channel COV LEFT		1.00006	2.01	2.01	150	104.311	1
Channel FLG LEFT		1.00006	1.74	2.01	150	133.629	2
Channel - rear		1.00006	2.62	3.06	22	10.2931	3
shell		18.001	18.3	18.8	145	142.207	4
shell cone -rea		18.001	18.3	18.8	145	142.115	5
Shell-2		18.001	18.3	18.8	145	142.207	6
shell cone- fro		18.001	18.3	18.8	145	142.115	7
shell- front		18.001	18.3	18.8	145	142.207	8
Channel front		1.00006	2.62	3.06	22	10.2931	9
Channel FLG RIGH		1.00006	1.72	2.01	150	134.087	10
Channel COV RIGH		1.00006	1.56	1.74	150	133.817	11

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 15.2 N/mm², < design pressure: 18.001 N/mm²
 Shell Side MAPnc = 18.5 N/mm²
 Channel Side MAWP = 1.56 N/mm²
 Channel Side MAPnc = 1.74 N/mm²

Elements Suitable for Design Internal Pressure.

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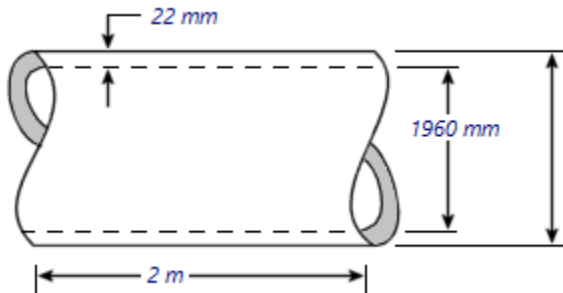
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Element: Channel - rear

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Channel - rear

Cylindrical Shell Design Information:			
Design Pressure P	1.019 N/mm ²	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	2.000 m	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	137989.5 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	137989.5 kPa
Inside Diameter Di	1960.000 mm	Actual Stress	53343.9 kPa
Spec. Min. Thk t,e, or sa	22.0000 mm	Spec. Nominal Thk.	22.000 mm
Surface Area	12.6 m ²	SG of Contents	1.000
Element Volume	5.9 m ³	Weight of Contents	59147.5 N
Empty Weight	20822.2 N	Operating Weight	79969.8 N
		Comp. All. Ope. Stress	128882.0 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1.02 \cdot 983) / (137990 \cdot 1 - 0.6 \cdot 1.02) \\
 &= 7.2931 + 3.0000 = 10.2931 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &\text{Less Operating Hydrostatic Head Pressure of } 0.019 \text{ N/mm}^2 \\
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137990 \cdot 1 \cdot 19) / (983 + 0.6 \cdot 19) \\
 &= 2.64 - 0.019 = 2.62 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137990 \cdot 1 \cdot 22) / (980 + 0.6 \cdot 22) \\
 &= 3.06 \text{ N/mm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1.02 \cdot (983 + 0.6 \cdot 19)) / (1 \cdot 19) \\
 &= 53343.879 \text{ kPa}
 \end{aligned}$$



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FileName : 124-C-NVS

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Element: Channel - rear

Step: 10 4:31pm Apr 10,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f (1 - R_f / R_o)$) 1.110 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 22$, $t_r = 11.2$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.59$, Temp. Reduction = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C

Min Metal Temp. at Required thickness (UCS 66.1) -28 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.555 N/mm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	22.000 mm	Basic MDMT	-5 °C
Stress/Thk. Ratio	0.587	Temperature Difference	24 °C
UCS-66(c) red.	No	Computed Min. Temp.	-28 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: Channel - rear

Cylindrical Shell From 30 to Tsht Ext. Chart: CS-2 at 82 °C

Maximum External Pressure per Code Case 2286:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 2000 / \sqrt{(1002 \cdot 19)}$$

$$= 14.495$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 13 \text{ and } M_x < 2 \cdot (D_o/t)^{0.94}, Ch = 1.12 \cdot M_x - 1.058$$

$$= 1.12 \cdot 14.5 - 1.058$$

$$= 0.0662$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.066 \cdot 199226 (19/2004)$$

$$= 199963.578 \text{ kPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 199963.6 / 242583.1$$

$$= 0.824$$

Predicted Buckling Stress [Fic]:

$$\text{Since } Fr > 0.552 \text{ and } Fr < 2.439, Fic = 0.7 \cdot F_y \cdot Fr^{0.4}$$

$$= 0.7 \cdot 242583 \cdot 0.820.4$$

$$= 157179.078 \text{ kPa}$$

Factor of Safety [FS]:

$$\text{Since } Fic > 0.55 F_y \text{ and } Fic < F_y$$

$$= 2.407 - 0.741 \cdot Fic / F_y$$

$$= 2.407 - 0.741 \cdot 157179 / 242583$$

$$= 1.927$$

Allowable Circumferential Compressive Stress [Fha]:

$$\text{Since } Fr > 0.552 \text{ and } Fr < 2.439, Fha = 0.7 \cdot F_y / FS \cdot Fr^{0.4}$$

$$= 0.7 \cdot 242583 / 1.93 \cdot 0.820.4$$

$$= 81571.914 \text{ kPa}$$



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Element: Channel - rear

Step: 10 4:31pm Apr 10, 2026

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha}(t / D_o)$$

$$= 2 \cdot 81.6 \text{ (19/2004)}$$

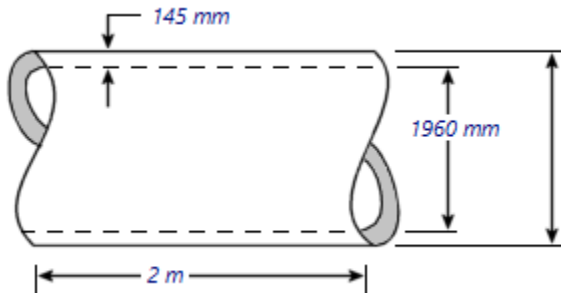
$$= 1.547 \text{ N/mm}^2$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: shell

Cylindrical Shell Design Information:			
Design Pressure P	18.012 N/mm ²	Int. Design Temperature	150 °C
Material	SA-516 70	Ext. Design Temperature	150 °C
Length	2.000 m	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	137989.5 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	137989.5 kPa
Inside Diameter Di	1960.000 mm	Actual Stress	135487.6 kPa
Spec. Min. Thk t _e , or sa	145.0000 mm	Spec. Nominal Thk.	145.000 mm
Surface Area	14.1 m ²	SG of Contents	0.566
Element Volume	3.5 m ³	Weight of Contents	19983.1 N
Empty Weight	145754.3 N	Operating Weight	165737.4 N
		Comp. All. Ope. Stress	147203.3 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (18 \cdot 983) / (137990 \cdot 1 - 0.6 \cdot 18) \\
 &= 139.2066 + 3.0000 = 142.2066 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &\text{Less Operating Hydrostatic Head Pressure of } 0.011 \text{ N/mm}^2 \\
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137990 \cdot 1 \cdot 142) / (983 + 0.6 \cdot 142) \\
 &= 18.3 - 0.011 = 18.3 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137990 \cdot 1 \cdot 145) / (980 + 0.6 \cdot 145) \\
 &= 18.8 \text{ N/mm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (18 \cdot (983 + 0.6 \cdot 142)) / (1 \cdot 142) \\
 &= 135487.578 \text{ kPa}
 \end{aligned}$$



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FileName : 124-C-NVS

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Element: shell

Step: 11 4:31pm Apr 10,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 6.888 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	15.212 N/mm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	145.000 mm	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

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FileName : 124-C-NVS

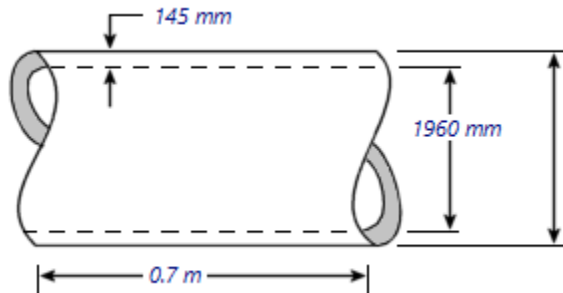
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Element: shell cone -rea

Step: 12 4:31pm Apr 10,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: shell cone -rea

Cylindrical Shell Design Information:			
Design Pressure P	18.001 N/mm ²	Int. Design Temperature	150 °C
Material	SA-516 70	Ext. Design Temperature	150 °C
Length	0.700 m	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	137989.5 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	137989.5 kPa
Inside Diameter Di	1960.000 mm	Actual Stress	135405.6 kPa
Spec. Min. Thk t,e, or sa	145.0000 mm	Spec. Nominal Thk.	145.000 mm
Surface Area	4.9 m ²	SG of Contents	0.000
Element Volume	1.2 m ³	Weight of Contents	0.0 N
Empty Weight	51014.0 N	Operating Weight	51014.0 N
		Comp. All. Ope. Stress	147203.3 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (18 \cdot 983) / (137990 \cdot 1 - 0.6 \cdot 18) \\
 &= 139.1153 + 3.0000 = 142.1153 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137990 \cdot 1 \cdot 142) / (983 + 0.6 \cdot 142) \\
 &= 18.3 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137990 \cdot 1 \cdot 145) / (980 + 0.6 \cdot 145) \\
 &= 18.8 \text{ N/mm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (18 \cdot (983 + 0.6 \cdot 142)) / (1 \cdot 142) \\
 &= 135405.641 \text{ kPa}
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 6.888 %



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Element: shell cone -rea

Step: 12 4:31pm Apr 10,2026

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.
This component must be impact tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

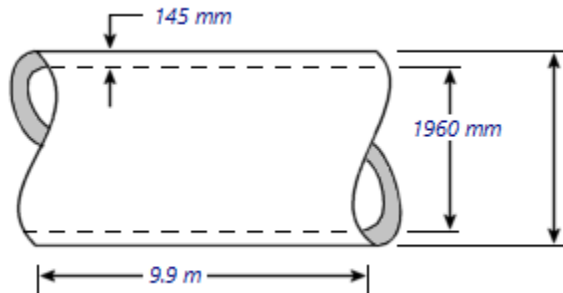
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	15.212 N/mm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	145.000 mm	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:			
Design Pressure P	18.012 N/mm ²	Int. Design Temperature	150 °C
Material	SA-516 70	Ext. Design Temperature	150 °C
Length	9.890 m	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	137989.5 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	137989.5 kPa
Inside Diameter Di	1960.000 mm	Actual Stress	135487.6 kPa
Spec. Min. Thk t _e , or sa	145.0000 mm	Spec. Nominal Thk.	145.000 mm
Surface Area	69.9 m ²	SG of Contents	0.566
Element Volume	17.4 m ³	Weight of Contents	98816.4 N
Empty Weight	720755.3 N	Operating Weight	819571.6 N
		Comp. All. Ope. Stress	147203.3 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (18 \cdot 983) / (137990 \cdot 1 - 0.6 \cdot 18) \\
 &= 139.2066 + 3.0000 = 142.2066 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.011 N/mm²

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137990 \cdot 1 \cdot 142) / (983 + 0.6 \cdot 142) \\
 &= 18.3 - 0.011 = 18.3 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137990 \cdot 1 \cdot 145) / (980 + 0.6 \cdot 145) \\
 &= 18.8 \text{ N/mm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (18 \cdot (983 + 0.6 \cdot 142)) / (1 \cdot 142) \\
 &= 135487.578 \text{ kPa}
 \end{aligned}$$



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Element: Shell-2

Step: 13 4:31pm Apr 10,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 6.888 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	15.212 N/mm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	145.000 mm	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

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FileName : 124-C-NVS

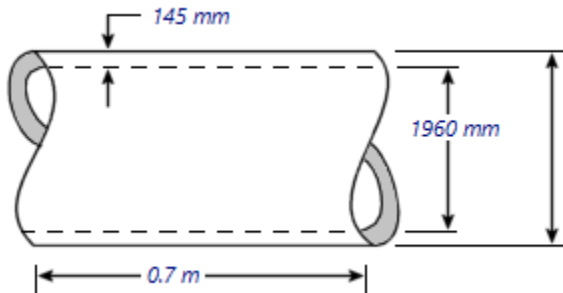
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Element: shell cone- fro

Step: 14 4:31pm Apr 10,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: shell cone- fro

Cylindrical Shell Design Information:			
Design Pressure P	18.001 N/mm ²	Int. Design Temperature	150 °C
Material	SA-516 70	Ext. Design Temperature	150 °C
Length	0.700 m	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	137989.5 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	137989.5 kPa
Inside Diameter Di	1960.000 mm	Actual Stress	135405.6 kPa
Spec. Min. Thk t,e, or sa	145.0000 mm	Spec. Nominal Thk.	145.000 mm
Surface Area	4.9 m ²	SG of Contents	0.000
Element Volume	1.2 m ³	Weight of Contents	0.0 N
Empty Weight	51014.0 N	Operating Weight	51014.0 N
		Comp. All. Ope. Stress	147203.3 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (18 \cdot 983) / (137990 \cdot 1 - 0.6 \cdot 18) \\
 &= 139.1153 + 3.0000 = 142.1153 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137990 \cdot 1 \cdot 142) / (983 + 0.6 \cdot 142) \\
 &= 18.3 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137990 \cdot 1 \cdot 145) / (980 + 0.6 \cdot 145) \\
 &= 18.8 \text{ N/mm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (18 \cdot (983 + 0.6 \cdot 142)) / (1 \cdot 142) \\
 &= 135405.641 \text{ kPa}
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 6.888 %



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FileName : 124-C-NVS

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Element: shell cone- fro

Step: 14 4:31pm Apr 10,2026

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm.
This component must be impact tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	15.212 N/mm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	145.000 mm	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

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FileName : 124-C-NVS

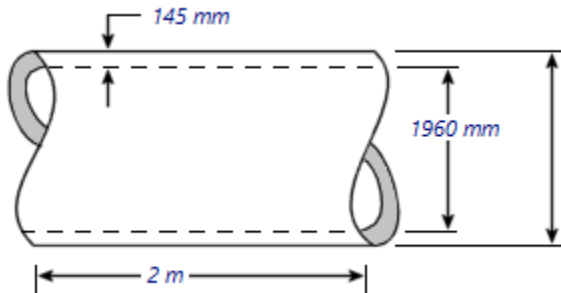
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Element: shell- front

Step: 15 4:31pm Apr 10,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: shell- front

Cylindrical Shell Design Information:			
Design Pressure P	18.012 N/mm ²	Int. Design Temperature	150 °C
Material	SA-516 70	Ext. Design Temperature	150 °C
Length	2.000 m	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	137989.5 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	137989.5 kPa
Inside Diameter Di	1960.000 mm	Actual Stress	135487.6 kPa
Spec. Min. Thk t _e , or sa	145.0000 mm	Spec. Nominal Thk.	145.000 mm
Surface Area	14.1 m ²	SG of Contents	0.566
Element Volume	3.5 m ³	Weight of Contents	19983.1 N
Empty Weight	145754.3 N	Operating Weight	165737.4 N
		Comp. All. Ope. Stress	147203.3 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (18 \cdot 983) / (137990 \cdot 1 - 0.6 \cdot 18) \\
 &= 139.2066 + 3.0000 = 142.2066 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &\text{Less Operating Hydrostatic Head Pressure of } 0.011 \text{ N/mm}^2 \\
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137990 \cdot 1 \cdot 142) / (983 + 0.6 \cdot 142) \\
 &= 18.3 - 0.011 = 18.3 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137990 \cdot 1 \cdot 145) / (980 + 0.6 \cdot 145) \\
 &= 18.8 \text{ N/mm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (18 \cdot (983 + 0.6 \cdot 142)) / (1 \cdot 142) \\
 &= 135487.578 \text{ kPa}
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 124-C-NVS

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Element: shell- front

Step: 15 4:31pm Apr 10,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 6.888 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	15.212 N/mm ²
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	145.000 mm	Basic MDMT	-17 °C
Stress/Thk. Ratio	-1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		Joint Efficiency E*	100 %

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FileName : 124-C-NVS

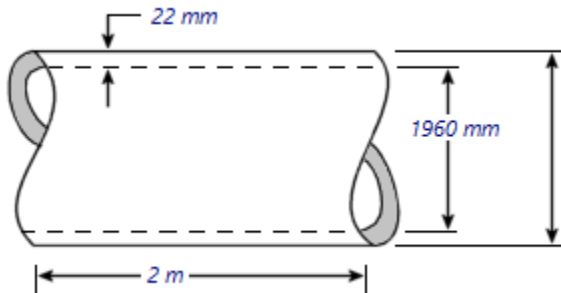
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Element: Channel front

Step: 16 4:31pm Apr 10,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Channel front

Cylindrical Shell Design Information:			
Design Pressure P	1.019 N/mm ²	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	2.000 m	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	137989.5 kPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	137989.5 kPa
Inside Diameter Di	1960.000 mm	Actual Stress	53343.9 kPa
Spec. Min. Thk t,e, or sa	22.0000 mm	Spec. Nominal Thk.	22.000 mm
Surface Area	12.6 m ²	SG of Contents	1.000
Element Volume	5.9 m ³	Weight of Contents	59147.5 N
Empty Weight	20822.2 N	Operating Weight	79969.8 N
		Comp. All. Ope. Stress	147203.3 kPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1.02 \cdot 983) / (137990 \cdot 1 - 0.6 \cdot 1.02) \\
 &= 7.2931 + 3.0000 = 10.2931 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &\text{Less Operating Hydrostatic Head Pressure of } 0.019 \text{ N/mm}^2 \\
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137990 \cdot 1 \cdot 19) / (983 + 0.6 \cdot 19) \\
 &= 2.64 - 0.019 = 2.62 \text{ N/mm}^2
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (137990 \cdot 1 \cdot 22) / (980 + 0.6 \cdot 22) \\
 &= 3.06 \text{ N/mm}^2
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1.02 \cdot (983 + 0.6 \cdot 19)) / (1 \cdot 19) \\
 &= 53343.879 \text{ kPa}
 \end{aligned}$$



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Element: Channel front

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f (1 - R_f / R_o)$) 1.110 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 22$, $t_r = 11.2$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.59$, Temp. Reduction = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C

Min Metal Temp. at Required thickness (UCS 66.1) -28 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.555 N/mm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	22.000 mm	Basic MDMT	-5 °C
Stress/Thk. Ratio	0.587	Temperature Difference	24 °C
UCS-66(c) red.	No	Computed Min. Temp.	-28 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: Channel front

Cylindrical Shell From 90 to 100 Ext. Chart: CS-2 at 82 °C

Maximum External Pressure per Code Case 2286:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 2000 / \sqrt{(1002 \cdot 19)}$$

$$= 14.495$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 13 \text{ and } M_x < 2 \cdot (D_o/t)^{0.94}, Ch = 1.12 \cdot M_x - 1.058$$

$$= 1.12 \cdot 14.5 - 1.058$$

$$= 0.0662$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.066 \cdot 197847 (19/2004)$$

$$= 198573.094 \text{ kPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 198573.1 / 242583.1$$

$$= 0.819$$

Predicted Buckling Stress [Fic]:

$$\text{Since } Fr > 0.552 \text{ and } Fr < 2.439, Fic = 0.7 \cdot F_y \cdot Fr^{0.4}$$

$$= 0.7 \cdot 242583 \cdot 0.820.4$$

$$= 156740.953 \text{ kPa}$$

Factor of Safety [FS]:

$$\text{Since } Fic > 0.55 F_y \text{ and } Fic < F_y$$

$$= 2.407 - 0.741 \cdot Fic / F_y$$

$$= 2.407 - 0.741 \cdot 156741 / 242583$$

$$= 1.928$$

Allowable Circumferential Compressive Stress [Fha]:

$$\text{Since } Fr > 0.552 \text{ and } Fr < 2.439, Fha = 0.7 \cdot F_y / FS \cdot Fr^{0.4}$$

$$= 0.7 \cdot 242583 / 1.93 \cdot 0.820.4$$

$$= 81288.094 \text{ kPa}$$



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Element: Channel front

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Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha}(t / D_o)$$

$$= 2 \cdot 81.3 \text{ (19/2004)}$$

$$= 1.541 \text{ N/mm}^2$$

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Shop Pressure Test Results: Step: 17 4:31pm Apr 10,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	19.776 N/mm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	23.401 N/mm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	24.079 N/mm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	16.733 N/mm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	25.741 N/mm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	25.741 N/mm ²
App 27-4	= M.A.W.P.	15.212 N/mm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	2.022 N/mm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1.300 N/mm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2.238 N/mm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	1.711 N/mm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	1.430 N/mm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1.430 N/mm ²
App 27-4	= M.A.W.P.	1.555 N/mm ²

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 18 • 1

= 23.401 N/mm²

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 1 • 1

= 1.300 N/mm²

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kPa & N/mm²):

From To	Stress	Allowable	Ratio	Pressure
Channel - rear	59556.4	233982.2	0.255	1.32
shell	172333.1	233982.2	0.737	23.42
shell cone -rea	172333.1	233982.2	0.737	23.42
Shell-2	172333.1	233982.2	0.737	23.42
shell cone- fro	172333.1	233982.2	0.737	23.42
shell- front	172333.1	233982.2	0.737	23.42
Channel front	59556.4	233982.2	0.255	1.32

Stress ratios for Nozzle and Pad Materials (kPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2- Tube outlet	Nozzle	117991.02	117991.02	1.000
N2- Tube outlet	Pad	137989.50	137989.50	1.000
N3- Shell inlet	Nozzle	137989.50	137989.50	1.000
N4- Shell outle	Nozzle	137989.50	137989.50	1.000
N1- Tube inlet	Nozzle	117991.02	117991.02	1.000
N1- Tube inlet	Pad	137989.50	137989.50	1.000

Minimum

1.000



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Shop Pressure Test Results: Step: 17 4:31pm Apr 10, 2026

Stress ratios for Jacket/Vapor Belt Materials:

Description	Cylinder/Closure	Ambient	Operating	Ratio
Jacket	Cylinder	137989.50	137989.50	1.000
Jacket	Closure	137989.50	137989.50	1.000
Jacket	Cylinder	137989.50	137989.50	1.000
Jacket	Closure	137989.50	137989.50	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (kPa):

Description	Ambient	Operating	Ratio
Channel COV LEFT	137989.50	137989.50	1.000
Channel FLG LEFT	137989.50	137989.50	1.000
Channel - rear	137989.50	137989.50	1.000
shell	137989.50	137989.50	1.000
shell cone -rea	137989.50	137989.50	1.000
Shell-2	137989.50	137989.50	1.000
shell cone- fro	137989.50	137989.50	1.000
shell- front	137989.50	137989.50	1.000
Channel front	137989.50	137989.50	1.000
Channel FLG RIGHT	137989.50	137989.50	1.000
Channel COV RIGHT	137989.50	137989.50	1.000

Minimum

1.000

Stress ratios for Exchanger Materials (kPa):

Description	Ambient	Operating	Ratio
Tube Material	92392.97	92392.97	1.000
Tubesheet Material	137989.50	137989.50	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (kPa):

Description	Ambient	Operating	Ratio
N2- Tube outlet	25922.52	215983.55	0.120
N3- Shell inlet	111851.36	224982.88	0.497
N4- Shell outle	147405.80	224982.88	0.655
N1- Tube inlet	25922.52	215983.55	0.120

Elements Suitable for Test Pressure.

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ASME TS Calc: Pressure Test

Case: 1 4:31p Apr 10,2026

Input Echo, Tubesheet Number 1, Description: tubesheet

Shell Data:

Main Shell Description: shell

Shell Test Pressure		23.40 N/mm ²
Shell Thickness	ts	145.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1960.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	137989.50 kPa
Shell Allowable Stress at Ambient		137989.50 kPa
Shell Allowable Stress at Operating Temperature		137989.50 kPa

Channel Data:

Channel Description: Channel - rear

Channel Type:		Cylinder
Channel Test Pressure		1.30 N/mm ²
Channel Thickness	tc	22.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1960.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	137989.50 kPa
Channel Allowable Stress at Ambient		137989.50 kPa
Channel Allowable Stress at Operating Temperature		137989.50 kPa

Tube Data:

Number of Tube Holes	Nt	4267
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	16000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		15290.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		92392.97 kPa
Tube Allowable Stress At Ambient		92392.97 kPa
Tube Allowable Stress at Operating Temperature		92392.97 kPa
Tube Yield Stress At design Temperature	Syt	179986.31 kPa
Tube Pitch (Center to Center Spacing)	P	26.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	3.0000 mm
Groove Weld Leg	ag	3.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	889.717 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm



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ASME TS Calc: Pressure Test

Case: 1 4:31p Apr 10,2026

Length of Expanded Portion of Tube	ltx	200.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	2.4 m ²
Perimeter of Tube Layout	Cp	5774.2 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	137989.50 kPa
Tubesheet Allowable Stress at Ambient	Tt	137989.50 kPa
Tubesheet Allowable Stress at Operating Temper.		137989.50 kPa
Thickness of Tubesheet	h	355.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	2250.000 mm

Area of the Untubed Lanes	AL	0.000 m ²
---------------------------	----	----------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Use Operating Temperatures for Thermal cases		Yes
Shell Operating Temperature	Tsop	21.11 °C
Channel Operating Temperature	Tcop	21.11 °C
Tube Operating Temperature	Ttop	21.11 °C
Tubesheet Operating Temperature	Tshop	21.11 °C
Mean Shell Metal Temp. along Shell len.	Tsm	21.11 °C
Mean Tube Metal Temp. along Tube length	Ttm	21.11 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



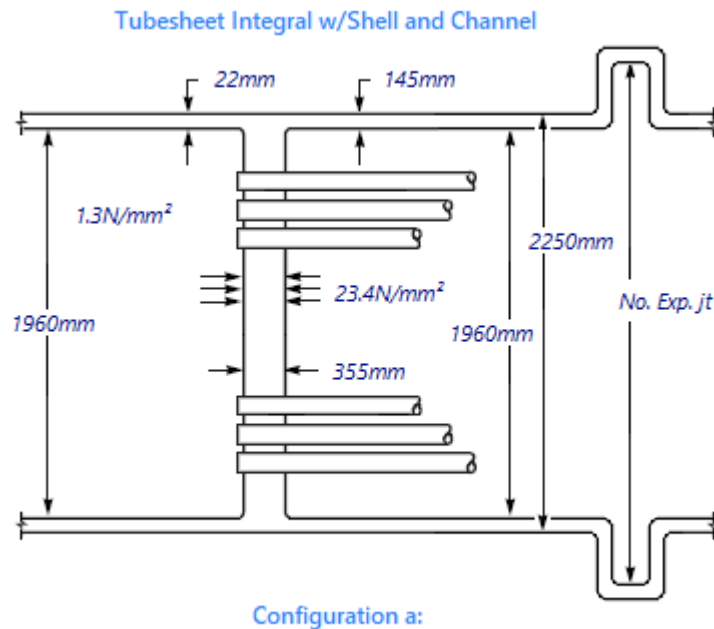
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ASME TS Calc: Pressure Test

Case: 1 4:31p Apr 10, 2026



Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C> 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20133E+06 N/mm²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20133E+06 N/mm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20133E+06 N/mm²
Elastic Mod. at Operating Temperature	21.1 °C	0.20133E+06 N/mm²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20271E+06 N/mm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20271E+06 N/mm²
Elastic Mod. at Operating Temperature	21.1 °C	0.20271E+06 N/mm²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20271E+06 N/mm²
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20271E+06 N/mm²
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20271E+06 N/mm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20271E+06 N/mm²
Elastic Mod. at Operating Temperature	21.1 °C	0.20271E+06 N/mm²
Elastic Mod. at Tubsht. Operat. Temp.	21.1 °C	0.20271E+06 N/mm²



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ASME TS Calc: Pressure Test

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TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20271E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20271E+06 N/mm ²
Elastic Mod. at Operating Temperature	21.1 °C	0.20271E+06 N/mm ²

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the

type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1.3 \cdot (19/2 - 0)) / (92393 \cdot 1 + 0.4 \cdot 1.3)$$

$$= 0.1333 + 0.0000 = 0.1333 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 15290 / \sqrt{(9.52 \cdot 2.11)}$$

$$= 3410.624$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 2 \cdot (D_o/t)^{0.94}, Ch = 0.55 / (D_o/t)$$

$$= 0.55 / (19/2.11)$$

$$= 0.0609$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.061 \cdot 202713 (2.11/19)$$

$$= 2188336.500 \text{ kPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 2188336.5 / 179986.3$$

$$= 12.158$$

Predicted Buckling Stress [Fic]:

$$\text{Since } Fr \geq 2.439, Fic = F_y$$

$$= 179986.312 \text{ kPa}$$

Factor of Safety [FS]:

$$\text{Since } Fic [179986] \geq F_y [179986], FS = 1.667$$

Allowable Circumferential Compressive Stress [Fha]:

$$\text{Since } Fr > 2.439, Fha = F_y / FS$$

$$= 179986.312 / 1.667$$

$$= 107970.180 \text{ kPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha (t / D_o)$$

$$= 2 \cdot 108 (2.11/19)$$

$$= 23.919 \text{ N/mm}^2$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	2.0643 mm
Allowable Internal Pressure at Corroded thickness	22.46 N/mm ²
Required Internal Design Pressure	1.30 N/mm ²
Allowable External Pressure at Corroded thickness	23.92 N/mm ²
Required External Design Pressure	23.40 N/mm ²



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ASME TS Calc: Pressure Test

Case: 1 4:31p Apr 10, 2026

Required Thickness due to Shell Side pressure 2.0643 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio $[\rho]$:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 200/355$$

$$= 0.5634 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole diameter $[d^*]$:

$$= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(19 - 2 \cdot 2.11 (202713/202713) \cdot (0.5634); 19 - 2 \cdot 2.11)$$

$$= 17.4581 \text{ mm}$$

Equivalent outer tube limit circle diameter $[Do]$:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 890 + 19$$

$$= 1798.484 \text{ mm}$$

Determine the basic ligament efficiency for shear $[\mu]$:

$$= (p - dt) / p$$

$$= (26 - 19) / 26$$

$$= 0.267$$

Equivalent outer tube limit radius $[ao]$:

$$= Do / 2$$

$$= 1798/2$$

$$= 899.2420 \text{ mm}$$

Effective tube pitch $[p^*]$:

$$= p / \sqrt{ 1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2) }$$

$$= 26 / \sqrt{ 1 - 4 \cdot \min(0 \cdot 1000248 ; 4 \cdot 1798 \cdot 26) / (\pi \cdot 1798^2) }$$

$$= 26.0000 \text{ mm}$$

Effective ligament efficiency for bending $[\mu^*]$:

$$= (p^* - d^*) / p^*$$

$$= (26 - 17.5) / 26$$

$$= 0.3285$$

Ratio $[ps]$:

$$= as / ao$$

$$= 980/899$$

$$= 1.0898$$

Ratio $[pc]$:

$$= ac / ao$$

$$= 980/899$$

$$= 1.0898$$

Compute parameter $[xt]$:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 4267 ((19 - 2 \cdot 2.11) / (2 \cdot 899))^2$$

$$= 0.7099$$

Determine parameter $[xs]$:



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ASME TS Calc: Pressure Test

Case: 1 4:31p Apr 10, 2026

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 4267 (19 / (2 \cdot 899))^2$$

$$= 0.5213$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts(Ds + ts)Es/L$$

$$= \pi \cdot 145 (1960 + 145)201334/15290$$

$$= 12625329.0000 \text{ N/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt(Dt - tt)Et/L$$

$$= \pi \cdot 2.11 (19 - 2.11)202713/15290$$

$$= 1488.6167 \text{ N/mm}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 12625329 / (4267 \cdot 1489)$$

$$= 1.98764$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks/Kt)$$

$$= 1 / (1 + 12625329/0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [βs]:

$$= ((12(1 - vs^2))^{1/4}) / ((Ds + ts)ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1960 + 145)145)$$

$$= 0.0033 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta s \cdot Es \cdot ts^3 / (6(1 - vs^2))$$

$$= 0.0033 \cdot 201334 \cdot 145^3 / (6(1 - 0.3^2))$$

$$= 369891296.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(Ds + ts) \cdot ks/h^3 \cdot (1 + h \cdot \beta s + (h \cdot \beta s)^2/2)$$

$$= 6(1960 + 145) \cdot 369891296/355^3 \cdot (1 + 355 \cdot 0.0033 + (355 \cdot 0.0033)^2/2)$$

$$= 297633.5625 \text{ N/mm}^2$$

Shell coefficient [δs]:

$$= Ds^2 / (4 \cdot Es \cdot ts) \cdot (1 - Ds \cdot vs / (2(Ds + ts)))$$

$$= 1960^2 / (4 \cdot 201334 \cdot 145) \cdot (1 - 1960 \cdot 0.3 / (2(1960 + 145)))$$

$$= 0.28305E-04 \text{ mm/kPa}$$

Channel coefficient [βc]:

$$= ((12(1 - vc^2))^{1/4}) / ((Dc + tc)tc)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1960 + 22)22)$$

$$= 0.0087 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta c \cdot Ec \cdot tc^3 / (6(1 - vc^2))$$

$$= 0.0087 \cdot 202713 \cdot 22^3 / (6(1 - 0.3^2))$$

$$= 3441513.7500 \text{ N/mm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:



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$$\begin{aligned}
 &= 6(Dc + tc/\cos(\theta)) \cdot kc/h^3 \cdot (Cm\theta + h \cdot \beta c \cdot (Cm\Delta + Cq\theta)/2 + (h \cdot \beta c)^2/2 \cdot Cq\Delta) \\
 &= 6(1960 + 22/\cos(0)) \cdot 3441513/355^3 \cdot (1 + 355 \cdot 0.0087 \cdot (1 + 1)/2 + (355 \cdot 0.0087)^2/2 \cdot 1) \\
 &= 8110.4 \text{ N/mm}^2
 \end{aligned}$$

Channel coefficient [δc]:

$$\begin{aligned}
 &= Dc^2/(4 \cdot Ec \cdot Tc) \cdot (1 - Dc \cdot vc/(2(Dc + tc))) \\
 &= 1960^2/(4 \cdot 202713 \cdot 22) \cdot (1 - 1960 \cdot 0.3/(2(1960 + 22))) \\
 &= 0.18342\text{E-}03 \text{ mm/kPa}
 \end{aligned}$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$\begin{aligned}
 h/p &= 13.653847 ; \mu^* = 0.328533 ; E^* = 64612 \text{ N/mm}^2 \\
 E^*/E &= 0.318741 ; \nu^* = 0.336622
 \end{aligned}$$

Note: As h/p (13.654) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [Xa]:

$$\begin{aligned}
 &= ((24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2)/(E^* \cdot L \cdot H^3))^{1/4} \\
 &= ((24(1 - 0.34^2)4267 \cdot 202713 \cdot 2.11 (19 - 2.11)899^2)/(64613 \cdot 15290 \cdot 355^3))^{1/4} \\
 &= 1.8626
 \end{aligned}$$

Values from Table 4.18.3

$$\begin{aligned}
 Zd &= 0.216484 ; Zv = 0.166592 ; Zm = 0.686789 \\
 Za &= 0.720801\text{E}+00 ; Zw = 0.166592
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$\begin{aligned}
 &= A / Do \\
 &= 2250/1798 \\
 &= 1.2511
 \end{aligned}$$

Compute coefficient [F]:

$$\begin{aligned}
 &= (1 - \nu^*)/ E^* \cdot (\lambda s + \lambda c + E \cdot \ln(K)) \\
 &= (1 - 0.34)/64613 \cdot (297634 + 8110 + 202713 \cdot \ln(1.25)) \\
 &= 3.6052
 \end{aligned}$$

Compute parameter [ϕ]:

$$\begin{aligned}
 &= (1 + \nu^*)F \\
 &= (1 + 0.34)3.61 \\
 &= 4.8188
 \end{aligned}$$

Compute parameter [Q1]:

$$\begin{aligned}
 &= (\rho s - 1 - \phi \cdot Zv)/(1 + \phi \cdot Zm) \\
 &= (1.09 - 1 - 4.82 \cdot 0.17)/(1 + 4.82 \cdot 0.69) \\
 &= -0.165441
 \end{aligned}$$

Compute Parameter [Qz1]:

$$\begin{aligned}
 &= (Zd + Q1 \cdot Zw)/2 \cdot Xa^4 \\
 &= (0.22 + -0.17 \cdot 0.17)/2 \cdot 1.86^4 \\
 &= 1.1370
 \end{aligned}$$

Compute parameter [Qz2]:

$$\begin{aligned}
 &= (Zv + Q1 \cdot Zm)/2 \cdot Xa^4 \\
 &= (0.17 + -0.17 \cdot 0.69)/2 \cdot 1.86^4 \\
 &= 0.3188
 \end{aligned}$$

Compute parameter [U]:



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$$\begin{aligned}
 &= (Zw + (ps - 1)Zm)Xa^4 / (1 + \phi \cdot Zm) \\
 &= (0.17 + (1.09 - 1)0.69)1.86^4 / (1 + 4.82 \cdot 0.69) \\
 &= 0.6376
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

$$= 0 \text{ (config a)}$$

Compute Parameter [y]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ws]:

$$\begin{aligned}
 &= ps \cdot ks \cdot \beta s \cdot \delta S (1 + h \cdot \beta s) \\
 &= 1.09 \cdot 369891296 \cdot 0.0033 \cdot 0.28305E-04 (1 + 355 \cdot 0.0033) \\
 &= 81391.4141 \text{ mm}^2
 \end{aligned}$$

Calculate parameter [w*s]:

$$\begin{aligned}
 &= ao^2 (ps^2 - 1) \cdot (ps - 1) / 4 - ws \\
 &= 899^2 (1.09^2 - 1) \cdot (1.09 - 1) / 4 - 81391 \\
 &= -77984.0625 \text{ mm}^2
 \end{aligned}$$

Parameter [wc]:

$$\begin{aligned}
 &= pc \cdot kc [\delta C (CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) - \\
 &\quad \zeta c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2)] \\
 &= 1.09 \cdot 3441513 [0.000183418 (1 + 0.0087 \cdot 1 \cdot 355) - \\
 &\quad 0.000 (1/355 + 1 \cdot 0.0087/2)] \\
 &= 24495.0566 \text{ mm}^2
 \end{aligned}$$

Parameter [w*c]:

$$\begin{aligned}
 &= ao^2 [(pc^2 + 1) \cdot (pc - 1) / 4 - (ps - 1) / 2] - wc \\
 &= 899^2 [(1.09^2 + 1) \cdot (1.09 - 1) / 4 - (1.09 - 1) / 2] - 24495 \\
 &= -21087.7070 \text{ mm}^2
 \end{aligned}$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$\begin{aligned}
 &= Ps \{ xs + 2(1 - xs)vt + [2/Kst \cdot (Ds/Do)^2] vs - \\
 &\quad [(ps^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(Dj^2 - (Ds)^2)/Do^2] \} \\
 &= 23.4 \{ 0.52 + 2(1 - 0.52)0.3 + [2/1.99 \cdot \\
 &\quad (1960/1798)^2]0.3 - [(1.09^2 - 1)/(1 \cdot 1.99)] - \\
 &\quad [(1 - 1)/(2 \cdot 1 \cdot 1.99)] [(0^2 - (1960)^2)/1798^2] \} \\
 &= 25.1003 \text{ N/mm}^2
 \end{aligned}$$

Pressure [P't]:

$$\begin{aligned}
 &= [xt + 2(1 - xt)vt + 1/(J \cdot Kst)] Pt \\
 &= [0.71 + 2(1 - 0.71)0.3 + 1/(1 \cdot 1.99)] 1.3 \\
 &= 1.8033 \text{ N/mm}^2
 \end{aligned}$$

Pressure [Py]:

$$\begin{aligned}
 &= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2) \\
 &= 4267 \cdot 1489 \cdot 0 / (3.14 \cdot 899^2) \\
 &= 0.0000 \text{ N/mm}^2
 \end{aligned}$$

Pressure [Pw]:

$$\begin{aligned}
 &= -yb \cdot U \cdot W* / (2 \cdot \pi \cdot ao^2) \\
 &= -0 \cdot 0.64 \cdot 0 / (2 \cdot \pi \cdot 899^2) \\
 &= -0.0000 \text{ N/mm}^2
 \end{aligned}$$



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Calculate the pressure [Prim]:

$$= -(U/ao^2) \cdot (\omega * s \cdot Ps - \omega * c \cdot Pt)$$

$$= -(0.64/899^2) \cdot (-77984 \cdot 23.4 - -21088 \cdot 1.3)$$

$$= 1.4173 \text{ N/mm}^2$$

Calculate the pressure [Pw]:

$$= (U/ao^2) \cdot (\omega s \cdot P * s - \omega c \cdot P * c)$$

$$= (0.64/899^2) \cdot (81391 \cdot 0 - 24495 \cdot 0)$$

$$= 0.0000 \text{ N/mm}^2$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst (Qz1 + (ps - 1) Qz2)) \cdot (Ps - Pt + Py + Pw + Prim)$$

$$= 0.1000E+01 \cdot 1.99 / (1 + 1 \cdot 1.99 (1.14 + (1.09 - 1) 0.32)) \cdot (25.1 - 1.8 + 0 + 0 + -0 + 1.42)$$

$$= 14.8098 \text{ N/mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega * s \cdot Ps - \omega * c \cdot Pt) - (\omega s \cdot P * s - \omega c \cdot P * c)) + W * \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$

$$= [((-77984 \cdot 23.4 - -21088 \cdot 1.3) - (81391 \cdot 0 - 24495 \cdot 0)) \cdot 1 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 4.82 \cdot 0.69)$$

$$= -417068.4062 \text{ N}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.17 + 2 \cdot -417068 / (14.8 \cdot 899^2)$$

$$= -0.23510$$

Fm Value from Table 4.18.3 = 0.117550

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu *) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.12 / 0.33) \cdot (2 \cdot 899 / (355 - 0))^2 \cdot 14.8$$

$$= 203992.3281 \text{ kPa}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot Sy = 1.5 \cdot 0.95 \cdot 249981 = 356223 \text{ kPa}, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot Fm / \mu *) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.11 / 0.33) \cdot (2 \cdot 899 / (254 - 0))^2 \cdot 13.9$$

$$= 356219.7812 \text{ kPa}$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + Cats + Catc$$

$$= 254 + 0 + 0$$

$$= 253.8484 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / ao) \cdot (0.8 \cdot \beta T \cdot Sy)$$

$$= (2 \cdot 0.27 \cdot 355 / 899) \cdot 189986$$

$$= 40097.1797 \text{ kPa}$$

Shear Stress is not required to be computed; [Pe] <= Tau[limit]

Tubesheet Shear Stress - final thickness [τ_f]:

$$= (Ap \cdot Pe) / (\mu \cdot h \cdot Cp)$$

$$= (2.44 \cdot 7.83) / (0.27 \cdot 67.2 \cdot 5774)$$

$$= 184515.1094 \text{ kPa}$$



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Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$= H + Cats + Catc$$

$$= 67.2 + 0 + 0$$

$$= 67.1995 \text{ mm}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(HreqB; HreqS)$$

$$= \max(254 ; 67.2)$$

$$= 253.8484 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$Ftmin = 0.87, Ftmax = 1.07$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot Ftmin) / (Xt - Xs)$$

$$= ((23.4 \cdot 0.52 - 1.3 \cdot 0.71) - (14.8) \cdot 0.87) /$$

$$(0.71 - 0.52)$$

$$= -8900.1953 \text{ kPa}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot Ftmax) / (Xt - Xs)$$

$$= ((23.4 \cdot 0.52 - 1.3 \cdot 0.71) - (14.8) \cdot 1.07) /$$

$$(0.71 - 0.52)$$

$$= -24014.8965 \text{ kPa}$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(| \sigma_{t,1} |; | \sigma_{t,2} |) = 24015 \text{ kPa}$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= \beta T \cdot Sy,t = 170987.0000 \text{ kPa}$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Zd + Q3 \cdot Zw)Xa^4); 1.25)$$

$$= \max((3.25 - 0.25(0.22 + -0.24 \cdot 0.17)1.86^4); 1.25)$$

$$= 2.0000 \text{ (Should be } \leq 2 \text{)}$$

Determine the factor [rt]:

$$= (\sqrt{ dt^2 + (dt - 2 \cdot tt)^2 }) / 4$$

$$= (\sqrt{ 19^2 + (19 - 2 \cdot 2.11)^2 }) / 4$$

$$= 6.0355 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Et / Syt)}$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 202713 / 179986)}$$

$$= 154.1158$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 0 \cdot 0 / 6.04$$

$$= 0.0000$$

Buckling Allowable Stress [Stb]:

$$= (Sy,t / Fs) \cdot (1 - Ft / (2 \cdot Ct))$$

$$= (179986 / 2) \cdot (1 - 0 / (2 \cdot 154))$$

$$= 170987.000 \text{ kPa (Never greater than } Sot \text{)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$



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$$= 24015 \cdot 0.00011$$

$$= 2696.59 \text{ N}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (19 - 2.11) 2.11$$

$$= 0.000 \text{ m}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (19 - 2.11) 170987$$

$$= 19199.818 \text{ N}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 3 (19 + 0.67 \cdot 3) 170987$$

$$= 19199.8184 \text{ N}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 170987 / 170987$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 3 (19 + 0.67 \cdot 3) 170987$$

$$= 18665.6699 \text{ N}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 19)^2 + 1.07 \cdot 2.11 \cdot (19 - 2.11) 1 \cdot 1} - 0.75 \cdot 19 \right)$$

$$= 2.562 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm

Minimum Required Groove Weld Leg [agr]: 1.2810 mm

Tube-Tubesheet Jt allowable, 19200 is $\geq$ tube strength 19200 N

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= \beta T \cdot S_{y,s} = 0.95 \cdot 259980 = 370472 \text{ kPa}$$

Axial Membrane Stress in Shell [σ_s, m]:

$$= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 899^2 / ((1960 + 145) 145) \cdot [14.8 + (1.09^2 - 1)$$

$$(23.4 - 1.3)] + 980^2 \cdot 1.3 / ((1960 + 145) 145)$$

$$= 54312.5352 \text{ kPa}$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 145 mm, the minimum Shell length adjacent to the tubesheet should be 959.587 mm

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):



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Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	203992.3	<= 356222.9 kPa	Pass
Tubesheet Shear Stress	66089.4	<= 189985.5 kPa	Pass
Maximum Tube Stress	24014.9	<= 170987.0 kPa	Pass
Minimum Tube Stress (Buckling)	-24014.9	<= -170987.0 kPa	Pass
Maximum Force on any one Tube	2696.6	<= 19199.8 N	Pass
Axial Membrane Stress in Shell	54312.5	<= 370471.8 kPa	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	253.848	355.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	3.000	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/Fail
D1uc	10.408 ...	13218	356223 4880 189986	Ps+Pt-Th D1	Ok
D2uc	263.606 ...	215514	356223 70969 189986	Ps+Pt-Th D2	Ok
D3uc	253.848 ...	203992	356223 66089 189986	Ps+Pt-Th D3	Ok
Max:	263.6063 ... mm	0.605	0.374 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and
Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	----	Shell Stresses Ten Allwd Cmp Allwd	:	----	Shell Band Stress Ten Allwd Cmp Allwd	:	Pass Fail
D1uc	547	370471	...	...	...	...	Ok
D2uc	53765	370471	...	...	...	...	Ok
D3uc	54313	370471	...	...	...	...	Ok
Max	RATIO	0.147	...	...	...	...	



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Tube, Shell and Channel Stress Summary:

---- Tube Stresses Tube Loads Shell Stress Channel Stress Pass											
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	1616	170987	-130	170987	181	19199	...	370471	...	370471	Ok
D2uc	25631	170987	-25631	170987	2878	19199	...	370471	...	370471	Ok
D3uc	24015	170987	-24015	170987	2697	19199	...	370471	...	370471	Ok
Max RATIO	0.150		0.150		0.150		...		...		

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	263.606	355.000	Ok
Tube thickness :	2.064	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	3.000	Ok
Tube-tubesheet groove weld leg :	1.281	3.000	Ok

Min Shell length of thk, (145.000) adj. to tubesheet: 959.587 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -2.090 mm
 Due to Pressure + Thermal Shell Compresses by : -2.090 mm

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0328	...	...	...	...	...
2	-2.0898	...	...	...	...	...
3	-2.0571	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...

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External Pressure Calculations: Step: 18 4:31pm Apr 10,2026

External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length m	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kPa
10	20	No Calc	...	147	No Calc	No Calc
20	30	No Calc	...	147	No Calc	No Calc
30	Tsht	1.99997	2004	19	No Calc	No Calc
Tsht	40	No Calc	...	19	No Calc	No Calc
40	Jckt	0.20003	2250	142	No Calc	No Calc
Jckt	Jckt	1.6	2250	142	No Calc	No Calc
Jckt	50	11.69	2250	142	No Calc	No Calc
50	60	11.69	2250	142	No Calc	No Calc
60	70	11.69	2250	142	No Calc	No Calc
70	80	11.69	2250	142	No Calc	No Calc
80	Jckt	11.69	2250	142	No Calc	No Calc
Jckt	Jckt	1.6	2250	142	No Calc	No Calc
Jckt	Tsht	0.19997	2250	142	No Calc	No Calc
Tsht	90	No Calc	...	142	No Calc	No Calc
90	100	2.00003	2004	19	No Calc	No Calc
100	110	No Calc	...	147	No Calc	No Calc
110	120	No Calc	...	147	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure N/mm ²	External M.A.W.P. N/mm ²
10	20	150	50.7492	0.10131	No Calc
20	30	150	126.619	0.10131	No Calc
30	Tsht	22	8.96634	0.10131	1.55
Tsht	40	22	No Calc	0.10131	No Calc
40	Jckt	145	No Calc	...	17.6
Jckt	Jckt	145	No Calc	...	17.6
Jckt	50	145	No Calc	...	17.6
50	60	145	No Calc	...	17.6
60	70	145	No Calc	...	17.6
70	80	145	No Calc	...	17.6
80	Jckt	145	No Calc	...	17.6
Jckt	Jckt	145	No Calc	...	17.6
Jckt	Tsht	145	No Calc	...	17.6
Tsht	90	145	No Calc	...	No Calc
90	100	22	8.98282	0.10131	1.54
100	110	150	127.152	0.10131	No Calc
110	120	150	99.7074	...	No Calc

Minimum

1.54

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners m	Allowable Length Bet. Stiffeners m	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	1.99997	1676.51	No Calc	No Calc



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External Pressure Calculations: Step: 18 4:31pm Apr 10,2026

Tsht	40	No Calc	No Calc	No Calc	No Calc
40	Jckt	0.20003	No Calc	No Calc	No Calc
Jckt	Jckt	1.6	No Calc	No Calc	No Calc
Jckt	50	11.69	No Calc	No Calc	No Calc
50	60	11.69	No Calc	No Calc	No Calc
60	70	11.69	No Calc	No Calc	No Calc
70	80	11.69	No Calc	No Calc	No Calc
80	Jckt	11.69	No Calc	No Calc	No Calc
Jckt	Jckt	1.6	No Calc	No Calc	No Calc
Jckt	Tsht	0.19997	No Calc	No Calc	No Calc
Tsht	90	No Calc	No Calc	No Calc	No Calc
90	100	2.00003	1544.68	No Calc	No Calc
100	110	No Calc	No Calc	No Calc	No Calc
110	120	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

For vessels with Jackets, the External Design Pressure includes the core vacuum pressure. The External MAWP is the total allowable external pressure for that segment minus the core vacuum pressure. The External MAWP can be less than the External Design pressure and this is acceptable given that the actual thickness is greater than the required thickness.

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Element and Detail Weights:

Step: 19

4:31pm Apr 10,2026

Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume m ³	Corroded Metal Wgt. kgm	Corroded ID Volume m ³	Extra due Misc % kgm
10	20	4830.22	...	4830.22	...	483.022
20	30	1369.09	0.5477	1342.56	0.54834	136.909
30	40	2123.42	5.89184	1836.64	5.92797	212.342
40	50	14863.8	3.51691	14577	3.55303	1486.38
50	60	5202.33	1.23092	5101.96	1.24356	520.233
60	70	73501.5	17.3911	72083.4	17.5698	7350.15
70	80	5202.33	1.23092	5101.96	1.24356	520.233
80	90	14863.8	3.51691	14577	3.55303	1486.38
90	100	2123.42	5.89184	1836.64	5.92797	212.342
100	110	1369.09	0.5477	1342.56	0.54834	136.909
110	120	4830.22	...	4830.22	...	483.022
Total		130279	39.77	127460	40.12	13027



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. m	Y Offset, Dtl. Cent. m	Z Offset Dtl. Cent. m	Description
30	Liqd	6031.77	1	...	...	Liquid: 20
30	Insl	206.733	1	...	...	Insulation- Tub
30	Nozl	359.945	1	-1.361	...	N2- Tube outlet
40	Liqd	2037.84	1	...	...	Liquid: 30
40	Nozl	2513.7	1	-1.22529	...	N3- Shell inlet
40	Wght	1520	1	...	1	Internal pipe w
40	Jckt	25007	1	...	...	Jacket
60	Sadl	735.537	0.45	1.1845	...	Left Saddle
60	Sadl	735.537	9.44	1.1845	...	Sdl 2 Fr60
60	Liqd	10077.1	4.945	...	...	Liquid: 50
60	Wght	450	0.45	...	-1	Fireproofing
60	Wght	450	9.44	...	-1	Fireproofing
80	Liqd	2037.84	1	...	...	Liquid: 70
80	Nozl	2440.01	1	1.23878	...	N4- Shell outle
80	Wght	1520	1.00005	...	1	Internal pipe w
80	Jckt	25007	1	...	...	Jacket
90	Liqd	6031.77	1	...	...	Liquid: 80
90	Insl	206.733	1	...	...	Tube side- Ins.
90	Nozl	304.51	1	1.361	...	N1- Tube inlet
30	FTsh	8352.99	2.1775	...	...	tubesheet
30	Tube	65359.8	10	...	...	Tubes
30	RTsh	8352.99	17.8225	...	...	Tubesheet
40	Baf1	141.415	0.8	...	...	Baffles
40	Baf1	141.415	1.34	...	...	Baffles
40	Baf1	141.415	1.88	...	...	Baffles
40	Baf1	141.415	2.42	...	...	Baffles
40	Baf1	141.415	2.96	...	...	Baffles
40	Baf1	141.415	3.5	...	...	Baffles
40	Baf1	141.415	4.04	...	...	Baffles
40	Baf1	141.415	4.58	...	...	Baffles
40	Baf1	141.415	5.12	...	...	Baffles
40	Baf1	141.415	5.66	...	...	Baffles
40	Baf1	141.415	6.2	...	...	Baffles



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Element and Detail Weights: Step: 19 4:31pm Apr 10,2026

40 Baf1	141.415	6.74	...	...	Baffles
40 Baf1	141.415	7.28	...	...	Baffles
40 Baf1	141.415	7.82	...	...	Baffles
40 Baf1	141.415	8.36	...	...	Baffles
40 Baf1	141.415	8.9	...	...	Baffles
40 Baf1	141.415	9.44	...	...	Baffles
40 Baf1	141.415	9.98	...	...	Baffles
40 Baf1	141.415	10.52	...	...	Baffles
40 Baf1	141.415	11.06	...	...	Baffles
40 Baf1	141.415	11.6	...	...	Baffles
40 Baf1	142.052	12.14	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	1471.1
Liquid	26216.4
Insulation	413.5
Nozzles	5618.2
Weights	3940.0
Exchanger Components	85177.6
Jacket, Closure, Insul	50013.9

Sum of the Detail Weights 172850.6 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	143307.1	143307.1	143307.1	143307.1	143307.1	143307.1
Saddles	1471.1	1471.1	1471.1	1471.1	1471.1	1471.1
Nozzles	5618.2	5618.2	5618.2	5618.2	5618.2	5618.2
Exchanger	85177.6	85177.6	85177.6	85177.6	85177.6	85177.6
Jacket	50013.9	50013.9	50013.9	50013.9	50013.9	50013.9
Insulation	...	...	413.5	413.5	413.5	413.5
Empty Weights	...	...	...	3940.0	3940.0	...
Test Weights	...	3940.0	...	...	...	...
Ope Weights	...	...	...	...	...	3940.0
Ope. Liquid	...	...	...	...	...	26216.4
Test Liquid	...	40710.3	...	...	...	...
Tube Tst Lqd	...	11787.7	...	...	...	...
Totals	285587.8	342025.8	286001.3	289941.3	289941.3	316157.7

Field Installation Options:

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	285587.8 kgm
Shop Test Wt.	- Fabricated weight + water (full)	342025.8 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	286001.3 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	289941.3 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	289941.3 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	316157.7 kgm
Oper. Wt. + CA	- Corr wt. + operating liquid	313056.8 kgm

Exchanger Tube Data

Volume of Exchanger tubes :	11.5 m ³
Weight of Ope Liq in tubes :	0.0 kgm



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Element and Detail Weights: Step: 19 4:31pm Apr 10,2026

Weight of Water in tubes : 11787.7 kgm

The Corroded Weight and thickness are used in the Horizontal
Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight includes test liquid in the jacket only for Type 1 jackets specified as
vapor belts.

Surface Areas of Elements:

From	To	Outside Surface Area m ²	Inside Surface Area m ²
10	20	5.23731	...
20	30	2.44359	1.13886
30	40	12.5884	12.312
40	50	14.1337	12.312
50	60	4.94678	4.3092
60	70	69.891	60.8828
70	80	4.94678	4.3092
80	90	14.1337	12.312
90	100	12.5884	12.312
100	110	2.44359	1.13886
110	120	5.23731	...

Total	148.6	121.0 m ²
Total	1599.8	1303.0 ft ²

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Nozzle Flange MAWP:

Step: 20 4:31pm Apr 10,2026

Nozzle Flange MAWP Results: (N/mm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N2- Tube outle	1.82	1.96	82	150	GR 1.1	...	...	...	...
N1- Tube inlet	1.82	1.96	82	150	GR 1.1	...	...	...	...

Shellside Flange Rating**Channelside Flange Rating**Lowest Flange Pressure Rating was (Ope)[TubeSide]: 1.821 N/mm²Lowest Flange Pressure Rating was (Amb)[TubeSide]: 1.960 N/mm²

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange.

See ASME VIII-1, UG-44(b) for more information.

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Wind Load Calculation:

Step: 21 4:31pm Apr 10,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	27 m/sec
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0.3 m
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 m
Distance Upwind of Crest Lh	0 m
Distance from Crest to the Vessel x	0 m
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 m

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$\begin{aligned}
 &= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar}))) \\
 &= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.23 \cdot 0.98)/(1 + 1.7 \cdot 3.4 \cdot 0.23))) \\
 &= \min(0.85, 0.91) \\
 &= 0.850
 \end{aligned}$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.614
Structure Height to Diameter ratio	9.556

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Z_g = 274 \text{ m}$

Effective Height [z]:

$$\begin{aligned}
 &= \text{Centroid Height} + \text{Vessel Base Elevation} \\
 &= 1.55 + 0.3 = 1.85 \text{ m} \\
 &= 6.07 \text{ ft. Imperial Units}
 \end{aligned}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\begin{aligned}
 &\text{Because } z (6.07 \text{ ft.}) < 15 \text{ ft.} \\
 &= 2.01(15 / Z_g)^{2/\alpha} \\
 &= 2.01(15/900)^{2/9.5} \\
 &= 0.85
 \end{aligned}$$

Type of Hill: No Hill



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Wind Load Calculation:

Step: 21 4:31pm Apr 10,2026

Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0 • 0 • 0)²

= 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 0.85 • 1 • 1 • 1 • 60.4²)

= 16 psf [0.77] kPa

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16 • 0.85 • 0.61 • 4.46

= 37.2 lbs. [166] N

Element	Hgt (z) m	K1	K2	K3	Kz	Kzt	qz kPa
Channel COV LEF	1.9	0.000	0.000	0.000	0.849	1.000	0.766
Channel FLG LEF	1.9	0.000	0.000	0.000	0.849	1.000	0.766
Channel - rear shell	1.9	0.000	0.000	0.000	0.849	1.000	0.766
shell cone -rea	1.9	0.000	0.000	0.000	0.849	1.000	0.766
Shell-2	1.9	0.000	0.000	0.000	0.849	1.000	0.766
shell cone- fro	1.9	0.000	0.000	0.000	0.849	1.000	0.766
shell- front	1.9	0.000	0.000	0.000	0.849	1.000	0.766
Channel front	1.9	0.000	0.000	0.000	0.849	1.000	0.766
Channel FLG RIG	1.9	0.000	0.000	0.000	0.849	1.000	0.766
Channel COV RIG	1.9	0.000	0.000	0.000	0.849	1.000	0.766

Wind Loads on Masses/Equipment/Piping

ID	Wind Area m ²	Elevation m	Pressure kPa	Force N
Internal pipe	0.00	1.85	0.77	0.00
Fireproofing	0.00	1.85	0.77	0.00
Fireproofing	0.00	1.85	0.77	0.00
Internal pipe	0.00	1.85	0.77	0.00

Wind Load Calculation:

From	To	Wind Height m	Wind Diameter m	Wind Area m ²	Wind Pressure kPa	Element Wind Load N
10	20	1.85	2.76	0.4139	0.76606	182.128
20	30	1.85	2.352	0.43501	0.76606	191.419
30	40	1.85	2.4048	4.80841	0.76606	2115.85
40	50	1.85	2.97072	5.93997	0.76606	2613.77
50	60	1.85	2.7	1.88953	0.76606	831.454
60	70	1.85	2.7	26.6964	0.76606	11747.3
70	80	1.85	2.7	1.88953	0.76606	831.454
80	90	1.85	2.97072	5.93997	0.76606	2613.77
90	100	1.85	2.4048	4.80841	0.76606	2115.85
100	110	1.85	2.352	0.43501	0.76606	191.419
110	120	1.85	2.76	0.4139	0.76606	182.128



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Wind Load Calculation:

Step: 21 4:31pm Apr 10, 2026

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation: Step: 22 4:31pm Apr 10,2026

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot hnx$ where $Ct = 0.020$, $x = 0.75$ and hn = Structural Height (ft.)
 $= 0.020(8.860.75)$
 $= 0.103$ seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33)$
 $= 0.030$

Check the Value of T which is the smaller of $Cu \cdot Ta$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.1, 0.03) \text{ per 12.8.2}$
 $= 0.030$
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 $= 0.3 \cdot Sds \cdot W \cdot I$
 $= 0.3 \cdot 0.16 \cdot 3055409 \cdot 1.25$
 $= 179658.047$ N

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.16 \cdot 3055409$
 $= 95817.6$ N

Earthquake Load Calculation:

From	To	Earthquake Height m	Earthquake Weight N	Element Ope Load N
10	20	1.15	235031	15795.6



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Earthquake Load Calculation: Step: 22 4:31pm Apr 10, 2026

20	30	0.98	235031	13460.6
30	40	0.98	235031	13460.6
40	50	0.98	235031	13460.6
50	60	0.98	235031	13460.6
60	Sad1	0.98	235031	13460.6
Sad1	70	0.98	235031	13460.6
60	70	0.98	235031	13460.6
70	80	0.98	235031	13460.6
80	90	0.98	235031	13460.6
90	100	0.98	235031	13460.6
100	110	0.98	235031	13460.6
110	120	1.15	235031	15795.6

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 23 4:31pm Apr 10,2026

Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	10.338 m
Center of Gravity of the Liquid	10.338 m
Center of Gravity of the Insulation	10.338 m
Center of Gravity of the Nozzles	10.162 m
Center of Gravity of the Added Weights (Operating)	10.338 m
Center of Gravity of the Added Weights (Empty)	10.338 m
Center of Gravity of the Tubesheet(s)	10.338 m
Center of Gravity of the Tubes	10.338 m
Center of Gravity of the Jacket/Vapor Belt	10.338 m
Center of Gravity of the Baffles	9.164 m
Center of Gravity of Bare Shell New and Cold	10.338 m
Center of Gravity of Bare Shell Corroded	10.338 m
Vessel CG in the Operating Condition	10.325 m
Vessel CG in the Fabricated (Shop/Empty) Condition	10.322 m
Vessel CG in the Test Condition	10.325 m

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Saddle Calcs: Operating Case: Step: 24 4:31pm Apr 10,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (254.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

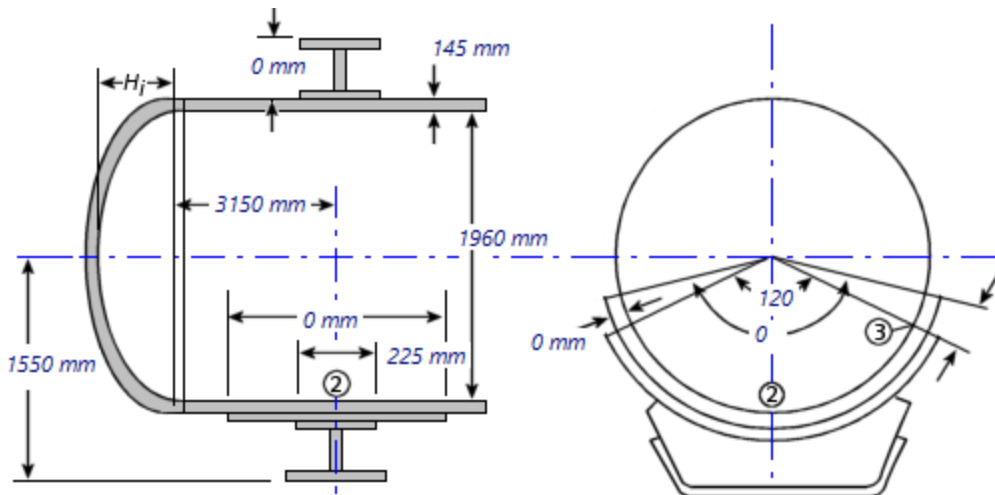
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(225 + 1.56 \cdot \sqrt{1054 \cdot 142}, 2 \cdot 3150)$$

$$= 828.5164 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1054.00	mm
Shell Thickness used in this Case	t	145.000	mm
Stiffened Vessel Length per 4.15.6	L	15.29	m
Distance from Saddle to Vessel tangent	a1 or a	3150.00	mm
Saddle Width	b1 or b	225.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		137989.50	kPa
Head Allowable Stress used in Calculation		137989.50	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1550.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	2051657.62	N
Pressure used in Analysis	P	18.006 N/mm ²

Horizontal Vessel Analysis Results:	Actual kPa	Allowable kPa
Long. Stress at Top of Midspan	63888.47	137989.50
Long. Stress at Bottom of Midspan	69757.53	137989.50
Long. Stress at Top of Saddles	90612.73	137989.50
Long. Stress at Bottom of Saddles	53637.27	137989.50



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Saddle Calcs: Operating Case: Step: 24 4:31pm Apr 10, 2026

Tangential Shear in Shell	9435.90	110391.59
Circ. Stress at Horn of Saddle	12426.40	172486.86
Circ. Compressive Stress in Shell	1325.83	137989.50

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (23617/2 + 0) \cdot 1550/1850$$

$$= 29680.2 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(4825, 0) \cdot 1550/8990$$

$$= 831.8 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(179658, 0) \cdot 1550/8990$$

$$= 30975.5 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (179658/2 + 0) \cdot 1550/1850$$

$$= 225786.5 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 1825871 + \max(832, 29680, 30976, 225786)$$

$$= 2051657.6 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(2.25 \cdot 1.2)^2)$$

$$= 4824.890 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	2058870.38	N
Transverse Shear Load	89829.02	N
Longitudinal Shear Load	179658.05	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 1850.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -2051657 \cdot 3.15 [1 - (1 - 3.15/15.3 + (1.05^2 - 0^2)/(2 \cdot 3.15 \cdot 15.3))/(1 + (4 \cdot 0)/(3 \cdot 15.3))]$$

$$= -1257406.5 \text{ N-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 2051657 \cdot 15.3/4((1 + 2(1.05^2 - 0^2)/(15.3^2))/(1 + (4 \cdot 0)/$$



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Saddle Calcs: Operating Case: Step: 24 4:31pm Apr 10, 2026

$$(3 \cdot 15.3) - 4 \cdot 3.15 / 15.3 \\ = 1454863.6 \text{ N-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 18 \cdot 1054 / (2 \cdot 142) - 1454863 / (\pi \cdot 1054^2 \cdot 142) \\ = 63888.47 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 18 \cdot 1054 / (2 \cdot 142) + 1454863 / (\pi \cdot 1054^2 \cdot 142) \\ = 69757.53 \text{ kPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 18 \cdot 1054 / (2 \cdot 142) - 1257406 / (0.11 \cdot \pi \cdot 1054^2 \cdot 142) \\ = 90612.73 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 18 \cdot 1054 / (2 \cdot 142) + 1257406 / (0.19 \cdot \pi \cdot 1054^2 \cdot 142) \\ = 53637.27 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3)) \\ = 2051657 (15.3 - 2 \cdot 3.15) / (15.3 + (4 \cdot 0 / 3)) \\ = 1206305.1 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t) \\ = 1.17 \cdot 1206305 / (1054 \cdot 142) \\ = 9435.90 \text{ kPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t} \\ = 0.78 \cdot \sqrt{1054 \cdot 142} \\ = 301.758 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2)) \\ = -0.76 \cdot 2051657 \cdot 0.1 / (142(225 + 302 + 302)) \\ = -1325.83 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2) \\ = -2051657 / (4 \cdot 142(225 + 302 + 302)) - \\ 3 \cdot 0.053 \cdot 2051657 / (2 \cdot 142^2) \\ = -12426.40 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a) \\ = \min(225 + 1.56 \cdot \sqrt{1054 \cdot 142}, 2 \cdot 3.15) \\ = 828.52 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 8990.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature}) \\ = 0.12426 \text{E-}04 \cdot 8990 (150 - 21.1) \\ = 14.398 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1850.0000	mm
Baseplate Thickness	Bpthk	50.0000	mm



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Saddle Calcs: Operating Case: Step: 24 4:31pm Apr 10,2026

Baseplate Width	Bpwid	460.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	261980.0	kPa
Height of Web at Center	Hw,c	627.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(983 \cdot 142)} + 254)142^3)/12$$

$$= 199674768.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 199674768 + 0.12(154 - 71)^2$$

$$= 1022601216.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ² mm ⁴
Shell	836.8	142.0	71.0	0.1	8436969.0	0.102E+10
Wearplate	254.0	16.0	150.0	0.0	609599.9	0.159E+06
Web	15.0	359.0	337.5	0.0	1817438.5	0.239E+09
BasePlate	460.0	50.0	542.0	0.0	12466004.0	0.346E+10
Totals	...	...	...	0.2	23330012.0	0.472E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 23330012/0.15$$

$$= 154.218 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120)\cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 2051657$$

$$= 417556.8438 \text{ N}$$

Tension Stress, St = (Fh/As)	=	12868.4424	kPa
Allowed Stress, Sa = 0.6 * Yield Str	=	157188.0312	kPa

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1550 - 983 \cdot \sin(120/2) / 1.05$$

$$= 737.066 \text{ mm}$$

Bending Moment, M = Fh * d	=	307891.5312	N-m
----------------------------	---	-------------	-----

Bending Stress, Sb = (M * C1 / I)	=	10045.5723	kPa
Allowed Stress, Sa = 2/3 * Yield Str	=	174653.3594	kPa

Minimum Thickness of Baseplate per Moss:



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$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(2051657 + 7213) \cdot 460}{4 \cdot 1850 \cdot 174653}}$$

$$= 46.887 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (980 + 145 + 16)$$

$$= 1976.270 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1976 / (4 - 1)$$

$$= 658.757 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 659 \cdot 460 / 2$$

$$= 0.151 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 2058870 / (1850 \cdot 460)$$

$$= 2419954.000 \text{ N/m}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 0.15 \cdot 2419954$$

$$= 366566.062 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 0.0031 + 0.0049$$

$$= 0.008 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 366566 / 0.0081$$

$$= 45308.453 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	15.0	225.0	...	...	...	0.142E+08

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 225 - 15$$

$$= 210.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 0.0081$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 225 / 2$$

$$= 112.500 \text{ mm}$$

Radius of Gyration [r]:



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Saddle Calcs: Operating Case: Step: 24 4:31pm Apr 10, 2026

$$= \sqrt{\frac{\text{Total Inertia}}{\text{Total Area}}}$$

$$= \sqrt{\frac{14238268}{0.0081}}$$

$$= 41.950 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1550 - \cos(120/2) (980 + 145 + 16) - 50$$

$$= 929.500 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120/2) (980 + 145 + 16)$$

$$= 1976.270 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 1850 - 50.8$$

$$= 1799.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (1976 - 1799) / 2$$

$$= 88.535 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{930^2 + 88.5^2}$$

$$= 933.707 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot \text{Elastic Modulus}}{\text{Yield Stress}}}$$

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot 199955}{261980}}$$

$$= 122.739$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 934 / 42$$

$$= 22.257$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 179658 / (2 \cdot 1850) \cdot 659 \cdot 934 / 2$$

$$= 14939.174 \text{ N-m}$$

Compressive Allowable, $KL/r < Cc$ ($22.3 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (22.3)^2 / (2 \cdot 123^2)) 261980 / (5/3 + 3 \cdot (22.3) / (8 \cdot 123) - (22.3)^3 / (8 \cdot 123^3))$$

$$= 148607 \text{ kPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 45308 / 148607 + (14939 \cdot 112 / 14238282) / 174653$$

$$= 0.980$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	210.0	0.0	0.0	0.0	0.142E+08
Web	658.8	15.0	0.0	0.0	0.0	0.185E+06
Totals	...	...	...	...	...	0.144E+08

Distance to Centroid from Datum [ytot]:



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= AY / A
= 0/0.013
= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
= 225/2
= 112.500 mm

Length of Inner Rib [L]:

= Saddle Height - $\sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2}$ - Bpthk
= 1550 - $\sqrt{(1141 + 16)^2 - (659/2)^2}$ - 50
= 407.576 mm

Radius of Gyration [r]:

= $\sqrt{(Total\ Inertia / Total\ Area)}$
= $\sqrt{(14419324/0.013)}$
= 33.264 mm

Slenderness ratio [KL/r]:

= KL/r
= 1 • 408/33.3
= 12.253

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)
= 179658/(2 • 1850)
= 48.556 N/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L
= 48.6 • 659 • 408
= 13042.305 N-m

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus
= 13042/128172
= 101717.859 kPa

Compressive Allowable, $KL/r < C_c$ (12.3 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
= $(1 - (12.3)^2 / (2 \cdot 123^2)) 261980 /$
 $(5/3 + 3 \cdot (12.3) / (8 \cdot 123) - (12.3)^3 / (8 \cdot 123^3))$
= 152980 kPa

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$
= 45325/152980 + $(13042 \cdot 112/14419324) / 174653$
= 0.879

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	283983.50 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0012 m²
BasePlate Allowable Stress	S	137989.20 kPa
Area Available in a Single Bolt	BltArea	0.0010 m²
Saddle Load Q0 (Weight)	Q0	1833083.9 N
Saddle Load QL (Wind/Seismic contribution)	QL	30975.5 N
Maximum Transverse Force	Ft	89829.0 N



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Saddle Calcs: Operating Case: Step: 24 4:31pm Apr 10, 2026

Maximum Longitudinal Force	F1	179658.0 N
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:
= 729201.4 N

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
= $\max(F_f, F_1) / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $729201 / (0.00098 \cdot 6)$
= 124080.6 kPa. Must be less than 170390.1 kPa.

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $89829 / (0.00098 \cdot 6)$
= 15285.3 kPa. Must be less than 170390.1 kPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
= $179658 / (170390 \cdot 6)$
= 0.0002 m²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1550 \cdot 89829 + 0$
= 139291.44 N-m

Eccentricity (e):
= R_{mom} / QO
= $139291 / 1833083$
= 75.96 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0, 0.00018, 0]$
= 0.0002 m²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 16, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = -1$, Temp. Reduction = 0 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
Wear Pad Width (254.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
= $\min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$



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$$= \min(225 + 1.56\sqrt{(1054 \cdot 142)}, 2 \cdot 3150)$$

$$= 828.5164 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1054.00	mm
Shell Thickness used in this Case	t	145.000	mm
Stiffened Vessel Length per 4.15.6	L	15.29	m
Distance from Saddle to Vessel tangent	a1 or a	3150.00	mm
Saddle Width	b1 or b	225.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		137989.50	kPa
Head Allowable Stress used in Calculation		137989.50	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1550.00	mm
Coefficient of Friction	mu	0.40	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		2041577.38	N
Pressure used in Analysis	P	18.006	N/mm ²

Horizontal Vessel Analysis Results:	Actual kPa	Allowable kPa
Long. Stress at Top of Midspan	63902.89	137989.50
Long. Stress at Bottom of Midspan	69743.11	137989.50
Long. Stress at Top of Saddles	90495.84	137989.50
Long. Stress at Bottom of Saddles	53702.05	137989.50
Tangential Shear in Shell	9389.54	110391.59
Circ. Stress at Horn of Saddle	12365.35	172486.86
Circ. Compressive Stress in Shell	1319.32	137989.50

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (23617/2 + 0) \cdot 1550/1850$$

$$= 29680.2 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l,\text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(4825, 0) \cdot 1550/8990$$

$$= 831.8 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(F_{l,\text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(179658, 0) \cdot 1550/8990$$

$$= 30975.5 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (179658/2 + 0) \cdot 1550/1850$$

$$= 225786.5 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 1815790 + \max(832, 29680, 30976, 225786)$$

$$= 2041577.4 \text{ N}$$



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Longitudinal Wind Force [F_L]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (2.25 \cdot 1.2)^2)$$

$$= 4824.890 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 2048790.00 N

Transverse Shear Load 89829.02 N

Longitudinal Shear Load 179658.05 N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= Baseplate Length

= 1850.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -2041577 \cdot 3.15 [1 - (1 - 3.15/15.3 + (1.05^2 - 0^2) /$$

$$(2 \cdot 3.15 \cdot 15.3) / (1 + (4 \cdot 0) / (3 \cdot 15.3))]$$

$$= -1251228.8 \text{ N-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 2041577 \cdot 15.3 / 4 ((1 + 2(1.05^2 - 0^2)/(15.3^2)) / (1 + (4 \cdot 0) /$$

$$(3 \cdot 15.3) - 4 \cdot 3.15 / 15.3)$$

$$= 1447715.6 \text{ N-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ₁]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 18 \cdot 1054 / (2 \cdot 142) - 1447715 / (\pi \cdot 1054^2 \cdot 142)$$

$$= 63902.89 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ₂]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 18 \cdot 1054 / (2 \cdot 142) + 1447715 / (\pi \cdot 1054^2 \cdot 142)$$

$$= 69743.11 \text{ kPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*₃]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= 18 \cdot 1054 / (2 \cdot 142) - 1251228 / (0.11 \cdot \pi \cdot 1054^2 \cdot 142)$$

$$= 90495.84 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*₄]:

$$= P \cdot Rm / (2t) + M1 / (K1* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 18 \cdot 1054 / (2 \cdot 142) + 1251228 / (0.19 \cdot \pi \cdot 1054^2 \cdot 142)$$

$$= 53702.05 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 2041577 (15.3 - 2 \cdot 3.15) / (15.3 + (4 \cdot 0 / 3))$$

$$= 1200378.2 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ₂]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 1200378 / (1054 \cdot 142)$$

$$= 9389.54 \text{ kPa}$$



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Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1054 \cdot 142)}$$

$$= 301.758 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 2041577 \cdot 0.1 / (142(225 + 302 + 302))$$

$$= -1319.32 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -2041577 / (4 \cdot 142(225 + 302 + 302)) -$$

$$3 \cdot 0.053 \cdot 2041577 / (2 \cdot 142^2)$$

$$= -12365.35 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(225 + 1.56 \cdot \sqrt{(1054 \cdot 142)}, 2 \cdot 3.15)$$

$$= 828.52 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1850.0000	mm
Baseplate Thickness	Bpthk	50.0000	mm
Baseplate Width	Bpwid	460.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	261980.0	kPa
Height of Web at Center	Hw,c	627.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(983 \cdot 142)} + 254)142^3) / 12$$

$$= 199674768.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 199674768 + 0.12(154 - 71)^2$$

$$= 1022601216.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	836.8	142.0	71.0	0.1	8436969.0	0.102E+10
Wearplate	254.0	16.0	150.0	0.0	609599.9	0.159E+06
Web	15.0	359.0	337.5	0.0	1817438.5	0.239E+09
BasePlate	460.0	50.0	542.0	0.0	12466004.0	0.346E+10
Totals	...	...	...	0.2	23330012.0	0.472E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 23330012 / 0.15$$

$$= 154.218 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$



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$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 2041577$$

$$= 415505.2500 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 12805.2168 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157188.0312 \text{ kPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1550 - 983 \cdot \sin(120/2) / 1.05$$

$$= 737.066 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 306378.7812 \text{ N-m}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 9996.2158 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174653.3594 \text{ kPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(2041577 + 7213)460 / (4 \cdot 1850 \cdot 174653)}$$

$$= 46.772 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (980 + 145 + 16)$$

$$= 1976.270 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 1976 / (4 - 1)$$

$$= 658.757 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 659 \cdot 460 / 2$$

$$= 0.151 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 2048790 / (1850 \cdot 460)$$

$$= 2408106.000 \text{ N/m}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 0.15 \cdot 2408106$$

$$= 364771.375 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 0.0031 + 0.0049$$

$$= 0.008 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 364771 / 0.0081$$



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= 45086.625 kPa

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	15.0	225.0	...	...	...	0.142E+08

Rib dimension [D]:

= Saddle Width - Web Thickness

= 225 - 15

= 210.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/0.0081

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 225/2

= 112.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{14238268/0.0081}$

= 41.950 mm

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk

= 1550 - cos(120/2)(980 + 145 + 16) - 50

= 929.500 mm

Length of Outer Rib, step 2, [Lw]:

= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)

= 2 • sin(120/2)(980 + 145 + 16)

= 1976.270 mm

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance

= 1850 - 50.8

= 1799.200 mm

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2

= (1976 - 1799)/2

= 88.535 mm

Length of Outer Rib [L]:

= $\sqrt{Lv^2 + Dd^2}$

= $\sqrt{930^2 + 88.5^2}$

= 933.707 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 199955/261980}$

= 122.739

Slenderness ratio [KL/r]:

= KL/r

= 1 • 934/42

= 22.257



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Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 179658 / (2 \cdot 1850) \cdot 659 \cdot 934 / 2$$

$$= 14939.174 \text{ N-m}$$

Compressive Allowable, $KL/r < C_c$ (22.3 < 123) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (22.3)^2 / (2 \cdot 123^2)) 261980 / (5/3 + 3 \cdot (22.3) / (8 \cdot 123) - (22.3^3) / (8 \cdot 123^3))$$

$$= 148607 \text{ kPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 45087 / 148607 + (14939 \cdot 112 / 14238282) / 174653$$

$$= 0.979$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	210.0	0.0	0.0	0.0	0.142E+08
Web	658.8	15.0	0.0	0.0	0.0	0.185E+06
Totals	...	...	...	...	...	0.144E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 0.013$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 225 / 2$$

$$= 112.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 1550 - \sqrt{ (1141 + 16)^2 - (659/2)^2 } - 50$$

$$= 407.576 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 14419324 / 0.013 }$$

$$= 33.264 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 408 / 33.3$$

$$= 12.253$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 179658 / (2 \cdot 1850)$$

$$= 48.556 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 48.6 \cdot 659 \cdot 408$$

$$= 13042.305 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 13042 / 128172$$

$$= 101717.859 \text{ kPa}$$



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Saddle Calcs: Operating Case: Step: 24 4:31pm Apr 10, 2026

Compressive Allowable, $KL/r < C_c$ ($12.3 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r)^4 / (8 \cdot C_c^4) - (KL/r)^6 / (8 \cdot C_c^6))$$

$$= (1 - (12.3)^2 / (2 \cdot 123^2)) 261980 / (5/3 + 3 \cdot (12.3)^4 / (8 \cdot 123^4) - (12.3)^6 / (8 \cdot 123^6))$$

$$= 152980 \text{ kPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 45103 / 152980 + (13042 \cdot 112 / 14419324) / 174653$$

$$= 0.877$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	283983.50 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0012 m²
BasePlate Allowable Stress	S	137989.20 kPa
Area Available in a Single Bolt	BltArea	0.0010 m²
Saddle Load Q0 (Weight)	Q0	1823003.6 N
Saddle Load QL (Wind/Seismic contribution)	QL	30975.5 N
Maximum Transverse Force	Ft	89829.0 N
Maximum Longitudinal Force	F1	179658.0 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 179658 / (0.00098 \cdot 6)$$

$$= 30570.5 \text{ kPa. Must be less than } 170390.1 \text{ kPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 89829 / (0.00098 \cdot 6)$$

$$= 15285.3 \text{ kPa. Must be less than } 170390.1 \text{ kPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 179658 / (170390 \cdot 6)$$

$$= 0.0002 \text{ m}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1550 \cdot 89829 + 0$$

$$= 139291.44 \text{ N-m}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 139291 / 1823003$$

$$= 76.38 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:



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= max[Bltarearl, Bltarears, Bltareart]
= max[0, 0.00018, 0]
= 0.0002 m²

MDMT Calculations for: Sdl 2 Fr60, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 16, c = 3 mm, E* = 1

Thickness Ratio = tr * E*/(tg - c) = -1, Temp. Reduction = 0 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Test Case:

Step: 25 4:31pm Apr 10, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (254.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

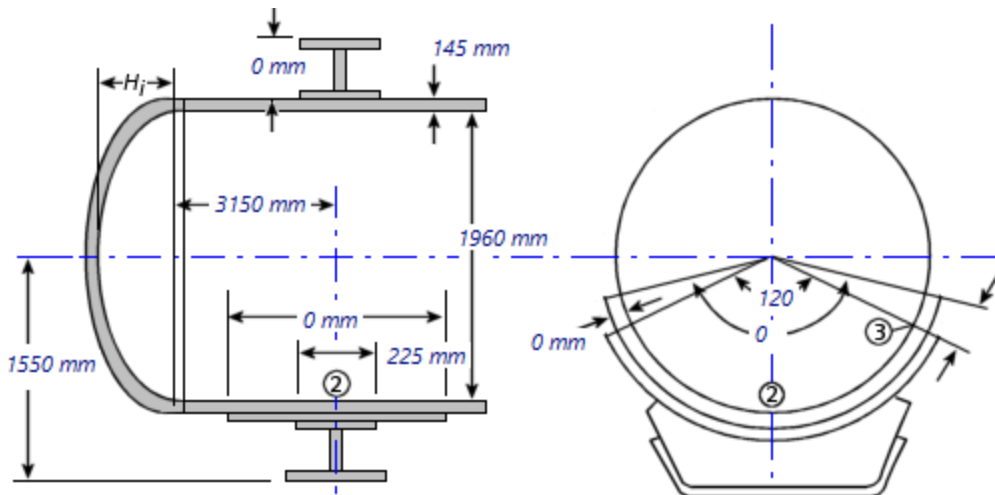
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(225 + 1.56 \cdot \sqrt{1052 \cdot 145}, 2 \cdot 3150)$$

$$= 834.4241 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1052.50	mm
Shell Thickness used in this Case	t	145.000	mm
Stiffened Vessel Length per 4.15.6	L	15.29	m
Distance from Saddle to Vessel tangent	a1 or a	3150.00	mm
Saddle Width	b1 or b	225.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		233982.17	kPa
Head Allowable Stress used in Calculation		233982.17	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1550.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	1724863.75	N
Pressure used in Analysis	P	23.412 N/mm ²

Horizontal Vessel Analysis Results:	Actual kPa	Allowable kPa
Long. Stress at Top of Midspan	82540.70	233982.17
Long. Stress at Bottom of Midspan	87385.91	233982.17
Long. Stress at Top of Saddles	104609.12	233982.17
Long. Stress at Bottom of Saddles	74074.38	233982.17



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Saddle Calcs: Test Case:

Step: 25

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Tangential Shear in Shell	7779.87	138569.45
Circ. Stress at Horn of Saddle	10068.12	350973.22
Circ. Compressive Stress in Shell	1083.86	233982.17

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (11808/2 + 0) \cdot 1550/1850$$

$$= 14840.1 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(2412, 0) \cdot 1550/8990$$

$$= 415.9 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 1710023 + \max(416, 14840, 0, 0)$$

$$= 1724863.8 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(2.25 \cdot 1.2)^2)$$

$$= 4824.890 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 1732076.50 N

Transverse Shear Load 5904.13 N

Longitudinal Shear Load 2412.32 N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 1850.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -1724863 \cdot 3.15 [1 - (1 - 3.15/15.3 + (1.05^2 - 0^2)/(2 \cdot 3.15 \cdot 15.3))/(1 + (4 \cdot 0)/(3 \cdot 15.3))]$$

$$= -1057301.8 \text{ N-m}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 1724863 \cdot 15.3/4 ((1 + 2(1.05^2 - 0^2)/(15.3^2))/(1 + (4 \cdot 0)/(3 \cdot 15.3)) - 4 \cdot 3.15/15.3)$$

$$= 1222951.0 \text{ N-m}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$

$$= 23.4 \cdot 1052/(2 \cdot 145) - 1222951/(\pi \cdot 1052^2 \cdot 145)$$

$$= 82540.70 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$



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Saddle Calcs: Test Case: Step: 25 4:31pm Apr 10, 2026

$$= 23.4 \cdot 1052 / (2 \cdot 145) + 1222951 / (\pi \cdot 1052^2 \cdot 145)$$

$$= 87385.91 \text{ kPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 23.4 \cdot 1052 / (2 \cdot 145) - 1057301 / (0.11 \cdot \pi \cdot 1052^2 \cdot 145)$$

$$= 104609.12 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 23.4 \cdot 1052 / (2 \cdot 145) + 1057301 / (0.19 \cdot \pi \cdot 1052^2 \cdot 145)$$

$$= 74074.38 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 1724863 (15.3 - 2 \cdot 3.15) / (15.3 + (4 \cdot 0 / 3))$$

$$= 1014161.4 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 1014161 / (1052 \cdot 145)$$

$$= 7779.87 \text{ kPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1052 \cdot 145)}$$

$$= 304.712 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 1724863 \cdot 0.1 / (145 (225 + 305 + 305))$$

$$= -1083.86 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -1724863 / (4 \cdot 145 (225 + 305 + 305)) - 3 \cdot 0.053 \cdot 1724863 / (2 \cdot 145^2)$$

$$= -10068.12 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(225 + 1.56 \cdot \sqrt{(1052 \cdot 145)}, 2 \cdot 3.15)$$

$$= 834.42 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1850.0000	mm
Baseplate Thickness	Bpthk	50.0000	mm
Baseplate Width	Bpwid	460.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	259980.2	kPa
Height of Web at Center	Hw,c	627.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(980 \cdot 145)} + 254)^3) / 12$$

$$= 213926912.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$



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Saddle Calcs: Test Case:

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$$= 213926912 + 0.12 (154 - 72.5)^2$$

$$= 1030379456.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ² (2) mm ⁴
Shell	842.1	145.0	72.5	0.1	8852155.0	0.103E+10
Wearplate	254.0	16.0	153.0	0.0	621791.9	0.933E+05
Web	15.0	359.0	340.5	0.0	1833593.5	0.245E+09
BasePlate	460.0	50.0	545.0	0.0	12535004.0	0.352E+10
Totals	...	...	...	0.2	23842544.0	0.479E+10

Distance to Centroid [C1]:

$$= AY / A$$

$$= 23842544 / 0.15$$

$$= 154.273 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 1724863$$

$$= 351047.1875 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 10818.7207 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155988.1250 \text{ kPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1550 - 980 \cdot \sin(120/2) / 1.05$$

$$= 739.547 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 259721.0000 \text{ N-m}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 8359.6367 \text{ kPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173320.1406 \text{ kPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(1724863 + 7213)460) / (4 \cdot 1850 \cdot 173320)}$$

$$= 43.171 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (980 + 145 + 16)$$

$$= 1976.270 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 1976 / (4 - 1)$$

$$= 658.757 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot B_{\text{pwid}} / 2$$

$$= 659 \cdot 460 / 2$$



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Saddle Calcs: Test Case: Step: 25 4:31pm Apr 10, 2026

$$= 0.151 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1732076 / (1850 \cdot 460)$$

$$= 2035847.125 \text{ N/m}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 0.15 \cdot 2035847$$

$$= 308382.938 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 0.0031 + 0.0049$$

$$= 0.008 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 308383 / 0.0081$$

$$= 38116.875 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	15.0	225.0	...	...	...	0.142E+08

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 225 - 15$$

$$= 210.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 0.0081$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 225 / 2$$

$$= 112.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(14238268 / 0.0081)}$$

$$= 41.950 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1550 - \cos(120/2) (980 + 145 + 16) - 50$$

$$= 929.500 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120/2) (980 + 145 + 16)$$

$$= 1976.270 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 1850 - 50.8$$

$$= 1799.200 \text{ mm}$$



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Saddle Calcs: Test Case: Step: 25 4:31pm Apr 10, 2026

Length of Outer Rib, step 4, [Dd]:

$$= (L_w - B_d) / 2$$

$$= (1976 - 1799) / 2$$

$$= 88.535 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{(L_v^2 + D_d^2)}$$

$$= \sqrt{(930^2 + 88.5^2)}$$

$$= 933.707 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955 / 259980)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 934 / 42$$

$$= 22.257$$

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 2412 / (2 \cdot 1850) \cdot 659 \cdot 934 / 2$$

$$= 200.593 \text{ N-m}$$

Compressive Allowable, $KL/r < C_c$ ($22.3 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (22.3)^2 / (2 \cdot 123^2)) 259980 / (5/3 + 3 \cdot (22.3) / (8 \cdot 123) - (22.3^3) / (8 \cdot 123^3))$$

$$= 147513 \text{ kPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 38117 / 147513 + (201 \cdot 112 / 14238282) / 173320$$

$$= 0.268$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	210.0	0.0	0.0	0.0	0.142E+08
Web	658.8	15.0	0.0	0.0	0.0	0.185E+06
Totals	...	...	...	...	...	0.144E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 0.013$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 225 / 2$$

$$= 112.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(R_o + W_{pdthk})^2 - (Pitch/2)^2} - B_{pthk}$$

$$= 1550 - \sqrt{(1141 + 16)^2 - (659/2)^2} - 50$$

$$= 407.576 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(14419324 / 0.013)}$$

$$= 33.264 \text{ mm}$$



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Saddle Calcs: Test Case:

Step: 25 4:31pm Apr 10, 2026

Slenderness ratio $[KL/r]$:

$$= KL/r$$

$$= 1 \cdot 408/33.3$$

$$= 12.253$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 2412 / (2 \cdot 1850)$$

$$= 0.652 \text{ N/mm}$$

Moment at base of inner Rib $[M_{base,c}]$:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.65 \cdot 659 \cdot 408$$

$$= 175.123 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load $[\sigma_B, base, c]$:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 175/128172$$

$$= 1365.795 \text{ kPa}$$

Compressive Allowable, $KL/r < C_c$ ($12.3 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.3)^2 / (2 \cdot 123^2)) 259980 / (5/3 + 3 \cdot (12.3) / (8 \cdot 123) - (12.3)^3 / (8 \cdot 123^3))$$

$$= 151831 \text{ kPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 38131/151831 + (175 \cdot 112/14419324) / 173320$$

$$= 0.259$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	171986.91 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0012 m ²
BasePlate Allowable Stress	S	137989.50 kPa
Area Available in a Single Bolt	BltArea	0.0010 m ²
Saddle Load Q0 (Weight)	Q0	1717236.4 N
Saddle Load QL (Wind/Seismic contribution)	QL	415.9 N
Maximum Transverse Force	Ft	5904.1 N
Maximum Longitudinal Force	F1	2412.3 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction $[\tau_b, l]$:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2412 / (0.00098 \cdot 6)$$

$$= 410.5 \text{ kPa. Must be less than } 103192.1 \text{ kPa.}$$

Shear Stress in a Single Bolt, Transverse Direction $[\tau_b, t]$:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 5904 / (0.00098 \cdot 6)$$

$$= 1004.6 \text{ kPa. Must be less than } 103192.1 \text{ kPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load $[Bltarearl]$:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load $[Bltarears]$:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$



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Saddle Calcs: Test Case: Step: 25 4:31pm Apr 10, 2026

$$= 2412 / (103192 \cdot 6)$$

$$= 0.0000 \text{ m}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1550 \cdot 5904 + 0$$

$$= 9155.11 \text{ N-m}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 9155 / 1717236$$

$$= 5.33 \text{ mm} < Bplen/6 \rightarrow \text{No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.38953E-05, 0]$$

$$= 0.0000 \text{ m}^2$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (254.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$ and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$$

$$= \min(225 + 1.56 \cdot \sqrt{1052 \cdot 145}, 2 \cdot 3150)$$

$$= 834.4241 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1052.50	mm
Shell Thickness used in this Case	t	145.000	mm
Stiffened Vessel Length per 4.15.6	L	15.29	m
Distance from Saddle to Vessel tangent	a1 or a	3150.00	mm
Saddle Width	b1 or b	225.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		233982.17	kPa
Head Allowable Stress used in Calculation		233982.17	kPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1550.00	mm
Coefficient of Friction	mu	0.40	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		1681568.25	N
Pressure used in Analysis	P	23.412	N/mm ²

Horizontal Vessel Analysis Results:		Actual kPa	Allowable kPa
Long. Stress at Top	of Midspan	82601.51	233982.17
Long. Stress at Bottom	of Midspan	87325.10	233982.17
Long. Stress at Top	of Saddles	104116.00	233982.17
Long. Stress at Bottom	of Saddles	74347.70	233982.17



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Saddle Calcs: Test Case:

Step: 25

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Tangential Shear in Shell	7584.59	138569.45
Circ. Stress at Horn of Saddle	9815.40	350973.22
Circ. Compressive Stress in Shell	1056.65	233982.17

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= Ftr(Ft/Num of Saddles + Z Force Load) • B / E

= 3 (11808/2 + 0) • 1550/1850

= 14840.1 N

Longitudinal Saddle Reaction Force [Fwl]:

= max(Fl,wind; Sum of X Forces) • B / Ls

= max(2412, 0) • 1550/8990

= 415.9 N

Load Combination Results for Q + Wind or Seismic [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= 1666728 + max(416, 14840, 0, 0)

= 1681568.2 N

Longitudinal Wind Force [Fl]:

= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))

= 1.1 • 766 (0 + $\pi/4$ (2.25 • 1.2)²)

= 4824.890 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 1688780.88 N

Transverse Shear Load 5904.13 N

Longitudinal Shear Load 2412.32 N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= Baseplate Length

= 1850.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L)) / (1 + (4hm)/3L)]

= -1681568•3.15 [1 - (1 - 3.15/15.3 + (1.05²-0²)/

(2•3.15•15.3) / (1 + (4•0)/(3•15.3))]

= -1030762.6 N-m

Moment per Equation 4.15.2 [M2]:

= Q•L/4 ((1 + 2(Rm²-hm²)/(L²)) / (1 + (4hm²)/(3L)) - 4a/L)

= 1681568•15.3/4 ((1 + 2(1.05²-0²)/(15.3²)) / (1 + (4•0)/

(3•15.3)) - 4•3.15/15.3)

= 1192254.0 N-m

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)

= 23.4 • 1052/(2•145) - 1192254/(π•1052²•145)

= 82601.51 kPa

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)



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Saddle Calcs: Test Case: Step: 25 4:31pm Apr 10, 2026

$$= 23.4 \cdot 1052 / (2 \cdot 145) + 1192254 / (\pi \cdot 1052^2 \cdot 145)$$

$$= 87325.10 \text{ kPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 23.4 \cdot 1052 / (2 \cdot 145) - 1030762 / (0.11 \cdot \pi \cdot 1052^2 \cdot 145)$$

$$= 104116.00 \text{ kPa}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 23.4 \cdot 1052 / (2 \cdot 145) + 1030762 / (0.19 \cdot \pi \cdot 1052^2 \cdot 145)$$

$$= 74347.70 \text{ kPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 1681568 (15.3 - 2 \cdot 3.15) / (15.3 + (4 \cdot 0 / 3))$$

$$= 988705.1 \text{ N}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 988705 / (1052 \cdot 145)$$

$$= 7584.59 \text{ kPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1052 \cdot 145)}$$

$$= 304.712 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 1681568 \cdot 0.1 / (145 (225 + 305 + 305))$$

$$= -1056.65 \text{ kPa}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -1681568 / (4 \cdot 145 (225 + 305 + 305)) - 3 \cdot 0.053 \cdot 1681568 / (2 \cdot 145^2)$$

$$= -9815.40 \text{ kPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(225 + 1.56 \cdot \sqrt{(1052 \cdot 145)}, 2 \cdot 3.15)$$

$$= 834.42 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1850.0000	mm
Baseplate Thickness	Bpthk	50.0000	mm
Baseplate Width	Bpwid	460.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	15.0000	mm
Web Thickness	Webtk	15.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	259980.2	kPa
Height of Web at Center	Hw, c	627.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(980 \cdot 145)} + 254)^3) / 12$$

$$= 213926912.000 \text{ mm}^4$$

Resolved Inertia for the Shell [I, shell]:



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Saddle Calcs: Test Case:

Step: 25 4:31pm Apr 10, 2026

$$\begin{aligned}
 &= \text{ShellInertia} + A\text{Shell}(C1 - Y)^2 \\
 &= 213926912 + 0.12 (154 - 72.5)^2 \\
 &= 1030379456.000 \text{ mm}^4
 \end{aligned}$$

	B mm	D mm	Y mm	A m ²	AY mm ³	I + AD ² mm ⁴
Shell	842.1	145.0	72.5	0.1	8852155.0	0.103E+10
Wearplate	254.0	16.0	153.0	0.0	621791.9	0.933E+05
Web	15.0	359.0	340.5	0.0	1833593.5	0.245E+09
BasePlate	460.0	50.0	545.0	0.0	12535004.0	0.352E+10
Totals	...	...	...	0.2	23842544.0	0.479E+10

Distance to Centroid [C1]:

$$\begin{aligned}
 &= AY / A \\
 &= 23842544 / 0.15 \\
 &= 154.273 \text{ mm}
 \end{aligned}$$

Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle} / 2 \\
 &= 180 - 120 / 2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.2 \cdot 1681568 \\
 &= 342235.5938 \text{ N}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 10547.1611 \text{ kPa} \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155988.1250 \text{ kPa}
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians}) \\
 &= 1550 - 980 \cdot \sin(120 / 2) / 1.05 \\
 &= 739.547 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 253201.7812 \text{ N-m}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 8149.8027 \text{ kPa} \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173320.1406 \text{ kPa}
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{(3(1681568 + 7213) \cdot 460) / (4 \cdot 1850 \cdot 173320)} \\
 &= 42.628 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120 / 2) (980 + 145 + 16) \\
 &= 1976.270 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (\text{Nr ribs} - 1) \\
 &= 1976 / (4 - 1) \\
 &= 658.757 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$



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Saddle Calcs: Test Case: Step: 25 4:31pm Apr 10, 2026

$$= 659 \cdot 460/2$$

$$= 0.151 \text{ m}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1688781 / (1850 \cdot 460)$$

$$= 1984958.625 \text{ N/m}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 0.15 \cdot 1984958$$

$$= 300674.500 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 0.0031 + 0.0049$$

$$= 0.008 \text{ m}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 300674 / 0.0081$$

$$= 37164.098 \text{ kPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	15.0	225.0	...	...	...	0.142E+08

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 225 - 15$$

$$= 210.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 0.0081$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 225 / 2$$

$$= 112.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(14238268 / 0.0081)}$$

$$= 41.950 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1550 - \cos(120/2) (980 + 145 + 16) - 50$$

$$= 929.500 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120/2) (980 + 145 + 16)$$

$$= 1976.270 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 1850 - 50.8$$

$$= 1799.200 \text{ mm}$$



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Saddle Calcs: Test Case: Step: 25 4:31pm Apr 10, 2026

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (1976 - 1799) / 2$$

$$= 88.535 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{930^2 + 88.5^2}$$

$$= 933.707 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 199955 / 259980}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 934 / 42$$

$$= 22.257$$

Bending Moment [Rm]:

$$= Fl / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 2412 / (2 \cdot 1850) \cdot 659 \cdot 934 / 2$$

$$= 200.593 \text{ N-m}$$

Compressive Allowable, $KL/r < Cc$ ($22.3 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (22.3)^2 / (2 \cdot 123^2)) 259980 / (5/3 + 3 \cdot (22.3) / (8 \cdot 123) - (22.3^3) / (8 \cdot 123^3))$$

$$= 147513 \text{ kPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc / Sca + (Rm \cdot C1 / I) / Sba$$

$$= 37164 / 147513 + (201 \cdot 112 / 14238282) / 173320$$

$$= 0.261$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	15.0	210.0	0.0	0.0	0.0	0.142E+08
Web	658.8	15.0	0.0	0.0	0.0	0.185E+06
Totals	...	...	...	...	...	0.144E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 0.013$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 225 / 2$$

$$= 112.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2} - Bpthk$$

$$= 1550 - \sqrt{(1141 + 16)^2 - (659/2)^2} - 50$$

$$= 407.576 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{14419324 / 0.013}$$

$$= 33.264 \text{ mm}$$



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Saddle Calcs: Test Case:

Step: 25 4:31pm Apr 10, 2026

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 408/33.3$$

$$= 12.253$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 2412 / (2 \cdot 1850)$$

$$= 0.652 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.65 \cdot 659 \cdot 408$$

$$= 175.123 \text{ N-m}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 175/128172$$

$$= 1365.795 \text{ kPa}$$

Compressive Allowable, $KL/r < C_c$ ($12.3 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.3)^2 / (2 \cdot 123^2)) 259980 / (5/3 + 3 \cdot (12.3) / (8 \cdot 123) - (12.3)^3 / (8 \cdot 123^3))$$

$$= 151831 \text{ kPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 37178/151831 + (175 \cdot 112/14419324) / 173320$$

$$= 0.253$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	171986.91 kPa
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	0.0012 m ²
BasePlate Allowable Stress	S	137989.50 kPa
Area Available in a Single Bolt	BltArea	0.0010 m ²
Saddle Load Q0 (Weight)	Q0	1673940.9 N
Saddle Load QL (Wind/Seismic contribution)	QL	415.9 N
Maximum Transverse Force	Ft	5904.1 N
Maximum Longitudinal Force	F1	2412.3 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2412 / (0.00098 \cdot 6)$$

$$= 410.5 \text{ kPa. Must be less than } 103192.1 \text{ kPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 5904 / (0.00098 \cdot 6)$$

$$= 1004.6 \text{ kPa. Must be less than } 103192.1 \text{ kPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:



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Saddle Calcs: Test Case: Step: 25 4:31pm Apr 10, 2026

= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $2412 / (103192 \cdot 6)$
= 0.0000 m²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot F_t + \text{Sum of X Moments}$
= $1550 \cdot 5904 + 0$
= 9155.11 N-m

Eccentricity (e):

= R_{mom} / QO
= $9155 / 1673940$
= 5.47 mm < $B_{plen} / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0, 0.38953E-05, 0]$
= 0.0000 m²

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: N2- Tube outlet

Nozl: 9 4:31pm Apr 10,2026

Input, Nozzle Desc: N2- Tube outlet

From: 30

Pressure for Reinforcement Calculations	P	1.5746	N/mm ²
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.10	N/mm ²
Temperature for External Pressure	Tempex	82	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137989.50	kPa
Parent Allowable Stress At Ambient	Sva	137989.50	kPa

Inside Diameter of Cylindrical Shell	D	1960.00	mm
Design Length of Section	L	1999.9696	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Note : User defined Limit(s) of Reinforcement specified below:

Physical Maximum for Diameter Limit	Dmax	1500.0000	mm
-------------------------------------	------	-----------	----

Distance from Bottom/Left Tangent		1.34	m
-----------------------------------	--	------	---

User Entered Minimum Design Metal Temperature		-28.89	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	117991.02	kPa
Allowable Stress At Ambient	Sna	117991.02	kPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		30.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137989.50	kPa
Pad Allowable Stress At Ambient	Spa	137989.50	kPa
Diameter of Pad along vessel surface	Dp	1302.0000	mm
Thickness of Pad	te	24.0000	mm
Weld leg size between Pad and Shell	Wp	20.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	24.0000	mm
Reinforcing Pad Width		250.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	



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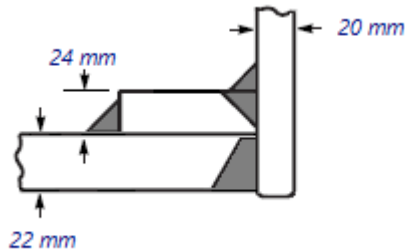
Nozzle Calcs.: N2- Tube outlet

Noz1: 9 4:31pm Apr 10,2026

Flange Series

B

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: N2- Tube outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	30.000 in.
Actual Thickness Used in Calculation	0.787 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.57 \cdot 983 / (137990 \cdot 1 - 0.6 \cdot 1.57)$
 $= 11.2938 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.57 \cdot 384 / (117991 \cdot 1 - 0.6 \cdot 1.57)$
 $= 5.1656 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.5711 mm

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1500.0000 mm
Parallel to Vessel Wall, opening length	d	750.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		47.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 117991 / 137990)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 117991 / 137990)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 137990 / 137990)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



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Nozzle Calcs.: N2- Tube outlet

Nozl: 9 4:31pm Apr 10,2026

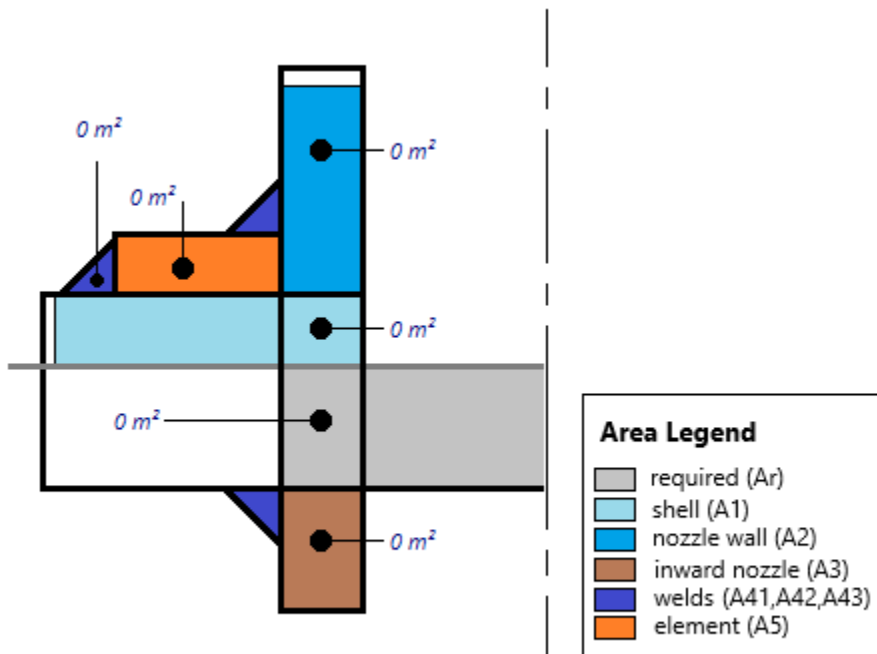


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.009	0.002	NA
Area in Shell	A1	0.006	0.009	NA
Area in Nozzle Wall	A2	0.001	0.001	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.001	0.001	NA
Area in Element	A5	0.012	0.012	NA
TOTAL AREA AVAILABLE	Atot	0.019	0.023	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 861.2912 24.0000 mm
Based on given Pad Diameter: 1302.0000 2.8460 mm
Based on Shell or Nozzle Thickness: 873.1494 20.0000 mm

Area Required [A]:
= (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)
= (768•11.3•1+2•17•11.3•1•(1-0.86))
= 0.009 m²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
= d(E1•t - F•tr) - 2 • tn (E1•t - F•tr) • (1 - fr1)
= 732 (1 • 19 - 1 • 11.3) - 2 • 17
(1 • 19 - 1 • 11.3) • (1 - 0.86)



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Nozzle Calcs.: N2- Tube outlet Nozl: 9 4:31pm Apr 10, 2026

$$= 0.006 \text{ m}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\ &= (2 \cdot 47.5) \cdot (17 - 5.17) \cdot 0.86 \\ &= 0.001 \text{ m}^2 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned} &= Wo^2 \cdot fr3 + (Wi-can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\ &= 20^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 20^2 \cdot 1 \\ &= 0.001 \text{ m}^2 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned} &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\ &= (1302 - 802) \cdot 24 \cdot 1 \\ &= 0.012 \text{ m}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 8.1656 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 14.2938 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 14.2938 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.7220 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[11.3, \max(14.3, 4.5)] \\ &= 11.3312 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(8.17, 11.3) \\ &= 11.3312 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 29874.4,	Allowable	: 117991.0 kPa	Passed
Expansion	: 0.0,	Allowable	: 265103.1 kPa	Passed
Occasional	: 17397.8,	Allowable	: 156928.1 kPa	Passed
Shear	: 9725.1,	Allowable	: 82593.7 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 20, tr = 5.17, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.3, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 20, tr = 5.17, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.3, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
--	-------



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Nozzle Calcs.: N2- Tube outlet Nozl: 9 4:31pm Apr 10, 2026

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 20, tr = 5.17, c = 3 mm, E* = 1
Thickness Ratio = $tr * E*/(tg - c) = 0.3$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 22, tr = 11.3, c = 3 mm, E* = 1
Thickness Ratio = $tr * E*/(tg - c) = 0.59$, Temp. Reduction = 23 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
Min Metal Temp. at Required thickness (UCS 66.1) -28 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 20, tr = 5.17, c = 3 mm, E* = 1
Thickness Ratio = $tr * E*/(tg - c) = 0.3$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -29 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $1.57/1.96 = 0.803$

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1152.0000 mm
Parallel to Vessel Wall, opening length d 576.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp 47.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (m²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		0.006	0.006
Area Available in Shell A1		0.003	0.003
Area Available in Nozzle Wall A2		0.001	0.001
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		0.000	0.000
Area Available in Pad A5		0.000	0.008
Area Available in Hub A6		0.000	0.000
TOTAL AREA AVAILABLE Atot		0.004	0.013

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.



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Nozzle Calcs.: N2- Tube outlet Nozl: 9 4:31pm Apr 10,2026

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: N2- Tube outlet

Intermediate Calc. for nozzle/shell Welds T_{min} 17.0000 mm
Intermediate Calc. for pad/shell Welds T_{min}Pad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = 0.7 * W _o mm
Pad Weld	9.5000 = 0.5*T _{min} Pad	14.1400 = 0.7 * W _p mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A_1+2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (0.0087 - 0.0056 + 2 \cdot 17 \cdot 0.86 \cdot (1 \cdot 19 - 11.3)) \cdot 137990)$$

$$= 462305.31 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2+A_5+A_4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (0.00096 + 0.012 + 0.00074 - 0 \cdot 0.86) \cdot 137990$$

$$= 1890868.38 \text{ N}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (0.00096 + 0 + 0.00034 + (0.00055)) \cdot 137990$$

$$= 256064.91 \text{ N}$$

Weld Load [W3]:

$$= (A_2+A_3+A_4+A_5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (0.00096 + 0 + 0.00074 + 0.012 + (0.00055)) \cdot 137990$$

$$= 1967088.50 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [S_{onw}]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 802 \cdot 20 \cdot 0.49 \cdot 117991$$

$$= 1456658. \text{ N}$$

Shear, Pad Element Weld [S_{pew}]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 1302 \cdot 20 \cdot 0.49 \cdot 137990$$

$$= 2765612. \text{ N}$$

Shear, Nozzle Wall [S_{nw}]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 392) \cdot (20 - 3) \cdot 0.7 \cdot 117991$$

$$= 1731305. \text{ N}$$

Tension, Pad Groove Weld [T_{pgw}]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 802 \cdot 24 \cdot 0.74 \cdot 137990$$

$$= 3087248. \text{ N}$$

Tension, Shell Groove Weld [T_{ngw}]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 802 \cdot (14 - 0) \cdot 0.74 \cdot 137990$$

$$= 1800895. \text{ N}$$



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Nozzle Calcs.: N2- Tube outlet

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See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (2765612 + 1731305) = 4496917 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (1456657 + 3087247 + 1800894 + 0) = 6344800 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (2765612 + 1800894 + 0) = 4566507 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 4496917 N	, must exceed W = 462305 N	or W1 = 1890868 N
Path 2-2 = 6344800 N	, must exceed W = 462305 N	or W2 = 256064 N
Path 3-3 = 4566507 N	, must exceed W = 462305 N	or W3 = 1967088 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.52 N/mm²

Nozzle is O.K. for the External Pressure: 0.1 N/mm²

The Drop for this Nozzle is : 85.7970 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 357.7970 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 634.78 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 750.00 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N2- Tube outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1960.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 137989.50	kPa
Vessel Hot S.I. Allowable	Smh 137989.50	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	762.000	mm
Nozzle Thickness	20.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 117991.02	kPa
Nozzle Hot S.I. Allowable	SNmh 117991.02	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	1302.000	mm
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Nozzle Calcs.: N2- Tube outlet Nozl: 9 4:31pm Apr 10,2026

Thickness of Reinforcing Pad 24.0000 mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	1.57	N/mm ²
Radial Load	P	-56227.61	N
Circumferential Shear	Vc	37488.32	N
Longitudinal Shear	Vl	37488.32	N
Circumferential Moment	Mc	78752.88	N-m
Longitudinal Moment	Ml	78752.88	N-m
Torsional Moment	Mt	112498.58	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 802.0 + 2 \cdot 1.65 \cdot \sqrt{(1004.5(22.0 - 3.0))}$$

$$= 1257.896 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2009.000 mm
Nozzle Outside Diameter	(d) =	802.000 mm
Vessel Thickness used	(T) =	43.000 mm
Nozzle Thickness used	(t) =	17.000 mm
T / t	=	2.529
d / t	=	47.176

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.729$$

Nr/P	=	0.042
Mr/P	=	0.061
M0/P	=	0.042
N0/P	=	0.167
MrD/Mc	=	0.151
NrDL/Mc	=	0.084
M0d/Mc	=	0.122
N0DL/Mc	=	0.247
MrD/Ml	=	0.071
NrDL/Ml	=	0.044
M0D/Ml	=	0.052
N0DL/Ml	=	0.314

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5088	5088	5088	5088
Outplane Bending (P)	7673	-7673	7673	-7673
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-16650	-16650	16650	16650
Outplane Bending (ML)	-16416	16416	16416	-16416
Normal Pressure Stress	35217	36796	35217	36796
Outplane Stress Summary	14913	33976	81046	34445



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Nozzle Calcs.: N2- Tube outlet

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Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1289	1289	1289	1289
Inplane Bending (P)	11196	-11196	11196	-11196
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2351	-2351	2351	2351
Inplane Bending (ML)	-22607	22607	22607	-22607
Inplane Pressure Stress	17608	17608	17608	17608
Inplane Stress Summary	5136	27957	55053	-12555

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	689	689	-689	-689
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2585	2585	2585	2585
Shear Stress Summary	3274	3274	1896	1896

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	15905	35410	81177	47152

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	5088	5088	5088	5088
Outplane Bending (P)	7673	-7673	7673	-7673
Outplane Membrane (Mc)	-13127	-13127	13127	13127
Outplane Bending (Mc)	-38995	38995	38995	-38995
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	17608	17608	17608	17608
Outplane Stress Summary	-21752	40891	82493	-10845

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1289	1289	1289	1289
Inplane Bending (P)	11196	-11196	11196	-11196
Inplane Membrane (Mc)	-4447	-4447	4447	4447
Inplane Bending (Mc)	-48048	48048	48048	-48048
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	35217	36796	35217	36796
Inplane Stress Summary	-4791	70490	100199	-16712

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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Nozzle Calcs.: N2- Tube outlet Nozl: 9 4:31pm Apr 10,2026

(Shear stress normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-689	-689	689	689
Torsional Shear (Mt)			2585	2585	2585	2585
Shear Stress Summary			1896	1896	3274	3274

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress			21959	70607	100785	18174

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)			5088	5088	5088	5088
Hoop Bending (P)			0	0	0	0
Hoop Membrane (Mc)			0	0	0	0
Hoop Bending (Mc)			0	0	0	0
Hoop Membrane (ML)			-16650	-16650	16650	16650
Hoop Bending (ML)			0	0	0	0
Hoop Pressure Stress			41753	43642	41753	43642
Hoop Stress Summary			30191	32080	63492	65381

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)			1337	1337	1337	1337
Axial Bending (P)			46876	-46876	46876	-46876
Axial Membrane (Mc)			0	0	0	0
Axial Bending (Mc)			0	0	0	0
Axial Membrane (ML)			-9769	-9769	9769	9769
Axial Bending (ML)			-99475	99475	99475	-99475
Axial Pressure Stress			20876	20876	20876	20876
Axial Stress Summary			-40154	65043	178335	-114367

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Shear stress)			Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)			1744	1744	-1744	-1744
Shear due to (Vl)			0	0	0	0
Shear due to Torsion			6542	6542	6542	6542
Shear Stress Summary			8287	8287	4798	4798

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)			Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress			72269	67008	178535	180004

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses in the			Cu Left	Cl Left	Du Right	Dl Right
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Nozzle Calcs.: N2- Tube outlet Nozl: 9 4:31pm Apr 10,2026

hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	5088	5088	5088	5088
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-13127	-13127	13127	13127
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	41753	43642	41753	43642
Hoop Stress Summary	33714	35603	59969	61858

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1337	1337	1337	1337
Axial Bending (P)	46876	-46876	46876	-46876
Axial Membrane (Mc)	-9769	-9769	9769	9769
Axial Bending (Mc)	-221957	221957	221957	-221957
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	20876	20876	20876	20876
Axial Stress Summary	-162636	187526	300818	-236850

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1744	-1744	1744	1744
Shear due to Torsion	6542	6542	6542	6542
Shear Stress Summary	4798	4798	8287	8287

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	196578	187671	301100	299163

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)} \\
 &= 802.0 + 2 \cdot 1.65 \cdot \sqrt{992.5(22.0 - 3.0)} \\
 &= 1255.165 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1985.000 mm
Nozzle Outside Diameter	(d) =	1302.000 mm
Vessel Thickness used	(T) =	19.000 mm
Nozzle Thickness used	(t) =	267.000 mm
T / t	=	0.071
d / T	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 6.704$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

$$Nr/P = 0.045$$



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Nozzle Calcs.: N2- Tube outlet Nozl: 9 4:31pm Apr 10,2026

Mr/P = 0.071
MØ/P = 0.019
NØ/P = 0.026
MrD/Mc = 0.239
NrDL/Mc = 0.092
MØd/Mc = 0.067
NØDL/Mc = 0.047
MrD/Ml = 0.038
NrDL/Ml = 0.036
MØD/Ml = 0.012
NØDL/Ml = 0.021

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3998	3998	3998	3998
Outplane Bending (P)	17912	-17912	17912	-17912
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3536	-3536	3536	3536
Outplane Bending (ML)	-12217	12217	12217	-12217
Normal Pressure Stress	80680	82252	80680	82252
Outplane Stress Summary	86837	77019	118345	59658

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	7032	7032	7032	7032
Inplane Bending (P)	66077	-66077	66077	-66077
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5950	-5950	5950	5950
Inplane Bending (ML)	-38347	38347	38347	-38347
Inplane Pressure Stress	40340	40340	40340	40340
Inplane Stress Summary	69152	13692	157748	-51102

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	958	958	-958	-958
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2220	2220	2220	2220
Shear Stress Summary	3178	3178	1261	1261

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	87389	77178	157782	110789

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	3998	3998	3998	3998
Outplane Bending (P)	17912	-17912	17912	-17912
Outplane Membrane (Mc)	-7908	-7908	7908	7908
Outplane Bending (Mc)	-67415	67415	67415	-67415
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	40340	40340	40340	40340

Outplane Stress Summary -13072 85934 137574 -33080

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	7032	7032	7032	7032
Inplane Bending (P)	66077	-66077	66077	-66077
Inplane Membrane (Mc)	-15381	-15381	15381	15381
Inplane Bending (Mc)	-239835	239835	239835	-239835
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	80680	82252	80680	82252

Inplane Stress Summary -101426 247660 409008 -201246

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-958	-958	958	958
Torsional Shear (Mt)	2220	2220	2220	2220

Shear Stress Summary 1261 1261 3178 3178

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress 101440 247667 409042 201301

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	35217	36796	35217	36796	35217	36796	35217	36796
Circ. Pl (SUS)	-11562	-11562	21738	21738	-3158	-3158	5736	5736
Circ. Q (SUS)	-8743	8743	24089	-24089	-36852	36852	59244	-59244
Long. Pm (SUS)	17608	17608	17608	17608	17608	17608	17608	17608
Long. Pl (SUS)	-1062	-1062	3640	3640	-8039	-8039	18215	18215
Long. Q (SUS)	-11411	11411	33803	-33803	-31322	31322	46668	-46668
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	689	689	-689	-689	-689	-689	689	689
Shear Q (SUS)	2585	2585	2585	2585	2585	2585	2585	2585
Pm (SUS)	35217	36796	35217	36796	35217	36796	35217	36796



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Pm+Pl (SUS)	23721	25288	56968	58546	32080	33657	41043	42602
Pm+Pl+Q (Total)	15907	35414	81181	47152	21962	70610	100783	18174

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	36796	137989	Passed
Pm+Pl (SUS)	58546	206984	Passed
Pm+Pl+Q (TOTAL)	100783	413968	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	80680	82252	80680	82252	80680	82252	80680	82252
Circ. Pl (SUS)	462	462	7534	7534	-8349	-8349	22413	22413
Circ. Q (SUS)	5695	-5695	30129	-30129	173758	173758	305912	-305912
Long. Pm (SUS)	40340	40340	40340	40340	40340	40340	40340	40340
Long. Pl (SUS)	1082	1082	12982	12982	-3910	-3910	11906	11906
Long. Q (SUS)	27730	-27730	104424	-104424	-49503	49503	85327	-85327
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	958	958	-958	-958	-958	-958	958	958
Shear Q (SUS)	2220	2220	2220	2220	2220	2220	2220	2220
Pm (SUS)	80680	82252	80680	82252	80680	82252	80680	82252
Pm+Pl (SUS)	81165	82736	88240	89811	72356	73927	103111	104682
Pm+Pl+Q (Total)	87390	77178	157786	110787	101445	247670	409042	201307

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	82252	137989	Passed
Pm+Pl (SUS)	104682	206984	Passed
Pm+Pl+Q (TOTAL)	409042	413968	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	41753	43642	41753	43642	41753	43642	41753	43642
Circ. Pl (SUS)	-11562	-11562	21738	21738	-8039	-8039	18215	18215
Circ. Q (SUS)	0	0	0	0	0	0	0	0



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Nozzle Calcs.: N2- Tube outlet

Nozl: 9 4:31pm Apr 10,2026

Long. Pm (SUS)	20876	20876	20876	20876	20876	20876	20876	20876
Long. Pl (SUS)	-8432	-8432	11106	11106	-8432	-8432	11106	11106
Long. Q (SUS)	-52599	52599	146351	-146351	-175081	175081	268833	-268833
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1744	1744	-1744	-1744	-1744	-1744	1744	1744
Shear Q (SUS)	6542	6542	6542	6542	6542	6542	6542	6542
Pm (SUS)	41753	43642	41753	43642	41753	43642	41753	43642
Pm+Pl (SUS)	30360	32233	63587	65470	33856	35733	60076	61958
Pm+Pl+Q (Total)	72271	67008	178533	180004	196585	187676	301099	299167

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	43642	117991	Passed
Pm+Pl (SUS)	65470	176986	Passed
Pm+Pl+Q (TOTAL)	301099	353973	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N3- Shell inlet

Nozl: 10

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Input, Nozzle Desc: N3- Shell inlet

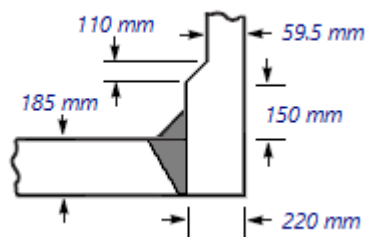
From: 40

Pressure for Reinforcement Calculations	P	15.2121	N/mm ²
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137989.50	kPa
Parent Allowable Stress At Ambient	Sva	137989.50	kPa
Inside Diameter of Cylindrical Shell	D	2532.00	mm
Shell Finished (Minimum) Thickness	t	185.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3.34	m
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137989.50	kPa
Allowable Stress At Ambient	Sna	137989.50	kPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Hub Height of Integral Nozzle	h	150.0000	mm
Height of Beveled Transition	L'	110.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		220.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	50.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	50.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: N3- Shell inlet



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Nozzle Calcs.: N3- Shell inlet Nozl: 10 4:31pm Apr 10,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	19.314 in.
Actual Thickness Used in Calculation	2.343 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 15.2 \cdot 1269 / (137990 \cdot 1 - 0.6 \cdot 15.2)$
 $= 149.7951 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 15.2 \cdot 248 / (137990 \cdot 1 - 0.6 \cdot 15.2)$
 $= 29.3083 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 160 / 0.5773$
 $= 277.9731 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	993.1512	mm
Parallel to Vessel Wall, opening length	d	496.5756	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	401.2805	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 137990 / 137990)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 137990 / 137990)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



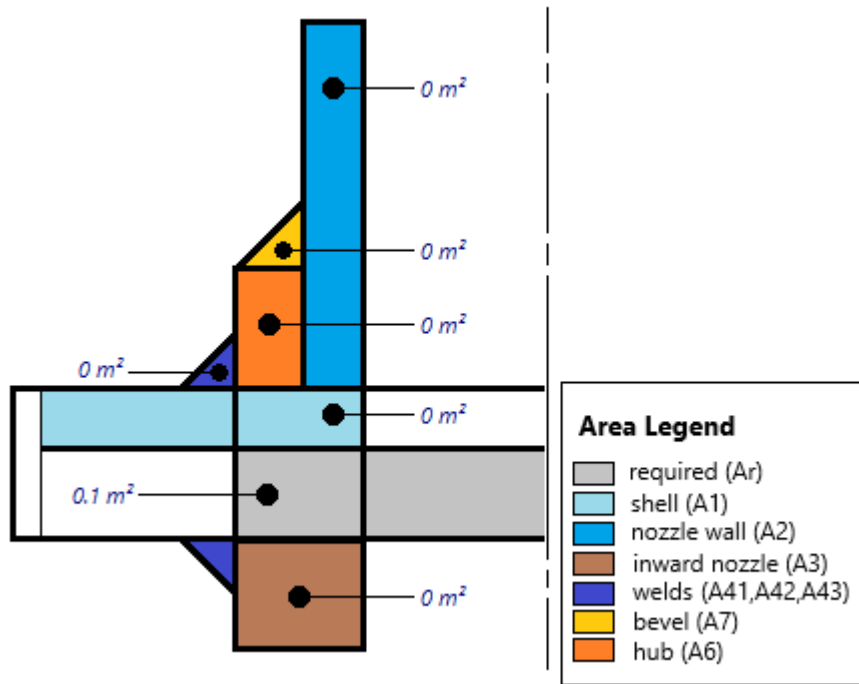
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Nozzle Calcs.: N3- Shell inlet

Noz1: 10 4:31pm Apr 10,2026



Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.074	NA	NA
Area in Shell	A1	0.016	NA	NA
Area in Nozzle Wall	A2	0.014	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.002	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	0.048	NA	NA
TOTAL AREA AVAILABLE	Atot	0.080	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (497 \cdot 150 \cdot 1 + 2 \cdot 217 \cdot 150 \cdot 1 \cdot (1 - 1))$$

$$= 0.074 \text{ m}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 497(1 \cdot 182 - 1 \cdot 150) - 2 \cdot 217$$

$$(1 \cdot 182 - 1 \cdot 150) \cdot (1 - 1)$$

$$= 0.016 \text{ m}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot ho)(tn - trn)fr2$$

$$= (2 \cdot 250)(56.5 - 29.3)1$$

$$= 0.014 \text{ m}^2$$



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Nozzle Calcs.: N3- Shell inlet Nozl: 10 4:31pm Apr 10, 2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= (W_o^2 - \text{Area Lost}) \cdot fr2 + (W_i - \text{can}/0.707)^2 - \text{Area Lost}) \cdot fr2$
 $= (50^2 - 0.00035) \cdot 1 + (0^2 - 0) \cdot 1$
 $= 0.002 \text{ m}^2$

Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(T_{lnp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr2$
 $= (2 \cdot \min(250, 250, 150)) \cdot (220 - 59.5) \cdot 1$
 $= 0.048 \text{ m}^2$ Includes Bevel Area of 0.000 m²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 32.3083 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 152.7951 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 152.7951 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[11.3, \max(153, 4.5)]$
 $= 11.3312 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(32.3, 11.3)$
 $= 32.3083 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 59.512 = 52.073 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 41503.0,	Allowable	: 137989.5 kPa	Passed
Expansion	: 0.0,	Allowable	: 303470.8 kPa	Passed
Occasional	: 30001.1,	Allowable	: 183526.0 kPa	Passed
Shear	: 7678.0,	Allowable	: 96592.6 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Weld Size Calculations, Description: N3- Shell inlet

Intermediate Calc. for nozzle/shell welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$35.3500 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) S_v)$
 $= \max(0, (0.074 - 0.016 + 2 \cdot 217 \cdot 1 \cdot (1 \cdot 182 - 150))) 137990$



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Nozzle Calcs.: N3- Shell inlet Nozl: 10 4:31pm Apr 10,2026

= 9985939.00 N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - C_{an}/.707)^2 \cdot fr2) \cdot S_v$$

$$= (0.062 + 0.0021 - 0 \cdot 1) \cdot 137990$$

$$= 8817031.00 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S_v$$

$$= (0.062 + 0 + 0.0021 + (0.079)) \cdot 137990$$

$$= 19716252.00 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (0.062 + 0 + 0.0021 + 0 + (0.079)) \cdot 137990$$

$$= 19716252.00 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 931 \cdot 50 \cdot 0.49 \cdot 137990$$

$$= 4941649. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - C_{an}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 357) \cdot (220 - 3) \cdot 0.7 \cdot 137990$$

$$= 23493726. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{g_{nvi}} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 931 \cdot (50 - 0) \cdot 0.74 \cdot 137990$$

$$= 7462898. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (4941649 + 23493726) = 28435374 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (4941649 + 0 + 7462898 + 0) = 12404547 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (4941649 + 7462898 + 0) = 12404547 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 28435374N	, must exceed W = 9985939 N	or W1 = 8817031 N
Path 2-2 = 12404547N	, must exceed W = 9985939 N	or W2 = 19716252 N
Path 3-3 = 12404547N	, must exceed W = 9985939 N	or W3 = 19716252 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 15.7 N/mm²

The Drop for this Nozzle is : 88.6032 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 523.6033 mm

Input Echo, WRC297 Item 1, Description: N3- Shell inle:



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Nozzle Calcs.: N3- Shell inlet Nozl: 10 4:31pm Apr 10,2026

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2532.000	mm
Shell Thickness	185.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	137989.50 kPa
Vessel Hot S.I. Allowable	Smh	137989.50 kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	490.576	mm
Nozzle Thickness	220.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc	137989.50 kPa
Nozzle Hot S.I. Allowable	SNmh	137989.50 kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	15.21	N/mm ²
Radial Load	P	-74967.03	N
Circumferential Shear	Vc	59973.27	N
Longitudinal Shear	Vl	59973.27	N
Circumferential Moment	Mc	107995.79	N-m
Longitudinal Moment	Ml	107995.79	N-m
Torsional Moment	Mt	125996.74	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_v/T_v (15.686) is < 20
check the limitations of WRC bulletin 297.

Warning - The ratio, D_n/T_n (4.230) is < 10
check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 930.576 + 2 \cdot 1.65 \cdot \sqrt{1360.0(185.0 - 3.0)}$$

$$= 2572.371 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2720.000 mm
Nozzle Outside Diameter	(d) =	930.576 mm
Vessel Thickness used	(T) =	182.000 mm
Nozzle Thickness used	(t) =	217.000 mm
T / t	=	0.839
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.323$$



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Nozzle Calcs.: N3- Shell inlet

Nozl: 10 4:31pm Apr 10,2026

Nr/P = 0.103
 Mr/P = 0.166
 MØ/P = 0.053
 NØ/P = 0.114
 MrD/Mc = 0.280
 NrDL/Mc = 0.129
 MØd/Mc = 0.099
 NØDL/Mc = 0.049
 MrD/Ml = 0.162
 NrDL/Ml = 0.093
 MØD/Ml = 0.058
 NØDL/Ml = 0.133

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	255	255	255	255
Outplane Bending (P)	723	-723	723	-723
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-461	-461	461	461
Outplane Bending (ML)	-1206	1206	1206	-1206
Normal Pressure Stress	98958	114174	98958	114174
Outplane Stress Summary	98268	114450	101605	112961

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	227	227	227	227
Inplane Bending (P)	2254	-2254	2254	-2254
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-324	-324	324	324
Inplane Bending (ML)	-3412	3412	3412	-3412
Inplane Pressure Stress	49475	49475	49475	49475
Inplane Stress Summary	48220	50537	55694	44359

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	220	220	-220	-220
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	503	503	503	503
Shear Stress Summary	723	723	282	282

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	98275	114457	101605	112961

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Nozzle Calcs.: N3- Shell inlet Nozl: 10 4:31pm Apr 10,2026

Outplane Membrane (P)	255	255	255	255
Outplane Bending (P)	723	-723	723	-723
Outplane Membrane (Mc)	-165	-165	165	165
Outplane Bending (Mc)	-2068	2068	2068	-2068
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	49475	49475	49475	49475

Outplane Stress Summary	48220	50909	52688	47103
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	227	227	227	227
Inplane Bending (P)	2254	-2254	2254	-2254
Inplane Membrane (Mc)	-448	-448	448	448
Inplane Bending (Mc)	-5874	5874	5874	-5874
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	98958	114174	98958	114174

Inplane Stress Summary	95117	117573	107762	106721
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-220	-220	220	220
Torsional Shear (Mt)	503	503	503	503

Shear Stress Summary	282	282	723	723
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	95117	117573	107769	106728
------------------------	-------	--------	--------	--------

Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	255	255	255	255
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-461	-461	461	461
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	14533	32783	14533	32783

Hoop Stress Summary	14326	32576	15250	33500
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	151	151	151	151
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Nozzle Calcs.: N3- Shell inlet Nozl: 10 4:31pm Apr 10,2026

Axial Bending (P)	1089	-1089	1089	-1089
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1482	-1482	1482	1482
Axial Bending (ML)	-1709	1709	1709	-1709
Axial Pressure Stress	7266	7266	7266	7266

Axial Stress Summary	5315	6556	11700	6101
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	186	186	-186	-186
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	420	420	420	420

Shear Stress Summary	606	606	234	234
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	14361	32590	15264	33500

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	255	255	255	255
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-165	-165	165	165
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	14533	32783	14533	32783

Hoop Stress Summary	14623	32873	14954	33204
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	151	151	151	151
Axial Bending (P)	1089	-1089	1089	-1089
Axial Membrane (Mc)	-1482	-1482	1482	1482
Axial Bending (Mc)	-3178	3178	3178	-3178
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7266	7266	7266	7266

Axial Stress Summary	3847	8025	13168	4633
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-186	-186	186	186
Shear due to Torsion	420	420	420	420

Shear Stress Summary	234	234	606	606
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Nozzle Calcs.: N3- Shell inlet

Nozl: 10 4:31pm Apr 10,2026

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE
(Stress Intensities Cu Cl Du Dl
Outside Inside Outside Inside

Two * Max Shear Stress 14623 32873 15140 33211

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	98958	114174	98958	114174	98958	114174	98958	114174
Circ. Pl (SUS)	-206	-206	716	716	-221	-221	675	675
Circ. Q (SUS)	-483	483	1929	-1929	-3620	3620	8128	-8128
Long. Pm (SUS)	49475	49475	49475	49475	49475	49475	49475	49475
Long. Pl (SUS)	-97	-97	551	551	90	90	420	420
Long. Q (SUS)	-1158	1158	5666	-5666	-1345	1345	2791	-2791
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	220	220	-220	-220	-220	-220	220	220
Shear Q (SUS)	503	503	503	503	503	503	503	503
Pm (SUS)	98958	114174	98958	114174	98958	114174	98958	114174
Pm+Pl (SUS)	98752	113968	99674	114890	98737	113953	99633	114849
Pm+Pl+Q (Total)	98279	114459	101604	112962	95118	117574	107770	106729

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	114174	137989	Passed
Pm+Pl (SUS)	114890	206984	Passed
Pm+Pl+Q (TOTAL)	117574	413968	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	14533	32783	14533	32783	14533	32783	14533	32783
Circ. Pl (SUS)	-206	-206	716	716	90	90	420	420
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7266	7266	7266	7266	7266	7266	7266	7266
Long. Pl (SUS)	-1331	-1331	1633	1633	-1331	-1331	1633	1633
Long. Q (SUS)	-620	620	2798	-2798	-2089	2089	4267	-4267
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	186	186	-186	-186	-186	-186	186	186
Shear Q (SUS)	420	420	420	420	420	420	420	420
Pm (SUS)	14533	32783	14533	32783	14533	32783	14533	32783



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Nozzle Calcs.: N3- Shell inlet Nozl: 10 4:31pm Apr 10,2026

Pm+Pl (SUS)	14331	32578	15254	33500	14626	32874	14958	33204
Pm+Pl+Q (Total)	14367	32591	15264	33501	14628	32875	15139	33215

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	32783	137989	Passed
Pm+Pl (SUS)	33500	206984	Passed
Pm+Pl+Q (TOTAL)	33501	413968	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Nozzle Calcs.: N4- Shell outle

Nozl: 11 4:31pm Apr 10,2026

Input, Nozzle Desc: N4- Shell outle

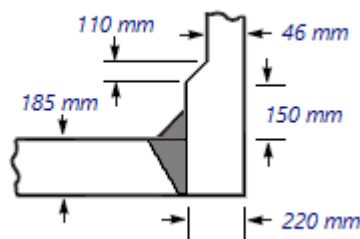
From: 80

Pressure for Reinforcement Calculations	P	15.2121	N/mm ²
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137989.50	kPa
Parent Allowable Stress At Ambient	Sva	137989.50	kPa
Inside Diameter of Cylindrical Shell	D	2532.00	mm
Shell Finished (Minimum) Thickness	t	185.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		16.98	m
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137989.50	kPa
Allowable Stress At Ambient	Sna	137989.50	kPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Hub Height of Integral Nozzle	h	150.0000	mm
Height of Beveled Transition	L'	110.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		220.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	50.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	50.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: N4- Shell outle



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Nozzle Calcs.: N4- Shell outle Nozl: 11 4:31pm Apr 10,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 20.376 in.
Actual Thickness Used in Calculation 1.812 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $15.2 \cdot 1269 / (137990 \cdot 1 - 0.6 \cdot 15.2)$
= 149.7951 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $15.2 \cdot 262 / (137990 \cdot 1 - 0.6 \cdot 15.2)$
= 30.9004 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $174 / 0.5773$
= 301.3340 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1047.1008	mm
Parallel to Vessel Wall, opening length	d	523.5504	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	367.5620	mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 137990 / 137990)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 137990 / 137990)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1, 1)$
= 1.000



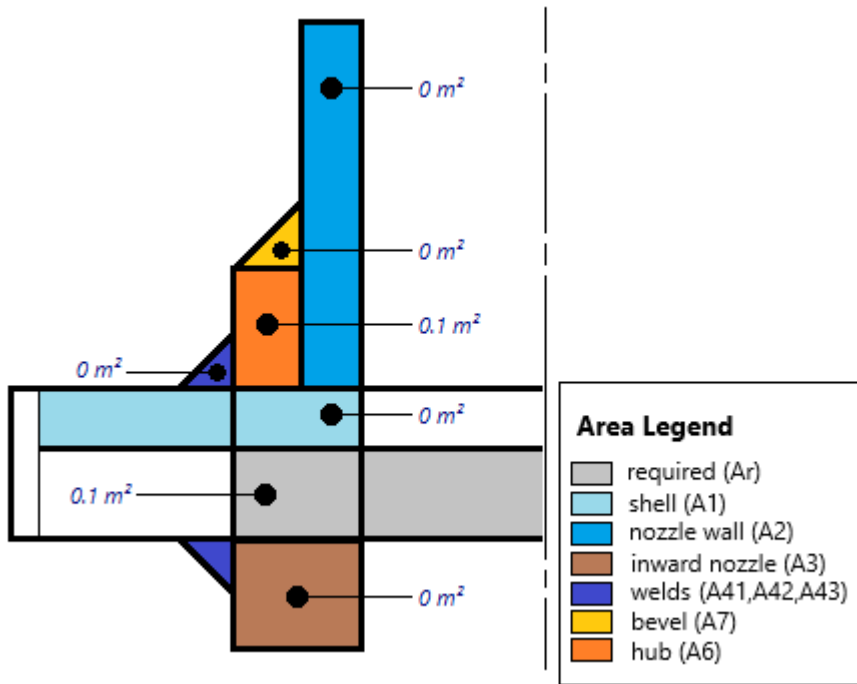
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Nozzle Calcs.: N4- Shell outle

Nozl: 11 4:31pm Apr 10,2026



Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.078	NA	NA
Area in Shell	A1	0.017	NA	NA
Area in Nozzle Wall	A2	0.006	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.002	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	0.053	NA	NA
TOTAL AREA AVAILABLE	Atot	0.078	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (524 \cdot 150 \cdot 1 + 2 \cdot 217 \cdot 150 \cdot 1 \cdot (1 - 1))$$

$$= 0.078 \text{ m}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 524(1 \cdot 182 - 1 \cdot 150) - 2 \cdot 217$$

$$(1 \cdot 182 - 1 \cdot 150) \cdot (1 - 1)$$

$$= 0.017 \text{ m}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot ho)(tn - trn)fr2$$

$$= (2 \cdot 250)(43 - 30.9)1$$

$$= 0.006 \text{ m}^2$$



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Nozzle Calcs.: N4- Shell outle Nozl: 11 4:31pm Apr 10, 2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= (W_o^2 - \text{Area Lost}) \cdot fr_2 + (W_i - \text{can}/0.707)^2 - \text{Area Lost}) \cdot fr_2$
 $= (50^2 - 0.27292E-04) \cdot 1 + (0^2 - 0) \cdot 1$
 $= 0.002 \text{ m}^2$

Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(T_{lnp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr_2$
 $= (2 \cdot \min(250, 250, 150)) \cdot (220 - 46) \cdot 1$
 $= 0.053 \text{ m}^2$ Includes Bevel Area of 0.001 m^2

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 33.9004 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 152.7951 \text{ mm}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 152.7951 \text{ mm}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[11.3, \max(153, 4.5)]$
 $= 11.3312 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(33.9, 11.3)$
 $= 33.9004 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 46.025 = 40.272 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 56838.2,	Allowable	: 137989.5 kPa	Passed
Expansion	: 0.0,	Allowable	: 288135.6 kPa	Passed
Occasional	: 42760.6,	Allowable	: 183526.0 kPa	Passed
Shear	: 9196.0,	Allowable	: 96592.6 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Weld Size Calculations, Description: N4- Shell outle

Intermediate Calc. for nozzle/shell Welds $T_{min} \quad 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$35.3500 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) S_v)$
 $= \max(0, (0.078 - 0.017 + 2 \cdot 217 \cdot 1 \cdot (1 \cdot 182 - 150)) 137990)$



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Nozzle Calcs.: N4- Shell outle Nozl: 11 4:31pm Apr 10,2026

= 10423625.00 N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot S_v$$

$$= (0.059 + 0.0025 - 0 \cdot 1) \cdot 137990$$

$$= 8495282.00 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S_v$$

$$= (0.059 + 0 + 0.0025 + (0.079)) \cdot 137990$$

$$= 19394500.00 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S$$

$$= (0.059 + 0 + 0.0025 + 0 + (0.079)) \cdot 137990$$

$$= 19394500.00 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 958 \cdot 50 \cdot 0.49 \cdot 137990$$

$$= 5084894. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 370) \cdot (220 - 3) \cdot 0.7 \cdot 137990$$

$$= 24381842. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{g_{nv}} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 958 \cdot (50 - 0) \cdot 0.74 \cdot 137990$$

$$= 7679228. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (5084894 + 24381842) = 29466736 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (5084894 + 0 + 7679227 + 0) = 12764121 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (5084894 + 7679227 + 0) = 12764121 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 29466736N , must exceed W = 10423625 N or W1 = 8495282 N
 Path 2-2 = 12764121N , must exceed W = 10423625 N or W2 = 19394500 N
 Path 3-3 = 12764121N , must exceed W = 10423625 N or W3 = 19394500 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 15.2 N/mm²

The Drop for this Nozzle is : 94.0229 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 529.0230 mm

Input Echo, WRC297 Item 1, Description: N4- Shell outl:



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Nozzle Calcs.: N4- Shell outle Nozl: 11 4:31pm Apr 10,2026

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2532.000	mm
Shell Thickness	185.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 137989.50	kPa
Vessel Hot S.I. Allowable	Smh 137989.50	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	517.550	mm
Nozzle Thickness	220.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 137989.50	kPa
Nozzle Hot S.I. Allowable	SNmh 137989.50	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	15.21	N/mm ²
Radial Load	P	-74967.03	N
Circumferential Shear	Vc	59973.27	N
Longitudinal Shear	Vl	59973.27	N
Circumferential Moment	Mc	107995.79	N-m
Longitudinal Moment	Ml	107995.79	N-m
Torsional Moment	Mt	125996.74	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_v/T_v (15.686) is < 20
check the limitations of WRC bulletin 297.

Warning - The ratio, D_n/T_n (4.353) is < 10
check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 957.55 + 2 \cdot 1.65 \cdot \sqrt{1360.0(185.0 - 3.0)}$$

$$= 2599.346 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2720.000	mm
Nozzle Outside Diameter	(d) =	957.550	mm
Vessel Thickness used	(T) =	182.000	mm
Nozzle Thickness used	(t) =	217.000	mm
T / t	=	0.839	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.361$$



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Nozzle Calcs.: N4- Shell outle

Nozl: 11 4:31pm Apr 10,2026

Nr/P = 0.101
 Mr/P = 0.163
 MØ/P = 0.052
 NØ/P = 0.113
 MrD/Mc = 0.278
 NrDL/Mc = 0.129
 MØd/Mc = 0.098
 NØDL/Mc = 0.050
 MrD/Ml = 0.159
 NrDL/Ml = 0.092
 MØD/Ml = 0.056
 NØDL/Ml = 0.134

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	255	255	255	255
Outplane Bending (P)	710	-710	710	-710
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-448	-448	448	448
Outplane Bending (ML)	-1144	1144	1144	-1144
Normal Pressure Stress	98958	114174	98958	114174
Outplane Stress Summary	98330	114415	101516	113023

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	227	227	227	227
Inplane Bending (P)	2213	-2213	2213	-2213
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-310	-310	310	310
Inplane Bending (ML)	-3240	3240	3240	-3240
Inplane Pressure Stress	49475	49475	49475	49475
Inplane Stress Summary	48365	50420	55467	44559

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	213	213	-213	-213
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	475	475	475	475
Shear Stress Summary	689	689	261	261

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	98337	114422	101516	113023

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Nozzle Calcs.: N4- Shell outle Nozl: 11 4:31pm Apr 10,2026

Outplane Membrane (P)	255	255	255	255
Outplane Bending (P)	710	-710	710	-710
Outplane Membrane (Mc)	-165	-165	165	165
Outplane Bending (Mc)	-1999	1999	1999	-1999
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	49475	49475	49475	49475

Outplane Stress Summary	48275	50854	52605	47186
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	227	227	227	227
Inplane Bending (P)	2213	-2213	2213	-2213
Inplane Membrane (Mc)	-434	-434	434	434
Inplane Bending (Mc)	-5681	5681	5681	-5681
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	98958	114174	98958	114174

Inplane Stress Summary	95283	117435	107514	106942
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-213	-213	213	213
Torsional Shear (Mt)	475	475	475	475

Shear Stress Summary	261	261	689	689
----------------------	-----	-----	-----	-----

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	95283	117435	107521	106949
------------------------	-------	--------	--------	--------

Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	255	255	255	255
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-448	-448	448	448
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15561	33818	15561	33818

Hoop Stress Summary	15368	33624	16264	34521
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	144	144	144	144
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Nozzle Calcs.: N4- Shell outle Nozl: 11 4:31pm Apr 10,2026

Axial Bending (P)	1068	-1068	1068	-1068
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1372	-1372	1372	1372
Axial Bending (ML)	-1620	1620	1620	-1620
Axial Pressure Stress	7777	7777	7777	7777
Axial Stress Summary	5998	7101	11982	6605

Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	179	179	-179	-179
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	399	399	399	399
Shear Stress Summary	579	579	220	220

Nozzle Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	15402	33631	16271	34521

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	255	255	255	255
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-165	-165	165	165
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15561	33818	15561	33818
Hoop Stress Summary	15650	33907	15981	34238

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	144	144	144	144
Axial Bending (P)	1068	-1068	1068	-1068
Axial Membrane (Mc)	-1372	-1372	1372	1372
Axial Bending (Mc)	-3068	3068	3068	-3068
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7777	7777	7777	7777
Axial Stress Summary	4550	8549	13430	5157

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-179	-179	179	179
Shear due to Torsion	399	399	399	399
Shear Stress Summary	220	220	579	579



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Nozzle Calcs.: N4- Shell outle

Nozl: 11 4:31pm Apr 10,2026

Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE
(Stress Intensities

	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	15650	33907	16105	34245

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	98958	114174	98958	114174	98958	114174	98958	114174
Circ. Pl (SUS)	-193	-193	703	703	-207	-207	661	661
Circ. Q (SUS)	-434	434	1854	-1854	-3468	3468	7894	-7894
Long. Pm (SUS)	49475	49475	49475	49475	49475	49475	49475	49475
Long. Pl (SUS)	-83	-83	537	537	90	90	420	420
Long. Q (SUS)	-1027	1027	5453	-5453	-1289	1289	2709	-2709
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	213	213	-213	-213	-213	-213	213	213
Shear Q (SUS)	475	475	475	475	475	475	475	475
Pm (SUS)	98958	114174	98958	114174	98958	114174	98958	114174
Pm+Pl (SUS)	98765	113981	99661	114877	98751	113967	99619	114835
Pm+Pl+Q (Total)	98340	114422	101516	113024	95284	117436	107521	106948

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	114174	137989	Passed
Pm+Pl (SUS)	114877	206984	Passed
Pm+Pl+Q (TOTAL)	117436	413968	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	15561	33818	15561	33818	15561	33818	15561	33818
Circ. Pl (SUS)	-193	-193	703	703	90	90	420	420
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7777	7777	7777	7777	7777	7777	7777	7777
Long. Pl (SUS)	-1228	-1228	1516	1516	-1228	-1228	1516	1516
Long. Q (SUS)	-552	552	2688	-2688	-2000	2000	4136	-4136
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	179	179	-179	-179	-179	-179	179	179
Shear Q (SUS)	399	399	399	399	399	399	399	399
Pm (SUS)	15561	33818	15561	33818	15561	33818	15561	33818



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Nozzle Calcs.: N4- Shell outle Nozl: 11 4:31pm Apr 10,2026

Pm+Pl (SUS)	15371	33626	16268	34522	15654	33909	15985	34239
Pm+Pl+Q (Total)	15403	33637	16275	34522	15655	33909	16105	34249

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	33818	137989	Passed
Pm+Pl (SUS)	34522	206984	Passed
Pm+Pl+Q (TOTAL)	34522	413968	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N1- Tube inlet

Nozl: 12 4:31pm Apr 10,2026

Input, Nozzle Desc: N1- Tube inlet

From: 90

Pressure for Reinforcement Calculations	P	1.5746	N/mm ²
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.10	N/mm ²
Temperature for External Pressure	Tempex	82	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	137989.50	kPa
Parent Allowable Stress At Ambient	Sva	137989.50	kPa

Inside Diameter of Cylindrical Shell	D	1960.00	mm
Design Length of Section	L	2000.0304	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		19.34	m
-----------------------------------	--	-------	---

User Entered Minimum Design Metal Temperature		-28.89	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	117991.02	kPa
Allowable Stress At Ambient	Sna	117991.02	kPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		30.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137989.50	kPa
Pad Allowable Stress At Ambient	Spa	137989.50	kPa
Diameter of Pad along vessel surface	Dp	1162.0000	mm
Thickness of Pad	te	24.0000	mm
Weld leg size between Pad and Shell	Wp	20.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	24.0000	mm
Reinforcing Pad Width		180.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was MAWP + static head (bottom of element).



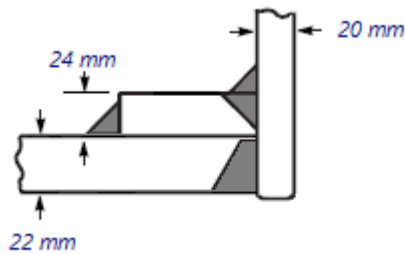
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Nozzle Calcs.: N1- Tube inlet

Nozl: 12 4:31pm Apr 10,2026



Reinforcement CALCULATION, Description: N1- Tube inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	30.000 in.
Actual Thickness Used in Calculation	0.787 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.57 \cdot 983 / (137990 \cdot 1 - 0.6 \cdot 1.57)$
 $= 11.2938 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.57 \cdot 384 / (117991 \cdot 1 - 0.6 \cdot 1.57)$
 $= 5.1656 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.5711 mm

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1536.0000 mm
Parallel to Vessel Wall, opening length	d	768.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		47.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 117991 / 137990)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 117991 / 137990)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 137990 / 137990)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



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Nozzle Calcs.: N1- Tube inlet

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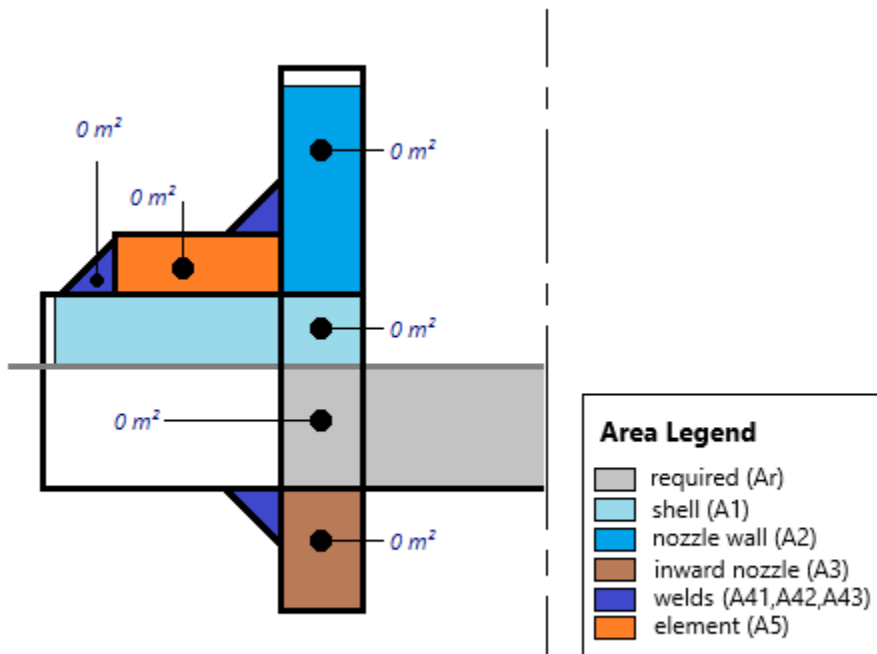


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (m²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	0.009	0.002	NA
Area in Shell	A1	0.006	0.010	NA
Area in Nozzle Wall	A2	0.001	0.001	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.001	0.001	NA
Area in Element	A5	0.009	0.009	NA
TOTAL AREA AVAILABLE	Atot	0.016	0.021	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness

Based on given Pad Thickness: 849.7318 24.0000 mm

Based on given Pad Diameter: 1162.0000 3.1821 mm

Based on Shell or Nozzle Thickness: 859.2783 20.0000 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (768 \cdot 11.3 \cdot 1 + 2 \cdot 17 \cdot 11.3 \cdot 1 \cdot (1 - 0.86))$$

$$= 0.009 \text{ m}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 768(1 \cdot 19 - 1 \cdot 11.3) - 2 \cdot 17$$

$$(1 \cdot 19 - 1 \cdot 11.3) \cdot (1 - 0.86)$$



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Nozzle Calcs.: N1- Tube inlet Nozl: 12 4:31pm Apr 10, 2026

$$= 0.006 \text{ m}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\ &= (2 \cdot 47.5) \cdot (17 - 5.17) \cdot 0.86 \\ &= 0.001 \text{ m}^2 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned} &= Wo^2 \cdot fr3 + (Wi-can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\ &= 20^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 20^2 \cdot 1 \\ &= 0.001 \text{ m}^2 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned} &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\ &= (1162 - 802) \cdot 24 \cdot 1 \\ &= 0.009 \text{ m}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 8.1656 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 14.2938 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 14.2938 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.7220 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[11.3, \max(14.3, 4.5)] \\ &= 11.3312 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(8.17, 11.3) \\ &= 11.3312 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 29874.4,	Allowable	: 117991.0 kPa	Passed
Expansion	: 0.0,	Allowable	: 265103.1 kPa	Passed
Occasional	: 17397.8,	Allowable	: 156928.1 kPa	Passed
Shear	: 9725.1,	Allowable	: 82593.7 kPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 20, tr = 5.17, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.3, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 20, tr = 5.17, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.3, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
--	-------



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Nozzle Calcs.: N1- Tube inlet Nozl: 12 4:31pm Apr 10, 2026

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 20, tr = 5.17, c = 3 mm, E* = 1
Thickness Ratio = $tr * E*/(tg - c) = 0.3$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 22, tr = 11.3, c = 3 mm, E* = 1
Thickness Ratio = $tr * E*/(tg - c) = 0.59$, Temp. Reduction = 23 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
Min Metal Temp. at Required thickness (UCS 66.1) -28 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 20, tr = 5.17, c = 3 mm, E* = 1
Thickness Ratio = $tr * E*/(tg - c) = 0.3$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -29 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $1.57/1.96 = 0.803$

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1152.0000 mm
Parallel to Vessel Wall, opening length d 576.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp 47.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (m²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	0.006	0.006
Area Available in Shell	A1	0.003	0.003
Area Available in Nozzle Wall	A2	0.001	0.001
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	0.000	0.000
Area Available in Pad	A5	0.000	0.008
Area Available in Hub	A6	0.000	0.000
TOTAL AREA AVAILABLE	Atot	0.004	0.013

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.



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Nozzle Calcs.: N1- Tube inlet Nozl: 12 4:31pm Apr 10,2026

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: N1- Tube inlet

Intermediate Calc. for nozzle/shell Welds Tmin 17.0000 mm
Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	14.1400 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (0.0087 - 0.0059 + 2 \cdot 17 \cdot 0.86 \cdot (1 \cdot 19 - 11.3)) \cdot 137990)$$

$$= 424024.84 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (0.00096 + 0.0086 + 0.00074 - 0 \cdot 0.86) \cdot 137990$$

$$= 1427236.12 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (0.00096 + 0 + 0.00034 + (0.00055)) \cdot 137990$$

$$= 256064.91 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (0.00096 + 0 + 0.00074 + 0.0086 + (0.00055)) \cdot 137990$$

$$= 1503456.25 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 802 \cdot 20 \cdot 0.49 \cdot 117991$$

$$= 1456658. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 1162 \cdot 20 \cdot 0.49 \cdot 137990$$

$$= 2468234. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 392) \cdot (20 - 3) \cdot 0.7 \cdot 117991$$

$$= 1731305. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 802 \cdot 24 \cdot 0.74 \cdot 137990$$

$$= 3087248. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 802 \cdot (14 - 0) \cdot 0.74 \cdot 137990$$

$$= 1800895. \text{ N}$$



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Nozzle Calcs.: N1- Tube inlet

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See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (2468234 + 1731305) = 4199539 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (1456657 + 3087247 + 1800894 + 0) = 6344800 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (2468234 + 1800894 + 0) = 4269129 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 4199539 N , must exceed W = 424024 N or W1 = 1427236 N

Path 2-2 = 6344800 N , must exceed W = 424024 N or W2 = 256064 N

Path 3-3 = 4269129 N , must exceed W = 424024 N or W3 = 1503456 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.23 N/mm²

Nozzle is O.K. for the External Pressure: 0.1 N/mm²

The Drop for this Nozzle is : 85.7970 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 357.7970 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N1- Tube inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1960.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 137989.50	kPa
Vessel Hot S.I. Allowable	Smh 137989.50	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	762.000	mm
Nozzle Thickness	20.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 117991.02	kPa
Nozzle Hot S.I. Allowable	SNmh 117991.02	kPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	1162.000	mm
Thickness of Reinforcing Pad	24.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:



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Nozzle Calcs.: N1- Tube inlet Nozl: 12 4:31pm Apr 10,2026

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	1.57	N/mm ²
Radial Load	P	-56227.61	N
Circumferential Shear	Vc	37488.32	N
Longitudinal Shear	Vl	37488.32	N
Circumferential Moment	Mc	78752.88	N-m
Longitudinal Moment	Ml	78752.88	N-m
Torsional Moment	Mt	112498.58	N-m

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 802.0 + 2 \cdot 1.65 \cdot \sqrt{1004.5(22.0 - 3.0)}$$

$$= 1257.896 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2009.000 mm
Nozzle Outside Diameter	(d) =	802.000 mm
Vessel Thickness used	(T) =	43.000 mm
Nozzle Thickness used	(t) =	17.000 mm
T / t	=	2.529
d / t	=	47.176

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.729$$

Nr/P	=	0.042
Mr/P	=	0.061
M0/P	=	0.042
N0/P	=	0.167
MrD/Mc	=	0.151
NrDL/Mc	=	0.084
M0d/Mc	=	0.122
N0DL/Mc	=	0.247
MrD/Ml	=	0.071
NrDL/Ml	=	0.044
M0D/Ml	=	0.052
N0DL/Ml	=	0.314

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5088	5088	5088	5088
Outplane Bending (P)	7673	-7673	7673	-7673
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-16650	-16650	16650	16650
Outplane Bending (Ml)	-16416	16416	16416	-16416
Normal Pressure Stress	35217	36796	35217	36796
Outplane Stress Summary	14913	33976	81046	34445

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Nozzle Calcs.: N1- Tube inlet Nozl: 12 4:31pm Apr 10,2026

Inplane Membrane (P)	1289	1289	1289	1289
Inplane Bending (P)	11196	-11196	11196	-11196
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2351	-2351	2351	2351
Inplane Bending (ML)	-22607	22607	22607	-22607
Inplane Pressure Stress	17608	17608	17608	17608

Inplane Stress Summary	5136	27957	55053	-12555
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Vessel Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	689	689	-689	-689
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2585	2585	2585	2585

Shear Stress Summary	3274	3274	1896	1896
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Vessel Stresses (kPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	15905	35410	81177	47152
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	5088	5088	5088	5088
Outplane Bending (P)	7673	-7673	7673	-7673
Outplane Membrane (Mc)	-13127	-13127	13127	13127
Outplane Bending (Mc)	-38995	38995	38995	-38995
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	17608	17608	17608	17608

Outplane Stress Summary	-21752	40891	82493	-10845
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	1289	1289	1289	1289
Inplane Bending (P)	11196	-11196	11196	-11196
Inplane Membrane (Mc)	-4447	-4447	4447	4447
Inplane Bending (Mc)	-48048	48048	48048	-48048
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	35217	36796	35217	36796

Inplane Stress Summary	-4791	70490	100199	-16712
------------------------	-------	-------	--------	--------

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
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Nozzle Calcs.: N1- Tube inlet Nozl: 12 4:31pm Apr 10,2026

Outplane Shear (V1)	-689	-689	689	689
Torsional Shear (Mt)	2585	2585	2585	2585

Shear Stress Summary	1896	1896	3274	3274
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	21959	70607	100785	18174
------------------------	-------	-------	--------	-------

Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	5088	5088	5088	5088
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-16650	-16650	16650	16650
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	41753	43642	41753	43642

Hoop Stress Summary	30191	32080	63492	65381
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	1337	1337	1337	1337
Axial Bending (P)	46876	-46876	46876	-46876
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-9769	-9769	9769	9769
Axial Bending (ML)	-99475	99475	99475	-99475
Axial Pressure Stress	20876	20876	20876	20876

Axial Stress Summary	-40154	65043	178335	-114367
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	1744	1744	-1744	-1744
Shear due to (V1)	0	0	0	0
Shear due to Torsion	6542	6542	6542	6542

Shear Stress Summary	8287	8287	4798	4798
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Nozzle Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	72269	67008	178535	180004
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	5088	5088	5088	5088
Hoop Bending (P)	0	0	0	0



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Nozzle Calcs.: N1- Tube inlet Nozl: 12 4:31pm Apr 10,2026

Hoop Membrane (Mc)	-13127	-13127	13127	13127
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	41753	43642	41753	43642

Hoop Stress Summary	33714	35603	59969	61858
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to pipe cross-section)	Left Outside	Left Inside	Right Outside	Right Inside

Axial Membrane (P)	1337	1337	1337	1337
Axial Bending (P)	46876	-46876	46876	-46876
Axial Membrane (Mc)	-9769	-9769	9769	9769
Axial Bending (Mc)	-221957	221957	221957	-221957
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	20876	20876	20876	20876

Axial Stress Summary	-162636	187526	300818	-236850
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1744	-1744	1744	1744
Shear due to Torsion	6542	6542	6542	6542

Shear Stress Summary	4798	4798	8287	8287
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Nozzle Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	196578	187671	301100	299163
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 802.0 + 2 \cdot 1.65 \cdot \sqrt{992.5(22.0 - 3.0)}$$

$$= 1255.165 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1985.000 mm
Nozzle Outside Diameter	(d) =	1162.000 mm
Vessel Thickness used	(T) =	19.000 mm
Nozzle Thickness used	(t) =	197.000 mm
T / t	=	0.096
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 5.983$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.044
Mr/P	=	0.070
M0/P	=	0.020
N0/P	=	0.030
MrD/Mc	=	0.240



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Nozzle Calcs.: N1- Tube inlet

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NrDL/Mc = 0.093
MØd/Mc = 0.069
NØDL/Mc = 0.055
MrD/Ml = 0.044
NrDL/Ml = 0.040
MØD/Ml = 0.014
NØDL/Ml = 0.026

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	4640	4640	4640	4640
Outplane Bending (P)	18394	-18394	18394	-18394
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4874	-4874	4874	4874
Outplane Bending (ML)	-15519	15519	15519	-15519
Normal Pressure Stress	80680	82252	80680	82252
Outplane Stress Summary	83321	79143	124109	57852

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	6915	6915	6915	6915
Inplane Bending (P)	65609	-65609	65609	-65609
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7439	-7439	7439	7439
Inplane Bending (ML)	-49454	49454	49454	-49454
Inplane Pressure Stress	40340	40340	40340	40340
Inplane Stress Summary	55970	23662	169758	-60369

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	1075	1075	-1075	-1075
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2785	2785	2785	2785
Shear Stress Summary	3860	3860	1709	1709

Vessel Stresses (kPa):

LONGITUDINAL PLANE (Stress Intensities	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	83852	79405	169820	118269

Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	4640	4640	4640	4640
Outplane Bending (P)	18394	-18394	18394	-18394
Outplane Membrane (Mc)	-10252	-10252	10252	10252
Outplane Bending (Mc)	-77991	77991	77991	-77991



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Nozzle Calcs.: N1- Tube inlet Nozl: 12 4:31pm Apr 10,2026

Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	40340	40340	40340	40340

Outplane Stress Summary	-24868	94325	151619	-41153
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	6915	6915	6915	6915
Inplane Bending (P)	65609	-65609	65609	-65609
Inplane Membrane (Mc)	-17443	-17443	17443	17443
Inplane Bending (Mc)	-270151	270151	270151	-270151
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	80680	82252	80680	82252

Inplane Stress Summary	-134389	276266	440799	-229148
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1075	-1075	1075	1075
Torsional Shear (Mt)	2785	2785	2785	2785

Shear Stress Summary	1709	1709	3860	3860
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Vessel Stresses (kPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	134410	276280	440847	229224
------------------------	--------	--------	--------	--------

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	35217	36796	35217	36796	35217	36796	35217	36796
Circ. Pl (SUS)	-11562	-11562	21738	21738	-3158	-3158	5736	5736
Circ. Q (SUS)	-8743	8743	24089	-24089	-36852	36852	59244	-59244
Long. Pm (SUS)	17608	17608	17608	17608	17608	17608	17608	17608
Long. Pl (SUS)	-1062	-1062	3640	3640	-8039	-8039	18215	18215
Long. Q (SUS)	-11411	11411	33803	-33803	-31322	31322	46668	-46668
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	689	689	-689	-689	-689	-689	689	689
Shear Q (SUS)	2585	2585	2585	2585	2585	2585	2585	2585
Pm (SUS)	35217	36796	35217	36796	35217	36796	35217	36796
Pm+Pl (SUS)	23721	25288	56968	58546	32080	33657	41043	42602
Pm+Pl+Q (Total)	15907	35414	81181	47152	21962	70610	100783	18174



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Nozzle Calcs.: N1- Tube inlet

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Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	36796	137989	Passed
Pm+Pl (SUS)	58546	206984	Passed
Pm+Pl+Q (TOTAL)	100783	413968	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	80680	82252	80680	82252	80680	82252	80680	82252
Circ. Pl (SUS)	-234	-234	9514	9514	-10528	-10528	24358	24358
Circ. Q (SUS)	2875	-2875	33913	-33913	-204542	204542	335760	-335760
Long. Pm (SUS)	40340	40340	40340	40340	40340	40340	40340	40340
Long. Pl (SUS)	-524	-524	14354	14354	-5612	-5612	14892	14892
Long. Q (SUS)	16155	-16155	115063	-115063	-59597	59597	96385	-96385
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1075	1075	-1075	-1075	-1075	-1075	1075	1075
Shear Q (SUS)	2785	2785	2785	2785	2785	2785	2785	2785
Pm (SUS)	80680	82252	80680	82252	80680	82252	80680	82252
Pm+Pl (SUS)	80474	82045	90226	91797	70184	71755	105061	106632
Pm+Pl+Q (Total)	83855	79410	169820	118271	134416	276282	440849	229229

Vessel Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	82252	137989	Passed
Pm+Pl (SUS)	106632	206984	Passed
Pm+Pl+Q (TOTAL)	440849	413968	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	41753	43642	41753	43642	41753	43642	41753	43642
Circ. Pl (SUS)	-11562	-11562	21738	21738	-8039	-8039	18215	18215
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	20876	20876	20876	20876	20876	20876	20876	20876
Long. Pl (SUS)	-8432	-8432	11106	11106	-8432	-8432	11106	11106
Long. Q (SUS)	-52599	52599	146351	-146351	-175081	175081	268833	-268833



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Nozzle Calcs.: N1- Tube inlet

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Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1744	1744	-1744	-1744	-1744	-1744	1744	1744
Shear Q (SUS)	6542	6542	6542	6542	6542	6542	6542	6542
Pm (SUS)	41753	43642	41753	43642	41753	43642	41753	43642
Pm+Pl (SUS)	30360	32233	63587	65470	33856	35733	60076	61958
Pm+Pl+Q (Total)	72271	67008	178533	180004	196585	187676	301099	299167

Nozzle Stress Summation Comparison (kPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	43642	117991	Passed
Pm+Pl (SUS)	65470	176986	Passed
Pm+Pl+Q (TOTAL)	301099	353973	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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ASME TS Calc: DESIGN CASE-1

Case: 1 4:31p Apr 10,2026

Input Echo, Tubesheet Number 1, Description: tubesheet

Shell Data:

Main Shell Description: shell

Shell Maximum Design Pressure	Psd,max	18.00 N/mm ²
Shell Maximum Operating Pressure	Psox,max	15.78 N/mm ²
Shell Minimum Operating Pressure	Psox,min	0.00 N/mm ²
Shell Thickness	ts	145.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1960.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	150.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	137989.50 kPa
Shell Allowable Stress at Ambient		137989.50 kPa
Shell Allowable Stress at Operating Temperature		137989.50 kPa

Channel Data:

Channel Description: Channel - rear

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	1.00 N/mm ²
Channel Maximum Operating Pressure	Ptox,max	0.50 N/mm ²
Channel Minimum Operating Pressure	Ptox,min	0.00 N/mm ²
Channel Thickness	tc	22.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1960.000 mm
Channel Design Temperature	TEMPC	82.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	137989.50 kPa
Channel Allowable Stress at Ambient		137989.50 kPa
Channel Allowable Stress at Operating Temperature		137989.50 kPa

Tube Data:

Number of Tube Holes	Nt	4267
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	16000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		15290.00 mm
Design Temperature of the Tubes		82.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92392.97 kPa
Tube Allowable Stress At Ambient		92392.97 kPa
Tube Allowable Stress at Operating Temperature		92392.97 kPa
Tube Yield Stress At design Temperature	Syt	165559.95 kPa
Tube Pitch (Center to Center Spacing)	P	26.0000 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	3.0000 mm
Groove Weld Leg	ag	3.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3



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Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	889.717 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	200.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	2.4 m ²
Perimeter of Tube Layout	Cp	5774.2 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	82.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	137989.50 kPa
Tubesheet Allowable Stress at Ambient	Tt	137989.50 kPa
Tubesheet Allowable Stress at Operating Temper.		137989.50 kPa
Thickness of Tubesheet	h	355.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	2250.000 mm

Area of the Untubed Lanes	AL	0.000 m ²
---------------------------	----	----------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Use Operating Temperatures for Thermal cases		Yes
Shell Operating Temperature	Tsop	40.00 °C
Channel Operating Temperature	Tcop	49.00 °C
Tube Operating Temperature	Ttop	49.00 °C
Tubesheet Operating Temperature	Tshop	49.00 °C
Mean Shell Metal Temp. along Shell len.	Tsm	40.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	49.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



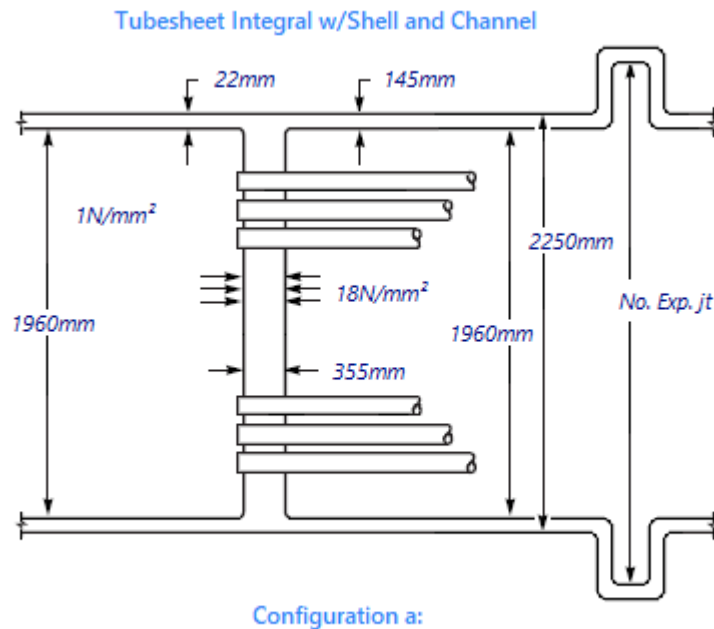
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ASME TS Calc: DESIGN CASE-1

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Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C> 0.3%

Th. Exp. Coeff. Metal Temp. along Len	40.0 °C	0.0000117134 /°C
Elastic Mod. at Design Temperature	150.0 °C	0.19370E+06 N/mm ²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	40.0 °C	0.20025E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20133E+06 N/mm ²
Elastic Mod. at Operating Temperature	40.0 °C	0.20025E+06 N/mm ²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	82.0 °C	0.19923E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20271E+06 N/mm ²
Elastic Mod. at Operating Temperature	49.0 °C	0.20112E+06 N/mm ²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	49.0 °C	0.0000117717 /°C
Elastic Mod. at Design Temperature	82.0 °C	0.19923E+06 N/mm ²
Elastic Mod. at Metal Temp. along Len	49.0 °C	0.20112E+06 N/mm ²
Elastic Mod. at Tubsht. Design Temp.	82.0 °C	0.19923E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20271E+06 N/mm ²
Elastic Mod. at Operating Temperature	49.0 °C	0.20112E+06 N/mm ²
Elastic Mod. at Tubsht. Operat. Temp.	49.0 °C	0.20112E+06 N/mm ²



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TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	82.0 °C	0.19923E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20271E+06 N/mm ²
Elastic Mod. at Operating Temperature	49.0 °C	0.20112E+06 N/mm ²

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the

type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1 \cdot (19/2 - 0)) / (92393 \cdot 1 + 0.4 \cdot 1)$$

$$= 0.1026 + 0.0000 = 0.1026 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$

$$= 15290 / \sqrt{(9.52 \cdot 2.11)}$$

$$= 3410.624$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 2 \cdot (D_o/t)^{0.94}, Ch = 0.55 / (D_o/t)$$

$$= 0.55 \cdot (19/2.11)$$

$$= 0.0609$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$

$$= 1.6 \cdot 0.061 \cdot 199226 (2.11/19)$$

$$= 2150689.750 \text{ kPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 2150689.8 / 165560.0$$

$$= 12.990$$

Predicted Buckling Stress [Fic]:

$$\text{Since } Fr \geq 2.439, Fic = F_y$$

$$= 165559.953 \text{ kPa}$$

Factor of Safety [FS]:

$$\text{Since } Fic [165560] \geq F_y [165560], FS = 1.667$$

Allowable Circumferential Compressive Stress [Fha]:

$$\text{Since } Fr > 2.439, Fha = F_y / FS$$

$$= 165559.953 / 1.667$$

$$= 99316.109 \text{ kPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha (t / D_o)$$

$$= 2 \cdot 99.3 (2.11/19)$$

$$= 22.002 \text{ N/mm}^2$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.7263 mm
Allowable Internal Pressure at Corroded thickness	22.46 N/mm ²
Required Internal Design Pressure	1.00 N/mm ²
Allowable External Pressure at Corroded thickness	22.00 N/mm ²
Required External Design Pressure	18.00 N/mm ²



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Required Thickness due to Shell Side pressure 1.7263 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio $[\rho]$:

= l_{tx} / h (modified for corrosion if present)

= 200/355

= 0.5634 (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter $[d^*]$:

= $\max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$

= $\max(19 - 2 \cdot 2.11 (199226/199226) \cdot$

$(92393/137990) \cdot (0.56); 19 - 2 \cdot 2.11)$

= 17.4581 mm

Equivalent outer tube limit circle diameter $[Do]$:

= $2 \cdot ro + dt$

= $2 \cdot 890 + 19$

= 1798.484 mm

Determine the basic ligament efficiency for shear $[\mu]$:

= $(p - dt) / p$

= $(26 - 19) / 26$

= 0.267

Equivalent outer tube limit radius $[ao]$:

= $Do / 2$

= 1798/2

= 899.2420 mm

Effective tube pitch $[p^*]$:

= $p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$

= $26 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 1000248; 4 \cdot 1798 \cdot 26) / (\pi \cdot 1798^2)) }$

= 26.0000 mm

Effective ligament efficiency for bending $[\mu^*]$:

= $(p^* - d^*) / p^*$

= $(26 - 17.5) / 26$

= 0.3285

Ratio $[ps]$:

= as / ao

= 980/899

= 1.0898

Ratio $[pc]$:

= ac / ao

= 980/899

= 1.0898

Compute parameter $[xt]$:

= $1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$

= $1 - 4267 ((19 - 2 \cdot 2.11) / (2 \cdot 899))^2$

= 0.7099

Determine parameter $[xs]$:



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$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 4267 (19 / (2 \cdot 899))^2$$

$$= 0.5213$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts(Ds + ts)Es/L$$

$$= \pi \cdot 145 (1960 + 145)193695/15290$$

$$= 12146300.0000 \text{ N/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt(Dt - tt)Et/L$$

$$= \pi \cdot 2.11 (19 - 2.11)199226/15290$$

$$= 1463.0074 \text{ N/mm}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 12146300 / (4267 \cdot 1463)$$

$$= 1.94570$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks/Kt)$$

$$= 1 / (1 + 12146300/0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [βs]:

$$= ((12(1 - vs^2))^{1/4}) / ((Ds + ts)ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1960 + 145)145)$$

$$= 0.0033 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta s \cdot Es \cdot ts^3 / (6(1 - vs^2))$$

$$= 0.0033 \cdot 193695 \cdot 145^3 / (6(1 - 0.3^2))$$

$$= 355856928.0000 \text{ N/mm}^2 \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(Ds + ts) \cdot ks/h^3 \cdot (1 + h \cdot \beta s + (h \cdot \beta s)^2/2)$$

$$= 6(1960 + 145) \cdot 355856928/355^3 \cdot (1 + 355 \cdot 0.0033 + (355 \cdot 0.0033)^2/2)$$

$$= 286340.8125 \text{ N/mm}^2$$

Shell coefficient [δs]:

$$= Ds^2 / (4 \cdot Es \cdot ts) \cdot (1 - Ds \cdot vs / (2(Ds + ts)))$$

$$= 1960^2 / (4 \cdot 193695 \cdot 145) \cdot (1 - 1960 \cdot 0.3 / (2(1960 + 145)))$$

$$= 0.29421\text{E-}04 \text{ mm/kPa}$$

Channel coefficient [βc]:

$$= ((12(1 - vc^2))^{1/4}) / ((Dc + tc)tc)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1960 + 22)22)$$

$$= 0.0087 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta c \cdot Ec \cdot tc^3 / (6(1 - vc^2))$$

$$= 0.0087 \cdot 199226 \cdot 22^3 / (6(1 - 0.3^2))$$

$$= 3382307.7500 \text{ N/mm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:



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$$\begin{aligned}
 &= 6(Dc + tc/\cos(\theta)) \cdot kc/h^3 \cdot (Cm\theta + h \cdot \beta c \cdot (Cm\Delta + Cq\theta)/2 + (h \cdot \beta c)^2/2 \cdot Cq\Delta) \\
 &= 6(1960 + 22/\cos(0)) \cdot 3382307/355^3 \cdot (1 + 355 \cdot 0.0087 \cdot (1 + 1)/2 + (355 \cdot 0.0087)^2/2 \cdot 1) \\
 &= 7970.8 \text{ N/mm}^2
 \end{aligned}$$

Channel coefficient [δc]:

$$\begin{aligned}
 &= Dc^2/(4 \cdot Ec \cdot Tc) \cdot (1 - Dc \cdot vc/(2(Dc + tc))) \\
 &= 1960^2/(4 \cdot 199226 \cdot 22) \cdot (1 - 1960 \cdot 0.3/(2(1960 + 22))) \\
 &= 0.18663\text{E-}03 \text{ mm/kPa}
 \end{aligned}$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$\begin{aligned}
 h/p &= 13.653847 ; \mu^* = 0.328533 ; E^* = 63501 \text{ N/mm}^2 \\
 E^*/E &= 0.318741 ; \nu^* = 0.336622
 \end{aligned}$$

Note: As h/p (13.654) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [Xa]:

$$\begin{aligned}
 &= ((24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2)/(E^* \cdot L \cdot H^3))^{1/4} \\
 &= ((24(1 - 0.34^2)4267 \cdot 199226 \cdot 2.11 (19 - 2.11)899^2)/(63501 \cdot 15290 \cdot 355^3))^{1/4} \\
 &= 1.8626
 \end{aligned}$$

Values from Table 4.18.3

$$\begin{aligned}
 Z_d &= 0.216484 ; Z_v = 0.166592 ; Z_m = 0.686789 \\
 Z_a &= 0.720801\text{E}+00 ; Z_w = 0.166592
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$\begin{aligned}
 &= A / Do \\
 &= 2250/1798 \\
 &= 1.2511
 \end{aligned}$$

Compute coefficient [F]:

$$\begin{aligned}
 &= (1 - \nu^*)/ E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\
 &= (1 - 0.34)/63501 \cdot (286341 + 7971 + 199226 \cdot \ln(1.25)) \\
 &= 3.5407
 \end{aligned}$$

Compute parameter [ϕ]:

$$\begin{aligned}
 &= (1 + \nu^*)F \\
 &= (1 + 0.34)3.54 \\
 &= 4.7326
 \end{aligned}$$

Compute parameter [Q1]:

$$\begin{aligned}
 &= (\rho_s - 1 - \phi \cdot Z_v)/(1 + \phi \cdot Z_m) \\
 &= (1.09 - 1 - 4.73 \cdot 0.17)/(1 + 4.73 \cdot 0.69) \\
 &= -0.164367
 \end{aligned}$$

Compute Parameter [Qz1]:

$$\begin{aligned}
 &= (Z_d + Q1 \cdot Z_w)/2 \cdot Xa^4 \\
 &= (0.22 + -0.16 \cdot 0.17)/2 \cdot 1.86^4 \\
 &= 1.1381
 \end{aligned}$$

Compute parameter [Qz2]:

$$\begin{aligned}
 &= (Z_v + Q1 \cdot Z_m)/2 \cdot Xa^4 \\
 &= (0.17 + -0.16 \cdot 0.69)/2 \cdot 1.86^4 \\
 &= 0.3232
 \end{aligned}$$

Compute parameter [U]:



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$$\begin{aligned}
 &= (Zw + (ps - 1)Zm)Xa^4 / (1 + \phi \cdot Zm) \\
 &= (0.17 + (1.09 - 1)0.69)1.86^4 / (1 + 4.73 \cdot 0.69) \\
 &= 0.6465
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

$$= 0 \text{ (config a)}$$

Compute Parameter [y]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ws]:

$$\begin{aligned}
 &= ps \cdot ks \cdot \beta s \cdot \delta S(1 + h \cdot \beta s) \\
 &= 1.09 \cdot 355856928 \cdot 0.0033 \cdot 0.29421E-04 (1 + 355 \cdot 0.0033) \\
 &= 81391.4141 \text{ mm}^2
 \end{aligned}$$

Calculate parameter [w*s]:

$$\begin{aligned}
 &= ao^2(ps^2 - 1) \cdot (ps - 1) / 4 - ws \\
 &= 899^2(1.09^2 - 1) \cdot (1.09 - 1) / 4 - 81391 \\
 &= -77984.0625 \text{ mm}^2
 \end{aligned}$$

Parameter [wc]:

$$\begin{aligned}
 &= pc \cdot kc[\delta C(CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) - \\
 &\quad \zeta c(CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2)) \\
 &= 1.09 \cdot 3382307 [0.000186629 (1 + 0.0087 \cdot 1 \cdot 355) - \\
 &\quad 0.000 (1/355 + 1 \cdot 0.0087/2)) \\
 &= 24495.0566 \text{ mm}^2
 \end{aligned}$$

Parameter [w*c]:

$$\begin{aligned}
 &= ao^2[(pc^2 + 1) \cdot (pc - 1) / 4 - (ps - 1) / 2] - wc \\
 &= 899^2[(1.09^2 + 1) \cdot (1.09 - 1) / 4 - (1.09 - 1) / 2] - 24495 \\
 &= -21087.7070 \text{ mm}^2
 \end{aligned}$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$\begin{aligned}
 &= Ps \{ xs + 2(1 - xs)vt + [2/Kst \cdot (Ds/Do)^2]vs - \\
 &\quad [(ps^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)][(Dj^2 - (Ds)^2)/Do^2] \} \\
 &= 18 \{ 0.52 + 2(1 - 0.52)0.3 + [2/1.95 \cdot \\
 &\quad (1960/1798)^2]0.3 - [(1.09^2 - 1)/(1 \cdot 1.95)] - \\
 &\quad [(1 - 1)/(2 \cdot 1 \cdot 1.95)][(0^2 - (1960)^2)/1798^2] \} \\
 &= 19.4104 \text{ N/mm}^2
 \end{aligned}$$

Pressure [P't]:

$$\begin{aligned}
 &= [xt + 2(1 - xt)vt + 1/(J \cdot Kst)]Pt \\
 &= [0.71 + 2(1 - 0.71)0.3 + 1/(1 \cdot 1.95)]1 \\
 &= 1.3980 \text{ N/mm}^2
 \end{aligned}$$

Pressure [Py]:

$$\begin{aligned}
 &= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2) \\
 &= 4267 \cdot 1463 \cdot 0 / (3.14 \cdot 899^2) \\
 &= 0.0000 \text{ N/mm}^2
 \end{aligned}$$

Pressure [Pw]:

$$\begin{aligned}
 &= -yb \cdot U \cdot W* / (2 \cdot \pi \cdot ao^2) \\
 &= -0 \cdot 0.65 \cdot 0 / (2 \cdot \pi \cdot 899^2) \\
 &= -0.0000 \text{ N/mm}^2
 \end{aligned}$$



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Calculate the pressure [Prim]:

$$= -(U/ao^2) \cdot (\omega * s \cdot Ps - \omega * c \cdot Pt)$$

$$= -(0.65/899^2) \cdot (-77984 \cdot 18 - -21088 \cdot 1)$$

$$= 1.1054 \text{ N/mm}^2$$

Calculate the pressure [Pw]:

$$= (U/ao^2) \cdot (\omega s \cdot P * s - \omega c \cdot P * c)$$

$$= (0.65/899^2) \cdot (81391 \cdot 0 - 24495 \cdot 0)$$

$$= 0.0000 \text{ N/mm}^2$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst (Qz1 + (ps - 1) Qz2)) \cdot (Ps - Pt + Py + Pw + Prim)$$

$$= 0.1000E+01 \cdot 1.95 / (1 + 1 \cdot 1.95 (1.14 + (1.09 - 1) 0.32)) \cdot (19.4 - 1.4 + 0 + 0 + -0 + 1.11)$$

$$= 11.3722 \text{ N/mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega * s \cdot Ps - \omega * c \cdot Pt) - (\omega s \cdot P * s - \omega c \cdot P * c)) + W * \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$

$$= [((-77984 \cdot 18 - -21088 \cdot 1) - (81391 \cdot 0 - 24495 \cdot 0)) \cdot 1 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 4.73 \cdot 0.69)$$

$$= -325289.7500 \text{ N}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.16 + 2 \cdot -325290 / (11.4 \cdot 899^2)$$

$$= -0.23512$$

Fm Value from Table 4.18.3 = 0.117560

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu *) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.12 / 0.33) \cdot (2 \cdot 899 / (355 - 0))^2 \cdot 11.4$$

$$= 156655.5781 \text{ kPa}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 137990 = 206984 \text{ kPa}$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot Fm / \mu *) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.11 / 0.33) \cdot (2 \cdot 899 / (301 - 0))^2 \cdot 11.1$$

$$= 206983.8594 \text{ kPa}$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + Cats + Catc$$

$$= 301 + 0 + 0$$

$$= 301.1911 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / ao) \cdot \min(0.8S; 0.533Sy)$$

$$= (2 \cdot 0.27 \cdot 355 / 899) \cdot 110392$$

$$= 23298.5684 \text{ kPa}$$

Shear Stress is not required to be computed; [Pe] <= Tau[limit]

Tubesheet Shear Stress - final thickness [τ_f]:

$$= (Ap \cdot Pe) / (\mu \cdot h \cdot Cp)$$

$$= (2.44 \cdot 8.02) / (0.27 \cdot 120 \cdot 5774)$$

$$= 105919.6719 \text{ kPa}$$



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Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$\begin{aligned}
 &= H + C_{ats} + C_{atc} \\
 &= 120 + 0 + 0 \\
 &= 119.9229 \text{ mm}
 \end{aligned}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$\begin{aligned}
 &= \max(H_{reqB}; H_{reqS}) \\
 &= \max(301 ; 120) \\
 &= 301.1911 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$F_{tmin} = 0.87, F_{tmax} = 1.07$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$\begin{aligned}
 &= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s) \\
 &= ((18 \cdot 0.52 - 1 \cdot 0.71) - (11.4) \cdot 0.87) / \\
 &\quad (0.71 - 0.52) \\
 &= -6754.9238 \text{ kPa}
 \end{aligned}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$\begin{aligned}
 &= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s) \\
 &= ((18 \cdot 0.52 - 1 \cdot 0.71) - (11.4) \cdot 1.07) / \\
 &\quad (0.71 - 0.52) \\
 &= -18358.8848 \text{ kPa}
 \end{aligned}$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(| \sigma_{t,1} | ; | \sigma_{t,2} |) = 18359 \text{ kPa}$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= S_{ot} = 92392.9688 \text{ kPa}$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$\begin{aligned}
 &= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w) X_a^4); 1.25) \\
 &= \max((3.25 - 0.25(0.22 + -0.24 \cdot 0.17) 1.86^4); 1.25) \\
 &= 2.0000 \text{ (Should be } \leq 2 \text{)}
 \end{aligned}$$

Determine the factor [rt]:

$$\begin{aligned}
 &= (\sqrt{ dt^2 + (dt - 2 \cdot tt)^2 }) / 4 \\
 &= (\sqrt{ 19^2 + (19 - 2 \cdot 2.11)^2 }) / 4 \\
 &= 6.0355 \text{ mm}
 \end{aligned}$$

Determine the factor [Ct]:

$$\begin{aligned}
 &= \sqrt{ (2 \cdot \pi^2 \cdot E / S_{yt}) } \\
 &= \sqrt{ (2 \cdot \pi^2 \cdot 199226 / 165560) } \\
 &= 154.1158
 \end{aligned}$$

Determine the factor [Ft]:

$$\begin{aligned}
 &= k \cdot L / r \\
 &= 0 \cdot 0 / 6.04 \\
 &= 0.0000
 \end{aligned}$$

Buckling Allowable Stress [Stb]:

$$\begin{aligned}
 &= (S_y / F_s) \cdot (1 - F_t / (2 \cdot C_t)) \\
 &= (165560 / 2) \cdot (1 - 0 / (2 \cdot 154)) \\
 &= 82779.977 \text{ kPa} \text{ (Never greater than } S_{ot} \text{)}
 \end{aligned}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$



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$$= 18359 \cdot 0.00011$$

$$= 2061.49 \text{ N}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (19 - 2.11) 2.11$$

$$= 0.000 \text{ m}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (19 - 2.11) 92393$$

$$= 10374.639 \text{ N}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 3 (19 + 0.67 \cdot 3) 92393$$

$$= 10374.6387 \text{ N}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 92393 / 92393$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 3 (19 + 0.67 \cdot 3) 92393$$

$$= 10086.0107 \text{ N}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{\frac{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d}{0.75 \cdot d_o}} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{\frac{(0.75 \cdot 19)^2 + 1.07 \cdot 2.11 \cdot (19 - 2.11) 1 \cdot 1}{0.75 \cdot 19}} - 0.75 \cdot 19 \right)$$

$$= 2.562 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm

Minimum Required Groove Weld Leg [agr]: 1.2810 mm

Tube-Tubesheet Jt allowable, 10375 is $\geq$ tube strength 10375 N

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= S_s \cdot E_{sw} = 137990 \cdot 1 = 137990 \text{ kPa}$$

Axial Membrane Stress in Shell [σ_s ,m]:

$$= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 899^2 / ((1960 + 145) 145) \cdot [11.4 + (1.09^2 - 1)(18 - 1)] + 980^2 \cdot 1 / ((1960 + 145) 145)$$

$$= 41726.0078 \text{ kPa}$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 145 mm, the minimum Shell length adjacent to the tubesheet should be 959.587 mm

Shell Membrane Stress due to Joint Interaction [σ_s ,m]:



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$$= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1) (P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 899^2 / ((1960 + 145) 145) [11.4 + (1.09^2 - 1) (18 - 1)] + 980^2 \cdot 1 / (((1960 + 145) 145))$$

$$= 41726.0078 \text{ kPa}$$

Shell Bending Stress due to Joint Interaction $[\sigma_s, b]$:

$$= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P_s / (E_s \cdot t_s)] + 6(1 - \nu_s^2) / (E_s) (a_o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 355856928 / 145^2 \{ 0.0033 [0.29421E-04 \cdot 18 + 980^2 \cdot 0 / (28085776)] + 6(1 - 0.34^2) / (63501) (899 / 355)^3 (1 + 355 \cdot 0.0033 / 2) [11.4 (0.17 + 0.69 \cdot -0.16) + 2 / 899^2 \cdot 0.69 \cdot -325290] \}$$

$$= 189686.3906 \text{ kPa}$$

Shell Stress Summation vs. Allowable

$$:: |\sigma_{s,m}| + |\sigma_{s,b}| \leq 1.5 \cdot S_s$$

$$:: |41726| + |189686| \leq 206984 \text{ kPa}$$

$$:: 231412, \text{ must be } \leq 206984 \text{ kPa (See Elastic-Plastic iteration below)}$$

Note:

For a given Channel thickness of 22 mm, the minimum Channel length adjacent to the tubesheet should be 373.776 mm

Channel Membrane Stress due to Joint Interaction $[\sigma_c, m]$:

$$= a_c^2 \cdot P_t / ((D_c + t_c) t_c \cdot \cos(\theta))$$

$$= 980^2 \cdot 1 / ((1960 + 22) 22 \cdot \cos(0))$$

$$= 22025.502 \text{ kPa}$$

Channel Bending Stress due to Joint Interaction $[\sigma_c, b]$:

$$= 6 \cdot k_c / t_c^2 \cdot \{ (\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6(1 - \nu_c^2) / E_c \cdot (a_o / h)^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 3382307 / 22^2 \cdot \{ (0.000186629 \cdot 1 \cdot 0.0087 - 0.000 \cdot 1) \cdot 1 - 6(1 - 0.34^2) / 63501 \cdot (899 / 355)^3 \cdot (1 + 355 \cdot 0.0087 / 2 \cdot 1) \cdot [11.4 (0.17 + 0.69 \cdot -0.16) + 2 / 899^2 \cdot 0.69 \cdot -325290] \}$$

$$= 59666.2188 \text{ kPa}$$

Channel Stress Summation vs. Allowable

$$:: |\sigma_{c,m}| + |\sigma_{c,b}| \leq 1.5 \cdot S_c$$

$$:: |22026| + |59666| \leq 206984 \text{ kPa}$$

$$:: 81692 \text{ must be } \leq 206984 \text{ kPa}$$

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	156655.6	<= 206984.2 kPa	Pass
Tubesheet Shear Stress	50749.0	<= 110391.6 kPa	Pass
Maximum Tube Stress	18358.9	<= 92393.0 kPa	Pass
Minimum Tube Stress (Buckling)	-18358.9	<= -82780.0 kPa	Pass
Maximum Force on any one Tube	2061.5	<= 10374.6 N	Pass
Axial Membrane Stress in Shell	41726.0	<= 137989.5 kPa	Pass
Shell Stress (jt. inter.)	231412.4	<= 206984.2 kPa	See below
Channel Stress (jt. inter.)	81691.7	<= 206984.2 kPa	Pass

Additional Results for the 2nd Elastic-Plastic Iteration.



These calculations are not intended to be used in the creep range. Check Part 4.18 in Section VIII Division 2.



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ASME TS Calc: DESIGN CASE-1

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Compute Allowable [Sstars]:

= min(Sys, SPS,s/2)

= min(231995, 463990/2) = 231995 kPa

Compute Allowable [S*c]:

= min(Syc, SPS,c/2)

= min(242595, 485189/2) = 242595 kPa

Compute Factor [facts]:

= min([1.4 - 0.4• abs(σs,b) / Sstars], 1)

= min([1.4 - 0.4• abs(189686)/231995], 1) = 1

Compute Factor [factc]:

= min([1.4 - 0.4• abs(σc,b) / Sstarc], 1)

= min([1.4 - 0.4• abs(59666)/242595], 1) = 1

Note: Since facts & factc are both >= 1, the design Passes.

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Shell Stress (jt. inter.)	231412.4 <=	463989.8 kPa	Pass
Channel Stress (jt. inter.)	81691.7 <=	206984.2 kPa	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	301.191	355.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	3.000	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	16.628 ...	10208	206984	3756 110392	Ps+Pt-Th D1	Ok
D2uc	311.651 ...	165479	206984	54505 110392	Ps+Pt-Th D2	Ok
D3uc	301.191 ...	156656	206984	50749 110392	Ps+Pt-Th D3	Ok
O1uc	17.303 ...	26002	485192	8991 110392	Ps+Pt+Th O1	Ok
O2uc	184.280 ...	169026	485192	58748 110392	Ps+Pt+Th O2	Ok
O3uc	180.071 ...	164592	485192	56871 110392	Ps+Pt+Th O3	Ok
O4uc	23.353 ...	31083	485192	10868 110392	Ps+Pt+Th O4	Ok
D1c	22.713 ...	10566	206984	3796 110392	Ps+Pt-Th-c D1	Ok
D2c	315.563 ...	168434	206984	55061 110392	Ps+Pt-Th-c D2	Ok
D3c	305.272 ...	159538	206984	51265 110392	Ps+Pt-Th-c D3	Ok
O1c	23.183 ...	26504	485192	9019 110392	Ps+Pt+Th-c O1	Ok
O2c	188.770 ...	171875	485192	59287 110392	Ps+Pt+Th-c O2	Ok
O3c	184.609 ...	167403	485192	57389 110392	Ps+Pt+Th-c O3	Ok
O4c	29.212 ...	31763	485192	10916 110392	Ps+Pt+Th-c O4	Ok

Max: 315.5628 ... mm 0.814 0.537 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures

derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the

Load Case Tables



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ASME TS Calc: DESIGN CASE-1

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[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass Fail
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	420	137989	...	...	...	...	...	...	...	Ok
D2uc	41306	137989	...	...	...	...	...	...	...	Ok
D3uc	41726	137989	...	...	...	...	...	...	...	Ok
O1uc	6662	523984	...	...	...	...	...	...	...	Ok
O2uc	42723	523984	...	...	...	...	...	...	...	Ok
O3uc	42934	523984	...	...	...	...	...	...	...	Ok
O4uc	6452	523984	...	...	...	...	...	...	...	Ok
D1c	442	137989	...	...	...	...	...	...	...	Ok
D2c	42246	137989	...	...	...	...	...	...	...	Ok
D3c	42689	137989	...	...	...	...	...	...	...	Ok
O1c	6718	523984	...	...	...	...	...	...	...	Ok
O2c	43594	523984	...	...	...	...	...	...	...	Ok
O3c	43816	523984	...	...	...	...	...	...	...	Ok
O4c	6496	523984	...	...	...	...	...	...	...	Ok
Max RATIO	0.309	...	...	...	...	...	...	...	...	...

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Pass Fail
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	1249	92392	-102	-82779	140	10374	8474	206984	95487	206984	Ok
D2uc	19608	92392	-19608	-82779	2202	10374	239047	463989	13796	206984	Ok
D3uc	18359	92392	-18359	-82779	2061	10374	231412	463989	81692	206984	Ok
O1uc	14058	184785	-14058	-87518	1579	20749	31493	523984	28987	512471	Ok
O2uc	31979	184785	-31979	-87518	3591	20749	245390	523984	30773	512471	Ok
O3uc	31358	184785	-31358	-87518	3521	20749	241542	523984	16928	512471	Ok
O4uc	14679	184785	-14679	-87518	1648	20749	35341	523984	18714	512471	Ok
D1c	1258	92392	-176	-82779	141	10374	9122	206984	110447	206984	Ok
D2c	19479	92392	-19479	-82779	2187	10374	248517	463989	16019	206984	Ok
D3c	18220	92392	-18220	-82779	2046	10374	240280	463989	94428	206984	Ok
O1c	13974	184785	-13974	-87518	1569	20749	32438	523984	36187	512471	Ok
O2c	31786	184785	-31786	-87518	3569	20749	254990	523984	32975	512471	Ok
O3c	31160	184785	-31160	-87518	3499	20749	250838	523984	22203	512471	Ok
O4c	14599	184785	-14599	-87518	1639	20749	36590	523984	18991	512471	Ok
Max RATIO	0.212	0.365	0.212	0.536	0.534	...	...	...	...	...	...

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
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ASME TS Calc: DESIGN CASE-1

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Tubesheet thickness :	315.563	355.000	Ok
Tube thickness :	1.726	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	3.000	Ok
Tube-tubesheet groove weld leg :	1.281	3.000	Ok

Min Shell length of thk, (145.000) adj. to tubesheet: 959.587 mm

Min Channel length of thk, (22.000) adj. to tubesheet: 373.776 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +1.637 mm
 Due to Pressure Shell Compresses by : -1.636 mm
 Due to Pressure + Thermal Shell Extends by : +1.650 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	19.591	1.000	22.157	1.000
Tubesheet Shear Stress	29.080	1.000	32.462	1.000
Tube Tensile Stress	73.419	1.000	84.820	1.000
Tube Compressive Stress	31.973	0.180	66.462	1.000
Tube-Tubesheet Joint load	73.420	1.000	84.820	1.000
Shell Stress (Axial, Junction)	5.081	0.100	22.002	0.993
Tubesheet-Channel Junction	4.393	1.000	22.002	0.993
Tube Pressure Stress	22.458	1.000	22.002	1.000
Minimum MAWP	4.393		22.002	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	20.359	1.000	22.475	1.000
Tubesheet Shear Stress	29.409	1.000	36.400	1.000
Tube Tensile Stress	74.345	1.000	84.357	1.000
Tube Compressive Stress	900.900	1.000	82.166	1.000
Tube-Tubesheet Joint load	74.345	1.000	84.357	1.000
Shell Stress (Axial, Junction)	20.359	0.334	22.475	0.575
Tubesheet-Channel Junction	5.435	0.997	22.475	0.083
Tube Pressure Stress	22.458	1.000	23.919	1.000
Minimum MAPnc	5.435		22.475	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0256	0.0256	...	...	...	...
2	-1.6357	-1.6363	...	...	...	...
3	-1.6101	-1.6107	...	...	...	...
4	...	...	...	...	...	...
5	0.0127	0.0127	1.6367	1.6374	...	...
6	-1.4209	-1.4214	1.6367	1.6374	...	...
7	-1.4082	-1.4087	1.6367	1.6374	...	...
8	...	...	1.6367	1.6374	...	...



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ASME TS Calc: DESIGN CASE-1

Case: 1 4:31p Apr 10, 2026

Tubesheet MDMT Calculations:**Shell Side MDMT calculation:**

Governing thickness on the shell side per figure UCS-66.3 (e):

$$\begin{aligned} &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness})) \\ &= \max(355/4, \min(355, 145)) \\ &= 145.000 \text{ mm} \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$\begin{aligned} &= \min(\text{ps}/\text{Shellside MAPnc}, 1) \\ &= \min(18/22.5, 1) \\ &= 0.801 \end{aligned}$$

Warning:

The minimum metal temperature could not be computed because the governing thickness for this component was greater than 100.000 mm. This component must be impact tested.

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

$$\begin{aligned} &= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness})) \\ &= \max(355/4, \min(355, 22)) \\ &= 88.750 \text{ mm} \end{aligned}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$\begin{aligned} &= \min(\text{pt}/\text{Tubeside MAPnc}, 1) \\ &= \min(1/5.44, 1) \\ &= 0.184 \end{aligned}$$

Thickness Ratio = 0.18, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

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ASME TS Calc: DESIGN CASE-2

Case: 2 4:31p Apr 10,2026

Input Echo, Tubesheet Number 2, Description: tubesheet

Shell Data:

Main Shell Description: shell

Shell Maximum Design Pressure	Psd,max	18.00 N/mm ²
Shell Maximum Operating Pressure	Psox,max	15.78 N/mm ²
Shell Minimum Operating Pressure	Psox,min	0.00 N/mm ²
Shell Thickness	ts	145.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1960.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	150.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	137989.50 kPa
Shell Allowable Stress at Ambient		137989.50 kPa

Channel Data:

Channel Description: Channel - rear

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	1.00 N/mm ²
Channel Maximum Operating Pressure	Ptox,max	0.50 N/mm ²
Channel Minimum Operating Pressure	Ptox,min	0.00 N/mm ²
Channel Thickness	tc	22.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1960.000 mm
Channel Design Temperature	TEMPC	82.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	137989.50 kPa
Channel Allowable Stress at Ambient		137989.50 kPa

Tube Data:

Number of Tube Holes	Nt	4267
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	16000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		15290.00 mm
Design Temperature of the Tubes		82.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92392.97 kPa
Tube Allowable Stress At Ambient		92392.97 kPa
Tube Yield Stress At design Temperature	Syt	165559.95 kPa
Tube Pitch (Center to Center Spacing)	P	26.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	3.0000 mm
Groove Weld Leg	ag	3.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	889.717 mm



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Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	200.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	2.4 m ²
Perimeter of Tube Layout	Cp	5774.2 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	82.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	137989.50 kPa
Tubesheet Allowable Stress at Ambient	Tt	137989.50 kPa
Thickness of Tubesheet	h	355.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	2250.000 mm

Area of the Untubed Lanes	AL	0.000 m ²
---------------------------	----	----------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)	Yes
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Mean Shell Metal Temp. along Shell len.	Tsm	40.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	49.00 °C
Junction Stress Reduction option	Perform Plastic Calculation	

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No
---	----



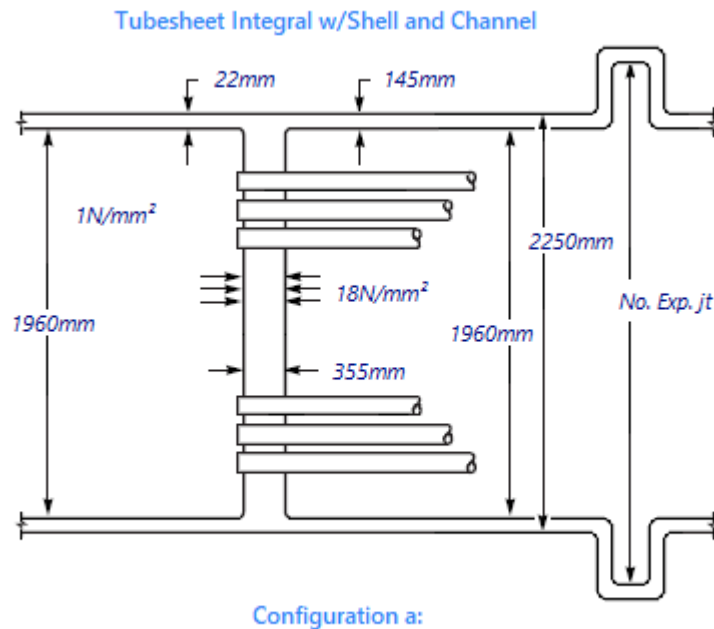
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Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C> 0.3%

Th. Exp. Coeff. Metal Temp. along Len	40.0 °C	0.0000117134 /°C
Elastic Mod. at Design Temperature	150.0 °C	0.19370E+06 N/mm²
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	40.0 °C	0.20025E+06 N/mm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20133E+06 N/mm²

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	82.0 °C	0.19923E+06 N/mm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20271E+06 N/mm²

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	49.0 °C	0.0000117717 /°C
Elastic Mod. at Design Temperature	82.0 °C	0.19923E+06 N/mm²
Elastic Mod. at Metal Temp. along Len	49.0 °C	0.20112E+06 N/mm²
Elastic Mod. at Tubsht. Design Temp.	82.0 °C	0.19923E+06 N/mm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20271E+06 N/mm²

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%



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Elastic Mod. at Design Temperature	82.0 °C	0.19923E+06 N/mm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20271E+06 N/mm ²

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$$
 per Appendix 1-1 (a)(1)

$$= (1 \cdot (19/2 - 0)) / (92393 \cdot 1 + 0.4 \cdot 1)$$

$$= 0.1026 + 0.0000 = 0.1026 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$$= L / \sqrt{(Ro \cdot t)}$$

$$= 15290 / \sqrt{(9.52 \cdot 2.11)}$$

$$= 3410.624$$

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (Do/t)0.94$, $Ch = 0.55 / (Do/t)$

$$= 0.55 \cdot (19/2.11)$$

$$= 0.0609$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot Ey (t / Do)$$

$$= 1.6 \cdot 0.061 \cdot 199226 (2.11/19)$$

$$= 2150689.750 \text{ kPa}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / Fy$$

$$= 2150689.8 / 165560.0$$

$$= 12.990$$

Predicted Buckling Stress [Fic]:

Since $Fr \geq 2.439$, $Fic = Fy$

$$= 165559.953 \text{ kPa}$$

Factor of Safety [FS]:

Since $Fic [165560] \geq Fy [165560]$, $FS = 1.667$

Allowable Circumferential Compressive Stress [Fha]:

Since $Fr > 2.439$, $Fha = Fy/FS$

$$= 165559.953 / 1.667$$

$$= 99316.109 \text{ kPa}$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha (t / Do)$$

$$= 2 \cdot 99.3 (2.11/19)$$

$$= 22.002 \text{ N/mm}^2$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.7263 mm
Allowable Internal Pressure at Corroded thickness	22.46 N/mm ²
Required Internal Design Pressure	1.00 N/mm ²
Allowable External Pressure at Corroded thickness	22.00 N/mm ²
Required External Design Pressure	18.00 N/mm ²
Required Thickness due to Shell Side pressure	1.7263 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:



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Case#	--Reqd. Tbsht	Thk. + CA Extnsn	----	Tubesheet Bend	Allwd	Stresses Shear	Allwd	Case Type	Pass/ Fail
D1uc	16.628	...	10208	206984	3756	110392	Ps+Pt-Th	D1	Ok
D2uc	311.651	...	165479	206984	54505	110392	Ps+Pt-Th	D2	Ok
D3uc	301.191	...	156656	206984	50749	110392	Ps+Pt-Th	D3	Ok
O1uc	18.393	...	25571	460139	8800	110392	Ps+Pt+Th	O1	Ok
O2uc	191.177	...	168149	460139	58472	110392	Ps+Pt+Th	O2	Ok
O3uc	186.960	...	163737	460139	56594	110392	Ps+Pt+Th	O3	Ok
O4uc	24.874	...	30675	460139	10678	110392	Ps+Pt+Th	O4	Ok
D1c	22.713	...	10566	206984	3796	110392	Ps+Pt-Th-c	D1	Ok
D2c	315.563	...	168434	206984	55061	110392	Ps+Pt-Th-c	D2	Ok
D3c	305.272	...	159538	206984	51265	110392	Ps+Pt-Th-c	D3	Ok
O1c	24.259	...	26061	460139	8826	110392	Ps+Pt+Th-c	O1	Ok
O2c	195.605	...	170956	460139	59006	110392	Ps+Pt+Th-c	O2	Ok
O3c	191.437	...	166508	460139	57108	110392	Ps+Pt+Th-c	O3	Ok
O4c	30.722	...	31344	460139	10724	110392	Ps+Pt+Th-c	O4	Ok
Max:	315.5628	...	mm	0.814	0.535	(Str. Ratio)			

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and
Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	----	Shell Stresses	----	:	----	Shell Band Stress	----	:	Pass
Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	: Fail
D1uc	420	137989	...	...	...	...	...	...	Ok
D2uc	41306	137989	...	...	...	...	...	...	Ok
D3uc	41726	137989	...	...	...	...	...	...	Ok
O1uc	6549	463989	...	...	...	...	...	...	Ok
O2uc	42560	463989	...	...	...	...	...	...	Ok
O3uc	42770	463989	...	...	...	...	...	...	Ok
O4uc	6339	463989	...	...	...	...	...	...	Ok
D1c	442	137989	...	...	...	...	...	...	Ok
D2c	42246	137989	...	...	...	...	...	...	Ok
D3c	42689	137989	...	...	...	...	...	...	Ok
O1c	6603	463989	...	...	...	...	...	...	Ok
O2c	43427	463989	...	...	...	...	...	...	Ok
O3c	43648	463989	...	...	...	...	...	...	Ok
O4c	6382	463989	...	...	...	...	...	...	Ok



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Max RATIO 0.309 ...

Tube, Shell and Channel Stress Summary:

---- Tube Stresses Tube Loads Shell Stress Channel Stress Pass											
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	1249	92392	-102	-82779	140	10374	8474	206984	95487	206984	Ok
D2uc	19608	92392	-19608	-82779	2202	10374	239047	463989	13796	206984	Ok
D3uc	18359	92392	-18359	-82779	2061	10374	231412	463989	81692	206984	Ok
O1uc	13813	184785	-13813	-82779	1551	20749	30668	463989	29070	485189	Ok
O2uc	31632	184785	-31632	-82779	3552	20749	244103	463989	30770	485189	Ok
O3uc	31007	184785	-31007	-82779	3482	20749	240286	463989	16973	485189	Ok
O4uc	14437	184785	-14437	-82779	1621	20749	34486	463989	18673	485189	Ok
D1c	1258	92392	-176	-82779	141	10374	9122	206984	110447	206984	Ok
D2c	19479	92392	-19479	-82779	2187	10374	248517	463989	16019	206984	Ok
D3c	18220	92392	-18220	-82779	2046	10374	240280	463989	94428	206984	Ok
O1c	13729	184785	-13729	-82779	1542	20749	31580	463989	36276	485189	Ok
O2c	31439	184785	-31439	-82779	3530	20749	253621	463989	32994	485189	Ok
O3c	30809	184785	-30809	-82779	3460	20749	249502	463989	22229	485189	Ok
O4c	14358	184785	-14358	-82779	1612	20749	35699	463989	18948	485189	Ok

Max RATIO 0.212 0.382 0.212 0.547 0.534

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	315.563	355.000	Ok
Tube thickness :	1.726	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	3.000	Ok
Tube-tubesheet groove weld leg :	1.281	3.000	Ok

Min Shell length of thk, (145.000) adj. to tubesheet: 959.587 mm

Min Channel length of thk, (22.000) adj. to tubesheet: 373.776 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +1.637 mm
 Due to Pressure Shell Compresses by : -1.636 mm
 Due to Pressure + Thermal Shell Extends by : +1.650 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	19.591	1.000	22.157	1.000
Tubesheet Shear Stress	29.080	1.000	32.584	1.000
Tube Tensile Stress	73.419	1.000	84.820	1.000
Tube Compressive Stress	31.905	0.188	62.741	1.000
Tube-Tubesheet Joint load	73.420	1.000	84.820	1.000
Shell Stress (Axial, Junction)	5.081	0.100	22.002	0.993
Tubesheet-Channel Junction	4.393	1.000	22.002	0.993
Tube Pressure Stress	22.458	1.000	22.002	1.000
Minimum MAWP	4.393		22.002	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	0 shellside Stress Rat.	Shellside MAPnc	0 tubeside Stress Rat.
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Tubesheet Bending Stress	20.359	1.000	22.475	1.000
Tubesheet Shear Stress	29.409	1.000	36.400	1.000
Tube Tensile Stress	74.345	1.000	84.357	1.000
Tube Compressive Stress	900.900	1.000	82.166	1.000
Tube-Tubesheet Joint load	74.345	1.000	84.357	1.000
Shell Stress (Axial, Junction)	20.359	0.334	22.475	0.575
Tubesheet-Channel Junction	5.435	0.997	22.475	0.083
Tube Pressure Stress	22.458	1.000	23.919	1.000
Minimum MAPnc	5.435		22.475	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0256	0.0256	...	...	...	...
2	-1.6357	-1.6363	...	...	...	...
3	-1.6101	-1.6107	...	...	...	...
4	...	...	...	...	...	...
5	0.0128	0.0128	1.6367	1.6374	...	...
6	-1.4343	-1.4349	1.6367	1.6374	...	...
7	-1.4215	-1.4221	1.6367	1.6374	...	...
8	...	...	1.6367	1.6374	...	...

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